

Parallel lines

Two lines in a plane that never meet — and how you can tell without following them.

LESSON 81

1 × 40 MIN

MATEMATIKA 7, §37

S8 5.1

LESSON CLOCK

8'

12'

12'

6'

2'

The definition	8 min
The axiom of parallels	12 min
Always the same distance	12 min
Around the room	6 min
Homework	2 min

LEARNING OBJECTIVES

- Define parallel lines and use the symbol $\parallel$.
- Distinguish parallel lines from lines that merely look apart.
- State the axiom of parallels and what it guarantees.
- Recognise parallel lines in the figures and objects around the room.

TEXTBOOK

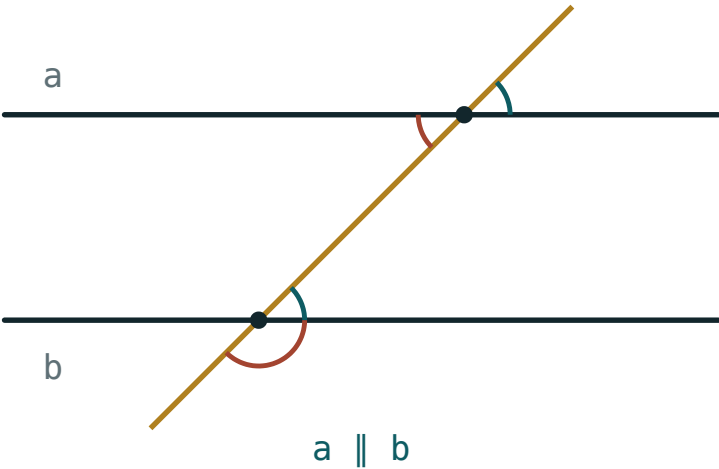
National	pp. 226–230
Cambridge	Stage 8, pp. 44–47
Workbook	Exercise 5.1

01

Explanation

The definition

Two lines in the same plane are **parallel** if they have no common point, however far they are produced in either direction. We write $a \parallel b$.



Two parallel lines and a transversal

TWO LINES IN A PLANE	COMMON POINTS	NAME
they cross	1	intersecting
they never meet	0	parallel
every point is shared	all	the same line

“THEY DO NOT MEET ON MY PAGE” PROVES NOTHING

A line has no end. Two lines drawn a little apart may cross far outside the page, so parallelism can never be checked by looking — it has to be established from angles, which is the next lesson.

The axiom of parallels

Through a point not on a line there passes exactly one line parallel to it.

“At least one” can be proved; “exactly one” cannot, and is taken as an **axiom**. It is the most famous statement in geometry — mathematicians tried for two thousand years to derive it, and the failure produced the non-Euclidean geometries.

CLAIM	STATUS
at least one parallel exists	a theorem
there is no more than one	the axiom
parallels stay the same distance apart	follows from the axiom

WHY THE AXIOM IS NEEDED

Without it a triangle's angles need not add to 180° . Every result of the next three lessons rests on it, which is why it is stated openly rather than assumed silently.

Always the same distance

The distance from a point to a line is measured along the perpendicular. If $a \parallel b$, that distance is the same wherever the point is taken on a .

LINES	DISTANCE BETWEEN THEM
parallel	constant
intersecting	falls to zero at the crossing

THIS IS HOW RAILS AND RULED PAPER WORK

A railway is built to a constant gauge precisely so that the rails are parallel; if the distance changed anywhere, the wheels would jam or drop. Constant distance and parallelism are the same fact stated two ways.

Around the room

OBJECT	PARALLEL PAIR	WHY IT MATTERS
ruled paper	the lines	writing stays level
a railway	the rails	constant gauge
a ladder	the two side rails	the rungs stay square
a window frame	opposite edges	the pane fits
a book's opposite edges	top and bottom	the pages stack

In each case the parallelism is not decoration: it is what makes the object work.

02 Worked examples

EXAMPLE 1 How many lines through a point M outside a line a are parallel to a ?

- ① At least one exists — this can be constructed.
- ② The axiom says there is no second.
- ③ So the number is exactly one.

ANSWER

1

EXAMPLE 2 Two lines in a plane are drawn 3 cm apart at one point and 3.4 cm apart at another. Are they parallel?

- ① Parallel lines keep a constant distance.
- ② Here the distance changes.
- ③ So they are not parallel — they meet somewhere.

ANSWER

No

EXAMPLE 3 If $a \parallel b$ and $b \parallel c$, what can be said about a and c ?

- ① Suppose a met c at a point P .
- ② Then through P there would be two lines parallel to b .
- ③ That contradicts the axiom.
- ④ So $a \parallel c$.

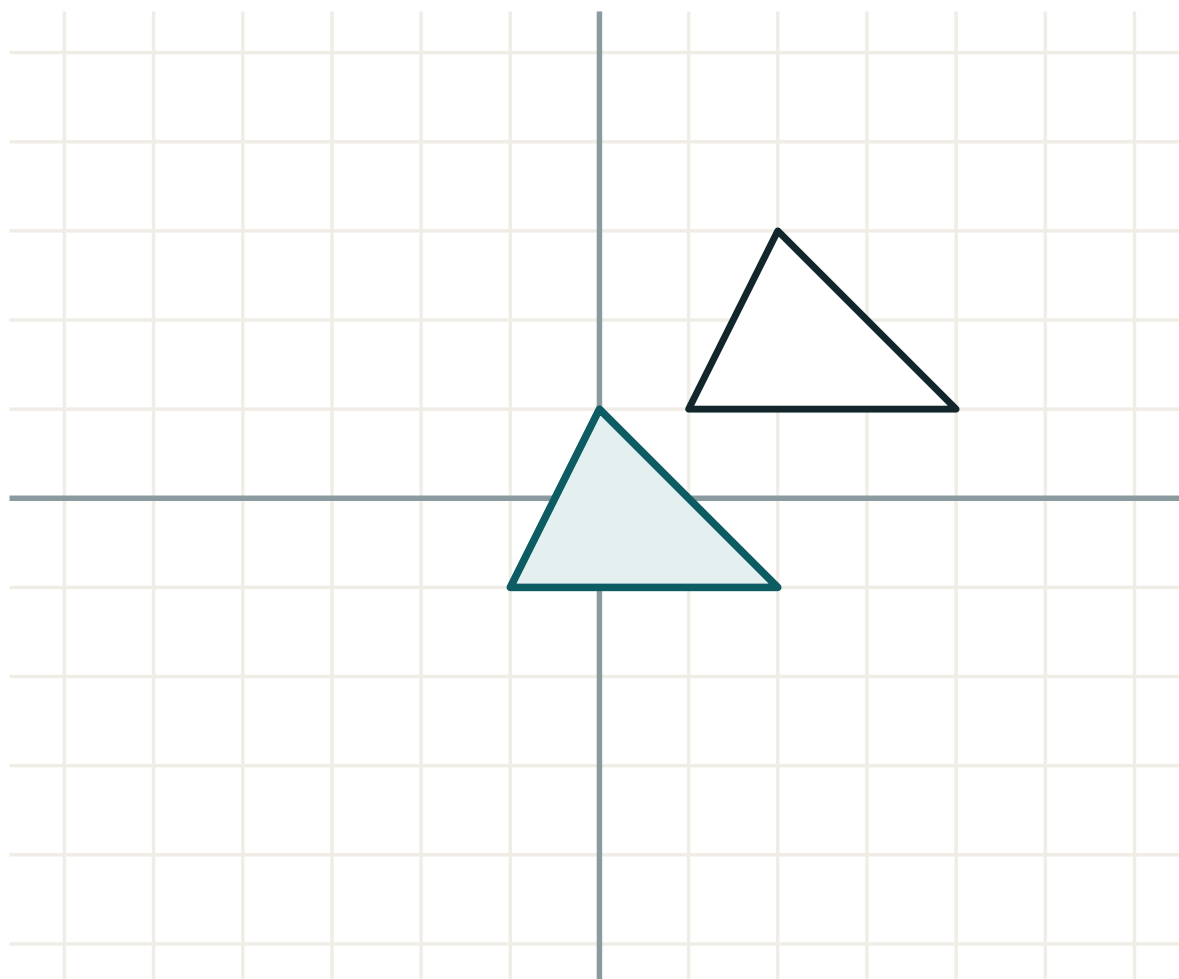
ANSWER

$a \parallel c$

03 Interactive model

Ask the class to draw two lines that “look parallel” and then measure the gap at both ends of the page; the ones that fail are the lesson.

A translated line is parallel to the original



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel lines	Parallel to'g'ri chiziqlar	Параллельные прямые
To intersect	Kesishmoq	Пересекаться
Plane	Tekislik	Плоскость
Axiom	Aksioma	Аксиома
Distance	Masofa	Расстояние
Direction	Yo'nalish	Направление
Unique	Yagona	Единственный
Symbol	Belgi	Обозначение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Parallel lines have how many common points?

The symbol for parallel is:

Through a point off a line, parallels number:

The distance between parallel lines is:

increasing

constant

decreasing

zero

Distance from a point to a line is measured:

any way

along the perpendicular

along the line

diagonally

If $a \parallel b$ and $b \parallel c$ then:

$a \perp c$

$a \parallel c$

$a = c$

nothing follows

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Two lines with no common point are

ANSWER parallel

2. Write “ a is parallel to b ” in symbols

ANSWER $a \parallel b$

3. Two lines crossing at one point are

ANSWER intersecting

4. Parallels through a point off a line

ANSWER 1

5. The distance between parallels is

ANSWER constant

6. Distance to a line is measured along the

ANSWER perpendicular

7. Name one parallel pair in the classroom

ANSWER e.g. the opposite edges of the board

Medium · the standard question

7 PROBLEMS

1. Gaps of 3 and 3.4 *cm*: parallel?

ANSWER No

2. $a \parallel b, b \parallel c$: is $a \parallel c$?

ANSWER Yes

3. Can two parallel lines lie in different planes?

ANSWER Not by this definition — the same plane is required

4. Is “exactly one parallel” a theorem or an axiom?

ANSWER An axiom

5. Is “at least one parallel” a theorem or an axiom?

ANSWER A theorem

6. Two rails 1520 *mm* apart everywhere are

ANSWER parallel

7. Why is looking at a drawing not a proof?

ANSWER The lines may meet beyond the page

Hard · stretch and reason

7 PROBLEMS

1. Prove that $a \parallel b$ and $b \parallel c$ force $a \parallel c$

ANSWER Two parallels to b through one point contradict the axiom

2. If $a \perp c$ and $b \perp c$ in a plane, then

ANSWER $a \parallel b$

3. How many lines are parallel to a in the whole plane?

ANSWER Infinitely many — one through each point off it

4. A quadrilateral with both pairs of opposite sides parallel is a

ANSWER parallelogram

5. Explain why the axiom cannot be checked by measurement

ANSWER Measurement covers a finite part of an endless line

6. What fails without the axiom of parallels?

ANSWER The angle sum of a triangle is no longer 180°

7. Two lines are 0 apart at one point: what are they?

ANSWER Intersecting there, or the same line

07 Homework

Homework — 5 tasks

Measure, do not eyeball: take the gap at both ends of the page.

1. Draw a line a and a point M off it, and construct the parallel through M .
2. Find five parallel pairs at home and say what each is for.
3. State the axiom of parallels in your own words.
4. Two lines are 4 cm apart at the left edge and 4 cm apart at the right. Are they parallel? What else would you check?

5. Explain why $a \parallel b$ and $b \parallel c$ give $a \parallel c$.

The criterion for two lines to be parallel

One pair of equal alternate angles is enough — parallelism becomes something you can check.

LESSON 82–83

2 × 40 MIN

MATEMATIKA 7, §38

S8 5.1

LESSON CLOCK

13'

23'

22'

18'

4'

The transversal and its eight angles	13 min
The three criteria	23 min
Using a criterion	22 min
Perpendiculars to one line	18 min
Homework	4 min

LEARNING OBJECTIVES

- Name the alternate, corresponding and co-interior angles at a transversal.
- State and use the three criteria for two lines to be parallel.
- Prove that two lines are parallel from an angle equality.
- Deduce that two perpendiculars to the same line are parallel.

TEXTBOOK

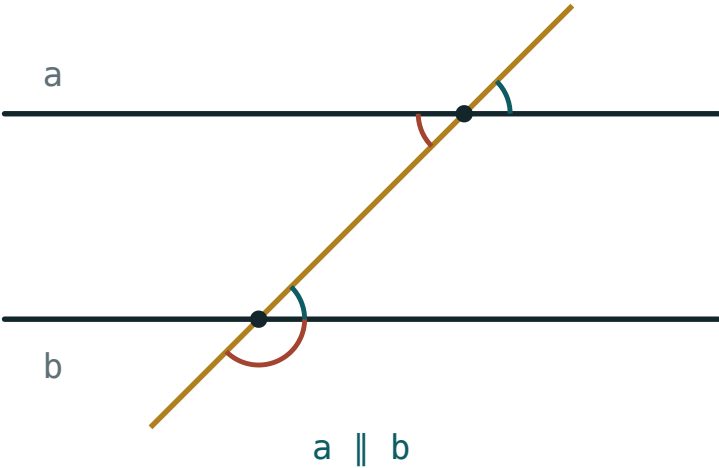
National	pp. 231–237
Cambridge	Stage 8, pp. 44–47
Workbook	Exercise 5.1

01

Explanation

The transversal and its eight angles

A line that cuts two others is a **transversal**. It makes eight angles, and three pairs of them have names.



The transversal and the named pairs

PAIR	WHERE THEY SIT	WHEN THE LINES ARE PARALLEL
alternate	between the lines, opposite sides of the transversal	equal
corresponding	same side, one above and one below	equal
co-interior	between the lines, same side	add to 180°

NAMES FIRST, FACTS SECOND

Half the errors in this chapter are naming errors, not reasoning errors. Point at a pair and say its name aloud before quoting any rule about it.

The three criteria

Each pair gives a criterion. Every one of them is a one-way street read this way: the angle fact comes first, and parallelism is the conclusion.

alternate equal $\implies a \parallel b$ corresponding equal $\implies a \parallel b$ co-interior sum $180^\circ \implies a \parallel b$

GIVEN	CONCLUSION
$\angle 1 = \angle 2$	alternate — the lines are parallel
$\angle 1 = \angle 3$	corresponding — the lines are parallel
$\angle 1 + \angle 4 = 180^\circ$	co-interior — the lines are parallel

ONE CORRECT PAIR IS ENOUGH – BUT IT MUST BE A NAMED PAIR

Two angles that happen to be equal prove nothing unless they are alternate or corresponding. Check the position first; the equality alone is not a criterion.

Using a criterion

The written proof is three lines: name the pair, quote the criterion, state the conclusion.

STATEMENT	REASON
$\angle 1 = 70^\circ, \angle 2 = 70^\circ$	given
they are alternate angles	position at the transversal
$a \parallel b$	the alternate-angle criterion

DATA	PAIR	PARALLEL?
$\angle 1 = 70^\circ, \angle 2 = 70^\circ$	alternate	yes
$\angle 1 = 110^\circ, \angle 4 = 70^\circ$	co-interior	yes
$\angle 1 = 70^\circ, \angle 4 = 80^\circ$	co-interior	no
$\angle 1 = 70^\circ, \angle 3 = 70^\circ$	corresponding	yes

Perpendiculars to one line

The neatest consequence: if two lines are both perpendicular to a third, they are parallel.

$$a \perp c \text{ and } b \perp c \implies a \parallel b$$

STATEMENT	REASON
$a \perp c, b \perp c$	given
the two angles at c are 90° each	definition of perpendicular
they are corresponding and equal	position at the transversal
$a \parallel b$	the corresponding-angle criterion

THIS IS WHAT A SET SQUARE DOES

Sliding a set square along a ruler draws parallel lines because every line it draws is perpendicular to the ruler. The tool is the theorem made of plastic.

02 Worked examples

EXAMPLE 1 A transversal makes $\angle 1 = 64^\circ$ and its alternate $\angle 2 = 64^\circ$. Are the lines parallel?

- ① The pair is alternate — check the position.
- ② They are equal.
- ③ By the alternate-angle criterion, $a \parallel b$.

ANSWER

Yes

EXAMPLE 2 Co-interior angles measure 115° and 65° . Are the lines parallel?

- ① $115^\circ + 65^\circ = 180^\circ$
- ② Co-interior angles adding to 180° is the criterion.
- ③ So the lines are parallel.

ANSWER

Yes

EXAMPLE 3 Prove that $a \perp c$ and $b \perp c$ give $a \parallel b$.

- ① Both lines meet c at 90° .
- ② Those two right angles are corresponding.
- ③ They are equal.
- ④ By the corresponding-angle criterion, $a \parallel b$.

ANSWER

$a \parallel b$

03 Interactive model

Draw one transversal on the board and have the class colour the three named pairs in three colours before any rule is stated; the naming errors disappear.

Name the pair, then decide

Angles between the lines, opposite sides of the transversal:

corresponding

alternate

co-interior

vertical

Angles on the same side, one above and one below:

corresponding

alternate

co-interior

adjacent

Angles between the lines on the same side:

corresponding

alternate

co-interior

vertical

Equal alternate angles give:

perpendicular lines

parallel lines

nothing

a triangle

Co-interior angles that add to 180° give:

parallel lines

intersecting lines

nothing

equal lines

$\angle 1 = 70^\circ$ and co-interior $\angle 4 = 80^\circ$:

parallel

not parallel

perpendicular

cannot say

Two lines perpendicular to a third are:

perpendicular

parallel

skew

equal

Two equal angles that are not a named pair prove:

parallelism

nothing

perpendicularity

congruence

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Transversal	Kesuvchi	Секущая
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Corresponding angles	Mos burchaklar	Соответственные углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Criterion	Alomat	Признак
To conclude	Xulosa chiqarmoq	Сделать вывод
Perpendicular	Perpendikulyar	Перпендикуляр
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A line cutting two others is a:

bisector

transversal

median

chord

A transversal makes how many angles?

4

6

8

12

Alternate angles equal $\Rightarrow$

the lines meet

the lines are parallel

nothing

a right angle

Co-interior angles of parallel lines add to:

90°

180°

270°

360°

Corresponding angles of 64° and 64° give:

parallel lines

perpendicular lines

nothing

a triangle

$a \perp c$ and $b \perp c$ give:

$a \perp b$

$a \parallel b$

$a = b$

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the pair between the lines, opposite sides

ANSWER alternate

2. Name the pair on the same side, one above one below

ANSWER corresponding

3. Name the pair between the lines, same side

ANSWER co-interior

4. Alternate angles 70° and 70° : parallel?

ANSWER Yes

5. Corresponding angles 50° and 60° : parallel?

ANSWER No

6. Co-interior 100° and 80° : parallel?

ANSWER Yes

7. A transversal makes how many angles?

ANSWER 8

Medium · the standard question

7 PROBLEMS

1. Co-interior 115° and 65°

ANSWER Parallel

2. Co-interior 70° and 80°

ANSWER Not parallel

3. Alternate $3x$ and 75° with the lines parallel: find x

ANSWER $x = 25$

4. Corresponding $2x + 10$ and 80° with the lines parallel

ANSWER $x = 35$

5. $a \perp c$, $b \perp c$: what follows?

ANSWER $a \parallel b$

6. Write the three-line proof for equal alternate angles

ANSWER given; alternate; criterion

7. Two equal angles that are vertical prove parallelism?

ANSWER No — not a named pair

Hard · stretch and reason

7 PROBLEMS

1. Co-interior angles are x and $2x$ with the lines parallel: find both

ANSWER 60° and 120°

2. Alternate angles are $4x - 10$ and $3x + 20$: find x for parallelism

ANSWER $x = 30$

3. Co-interior $5x$ and $4x$: find x

ANSWER $x = 20$

4. Why is each criterion stated as an implication, not an equality?

ANSWER The angle fact is the evidence; parallelism is the conclusion

5. A set square slid along a ruler draws parallels — why?

ANSWER Every line drawn is perpendicular to the ruler

6. Corresponding angles $x + 30$ and $2x - 10$: find x

ANSWER $x = 40$

7. Given $\angle 1 = 108^\circ$, find the co-interior angle that makes the lines parallel

ANSWER 72°

07 Homework

Homework — 5 tasks

Name the pair in words before you write a single number.

1. Draw a transversal across two lines and label all three named pairs.
2. Alternate angles are $5x$ and 65° . Find x if the lines are parallel.
3. Co-interior angles are $3x$ and $2x$. Find both.
4. Write the three-line proof that $a \perp c$ and $b \perp c$ give $a \parallel b$.
5. Explain why two equal vertical angles do not prove that two lines are parallel.

Angles formed when two parallel lines are cut by a transversal

The criteria read backwards: now parallelism is given, and the angles are the conclusion.

LESSON 84–85

2 × 40 MIN

MATEMATIKA 7, §39

S8 5.1

LESSON CLOCK

13'

23'

22'

18'

4'

The properties	13 min
All eight from one	23 min
Property or criterion?	22 min
Problems	18 min
Homework	4 min

LEARNING OBJECTIVES

- State the three properties of angles on parallel lines.
- Find every one of the eight angles from a single given one.
- Distinguish a property from a criterion, and use each in its own direction.
- Solve angle problems in which parallel lines are given.

TEXTBOOK

National	pp. 238–245
Cambridge	Stage 8, pp. 44–47
Workbook	Exercise 5.1

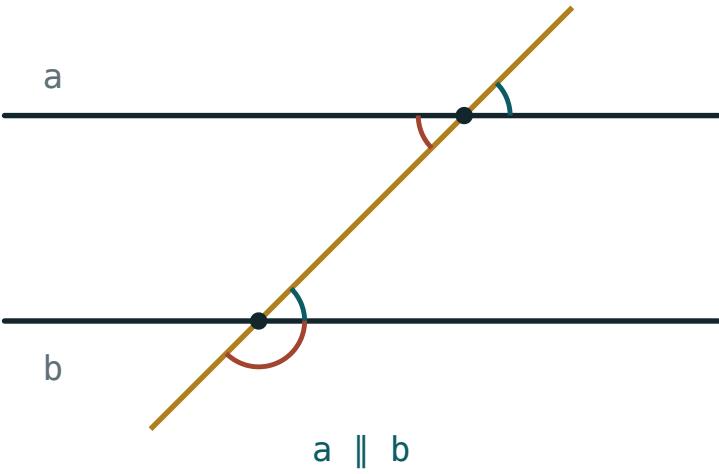
01

Explanation

The properties

If the lines **are** parallel, the same three pairs behave as follows.

$a \parallel b \implies$ alternate equal, corresponding equal, co-interior add to 180°



One angle decides all eight

PAIR	PROPERTY
alternate	equal
corresponding	equal
co-interior	sum 180°

TWO THEOREMS, ONE PICTURE

Last lesson the angle fact was the evidence and parallelism the conclusion. This lesson the arrow is reversed. The picture is identical; only the direction of the reasoning changes.

All eight from one

Given one angle at a transversal cutting parallel lines, every other angle follows. With $\angle 1 = 70^\circ$:

ANGLE	RELATION TO $\angle 1$	SIZE
$\angle 2$	supplementary	110°
$\angle 3$	vertical	70°
$\angle 4$	supplementary	110°
$\angle 5$	corresponding	70°
$\angle 6$	co-interior	110°
$\angle 7$	alternate	70°
$\angle 8$	supplementary to $\angle 7$	110°

ONLY TWO SIZES APPEAR

Every angle is either 70° or 110° , and the two add to 180° . Once you see that, the eight angles collapse into one question: which of the two is this?

Property or criterion?

Which one you may use depends entirely on what you were given.

GIVEN	WANTED	USE
$a \parallel b$	an angle	the property
an angle equality	$a \parallel b$	the criterion
$a \parallel b$	$a \parallel b$	nothing to prove

USING THE PROPERTY TO PROVE PARALLELISM IS CIRCULAR

Writing “the angles are alternate, so they are equal, so the lines are parallel” assumes what it sets out to prove. If parallelism is the goal, the angle equality must be given, not deduced.

Problems

PROBLEM	REASONING	ANSWER
$a \parallel b$, one co-interior angle is $3x$, the other $2x$	$5x = 180$	36° and 144°
$a \parallel b$, corresponding angles $4x - 20$ and $2x + 40$	$2x = 60$	$x = 30$
$a \parallel b$, alternate angles are equal and one is $5x$, the other 105°	$5x = 105$	$x = 21$

A PARALLEL DRAWN IN IS OFTEN THE WHOLE SOLUTION

Where a figure gives no transversal, drawing a line through a vertex parallel to a given side creates alternate angles out of nothing. The next lesson proves the angle sum of a triangle in exactly this way.

02 Worked examples

EXAMPLE 1 $a \parallel b$ and one angle at the transversal is 70° . Find the co-interior angle and the alternate angle.

- 1 Alternate angles are equal: 70° .
- 2 Co-interior angles add to 180° .
- 3 $180^\circ - 70^\circ = 110^\circ$

ANSWER

Alternate 70° , co-interior 110°

EXAMPLE 2 $a \parallel b$. Co-interior angles are $3x$ and $2x$. Find them.

- 1 $3x + 2x = 180^\circ$
The co-interior property.
- 2 $5x = 180^\circ$
- 3 $x = 36^\circ$
- 4 The angles are 108° and 72° .

ANSWER

108° and 72°

EXAMPLE 3 $a \parallel b$. Corresponding angles are $4x - 20$ and $2x + 40$. Find x .

- 1 Corresponding angles are equal.
- 2 $4x - 20 = 2x + 40$
- 3 $2x = 60$
- 4 $x = 30$

ANSWER

$x = 30$

03 Interactive model

Mark one angle on the board and ask the class to fill in the other seven in silence; the eight-angle picture is learnt in a single minute this way.

Which of the two sizes is it?

$a \parallel b$, $\angle 1 = 70^\circ$. Its vertical angle:

Its corresponding angle:

Its co-interior angle:

Its alternate angle:

70°

110°

35°

180°

How many distinct sizes appear among the eight?

1

2

4

8

Given $a \parallel b$ and asked for an angle, you use:

the criterion

the property

either

neither

Given equal alternate angles and asked to prove $a \parallel b$:

the criterion

the property

either

neither

Co-interior angles $3x$ and $2x$ on parallel lines:

$$x = 30$$

$$x = 36$$

$$x = 45$$

$$x = 60$$

04 Terminology

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Criterion	Alomat	Признак
Converse	Teskari teorema	Обратная теорема
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Corresponding angles	Mos burchaklar	Соответственные углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Vertical angles	Vertikal burchaklar	Вертикальные углы
Supplementary	Qo'shni (yig'indisi 180°)	Смежные
Angle	Burchak	Угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

On parallel lines, alternate angles are:

equal

supplementary

complementary

unrelated

On parallel lines, co-interior angles:

are equal

add to 180°

add to 90°

are right

If one angle is 70° , the others are:

70° only

70° or 110°

110° only

all different

The property is used when the given is:

an angle equality

parallelism

a length

nothing

Corresponding angles $4x - 20$ and $2x + 40$ give:

$$x = 10$$

$$x = 30$$

$$x = 20$$

$$x = 60$$

Proving parallelism from the property is:

valid

circular

quicker

the same thing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $a \parallel b$, alternate to 70°

ANSWER 70°

2. $a \parallel b$, corresponding to 70°

ANSWER 70°

3. $a \parallel b$, co-interior with 70°

ANSWER 110°

4. The vertical angle of 70°

ANSWER 70°

5. The supplement of 70°

ANSWER 110°

6. How many distinct sizes among the eight?

ANSWER 2

7. $a \parallel b$, co-interior with 90°

ANSWER 90°

Medium · the standard question

7 PROBLEMS

1. Co-interior $3x$ and $2x$

ANSWER 108° and 72°

2. Corresponding $4x - 20$ and $2x + 40$

ANSWER $x = 30$

3. Alternate $5x$ and 105°

ANSWER $x = 21$

4. One angle is 130° : its co-interior angle

ANSWER 50°

5. One angle is 130° : its alternate angle

ANSWER 130°

6. Co-interior angles are equal: how big is each?

ANSWER 90°

7. Given $a \parallel b$, which do you use, property or criterion?

ANSWER The property

Hard · stretch and reason

7 PROBLEMS

1. Co-interior $x + 40$ and $2x - 10$ on parallel lines

ANSWER $x = 50$

2. Alternate $7x - 15$ and $5x + 9$

ANSWER $x = 12$

3. A transversal cuts three parallel lines; one angle is 64° : how many angles equal 64° ?

ANSWER 6 — three crossings, two each

4. Why does a proof of parallelism need the criterion, not the property?

ANSWER The property starts from parallelism, which is what is being proved

5. $a \parallel b$ and a transversal is perpendicular to a : what are all eight angles?

ANSWER All 90°

6. Co-interior $4x$ and $5x$

ANSWER 80° and 100°

7. One angle is α : write every other angle in terms of α

ANSWER α or $180^\circ - \alpha$

07 Homework

Homework — 5 tasks

Write the name of the pair beside every number you find.

1. Mark one angle of 55° at a transversal cutting parallel lines and find the other seven.
2. Co-interior angles are $5x$ and $4x$. Find both.
3. Alternate angles are $3x + 12$ and $4x - 8$. Find x .
4. Explain in two sentences the difference between the property and the criterion.
5. A transversal is perpendicular to one of two parallel lines. What can you say about the other?

The sum of the interior angles of a triangle

180° for every triangle, proved with one line drawn parallel to a side — and the exterior angle that follows.

LESSON 86–87

2 × 40 MIN

MATEMATIKA 7, §40

S8 5.2

LESSON CLOCK

9'

23'

19'

24'

5'

The experiment	9 min
The proof	23 min
Finding a missing angle	19 min
The exterior angle	24 min
Homework	5 min

LEARNING OBJECTIVES

- Prove that the angles of a triangle add to 180°.
- Find a missing angle of a triangle.
- State and use the exterior-angle theorem.
- Deduce that a triangle has at most one right or obtuse angle.

TEXTBOOK

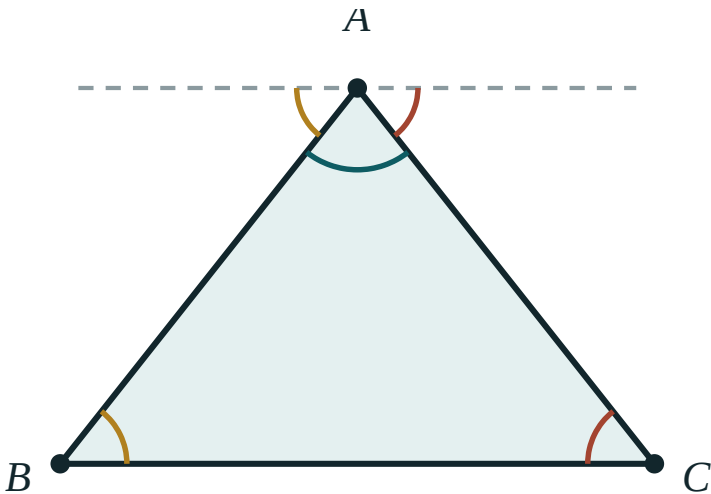
National	pp. 246–253
Cambridge	Stage 8, pp. 48–51
Workbook	Exercise 5.2

01

Explanation

The experiment

Tear the three corners off a paper triangle and lay them side by side: they form a straight angle. Every triangle in the class gives the same result, whatever its shape.



the three angles fill 180°

The three angles laid along a straight line

TRIANGLE	ANGLES	SUM
equilateral	$60^\circ, 60^\circ, 60^\circ$	180°
right	$90^\circ, 55^\circ, 35^\circ$	180°
obtuse	$120^\circ, 40^\circ, 20^\circ$	180°

THE EXPERIMENT IS NOT THE PROOF

Tearing paper checks three triangles; the theorem claims it for all of them. That gap is exactly what a proof closes, and here the proof needs the parallel lines of the last two lessons.

The proof

Through the vertex A draw the line parallel to BC . Now AB and AC are transversals of a pair of parallel lines.

STATEMENT	REASON
draw $\ell \parallel BC$ through A	the axiom of parallels
$\angle 1 = \angle B$	alternate angles, AB a transversal
$\angle 2 = \angle C$	alternate angles, AC a transversal
$\angle 1 + \angle A + \angle 2 = 180^\circ$	angles on a straight line at A
$\angle A + \angle B + \angle C = 180^\circ$	substituting the first two lines

$$\angle A + \angle B + \angle C = 180^\circ$$

WHERE THE AXIOM HIDES

The proof used the parallel through A — and needed it to be the only one. This theorem is therefore a consequence of the axiom of parallels, not an independent fact.

Finding a missing angle

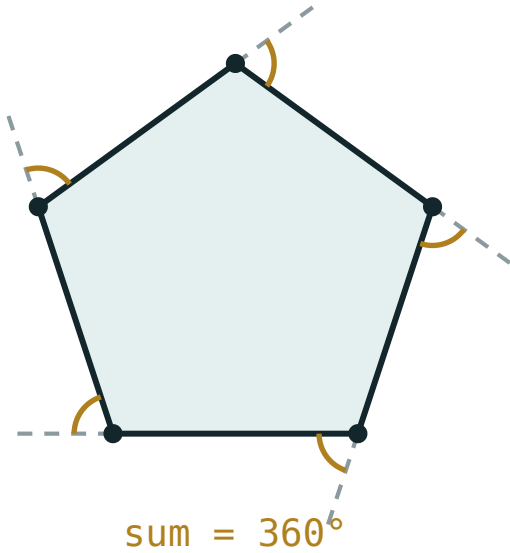
GIVEN	WORKING	THIRD ANGLE
$50^{\circ}, 60^{\circ}$	$180 - 110$	70°
$90^{\circ}, 35^{\circ}$	$180 - 125$	55°
isosceles, apex 40°	$(180 - 40) \div 2$	70° each
angles in ratio $2 : 3 : 4$	$9x = 180$	$40^{\circ}, 60^{\circ}, 80^{\circ}$

THREE CONSEQUENCES, FREE OF CHARGE

A triangle has at most one right angle and at most one obtuse angle, since two of them already reach 180° ; and an equiangular triangle has three angles of 60° .

The exterior angle

Produce one side. The angle outside the triangle beside $\angle C$ is an **exterior angle**, and it equals the sum of the two interior angles not next to it.



An exterior angle equals the two remote interior angles

$\angle_{ext} = \angle A + \angle B$

STATEMENT	REASON
$\angle_{ext} + \angle C = 180^\circ$	angles on a straight line
$\angle A + \angle B + \angle C = 180^\circ$	the angle sum
$\angle_{ext} = \angle A + \angle B$	comparing the two lines

THE TWO REMOTE ANGLES, NOT ALL THREE

An exterior angle equals the sum of the two interior angles at the other vertices. Adding $\angle C$ as well is the standard slip, and it always gives an answer over 180° .

02

Worked examples

EXAMPLE 1 Two angles of a triangle are 50° and 60° . Find the third.

1

$50^\circ + 60^\circ = 110^\circ$

2

$180^\circ - 110^\circ = 70^\circ$

ANSWER

70°

EXAMPLE 2 The angles of a triangle are in the ratio 2 : 3 : 4. Find them.

- ① Let them be $2x$, $3x$, $4x$.
- ② $2x + 3x + 4x = 180^\circ$
- ③ $9x = 180^\circ$, so $x = 20^\circ$.
- ④ The angles are 40° , 60° , 80° .

ANSWER

40° , 60° , 80°

EXAMPLE 3 An exterior angle of a triangle is 120° and one remote interior angle is 45° . Find the other two interior angles.

- ① The remote pair sums to the exterior angle.
- ② $120^\circ - 45^\circ = 75^\circ$
The second remote angle.
- ③ The adjacent angle is $180^\circ - 120^\circ = 60^\circ$.
- ④ Check: $45^\circ + 75^\circ + 60^\circ = 180^\circ$ ✓

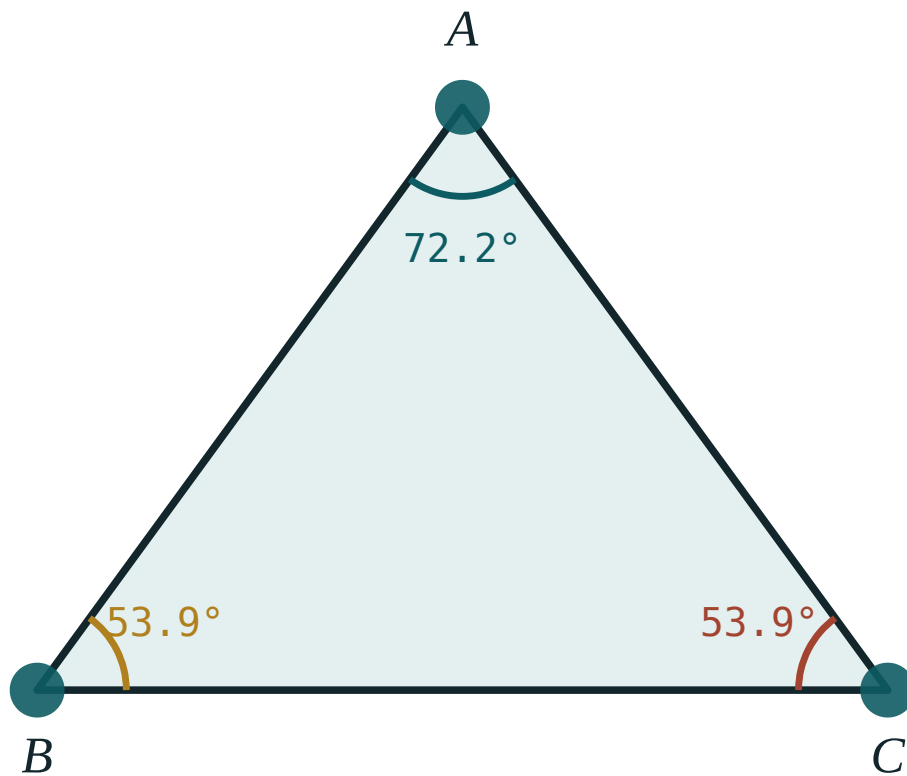
ANSWER

75° and 60°

03 Interactive model

Have the class tear the corners first and only then draw the parallel through the vertex; the proof lands as the explanation of something they have already seen happen.

Drag a vertex — the sum never moves



$\angle A$ 72.2°

$\angle B$ 53.9°

$\angle C$ 53.9°

sum 180°

longest side **BC**

largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Interior angle	Ichki burchak	Внутренний угол
Exterior angle	Tashqi burchak	Внешний угол
Angle sum	Burchaklar yig'indisi	Сумма углов
Auxiliary line	Yordamchi chiziq	Вспомогательная линия
Theorem	Teorema	Теорема
Acute-angled	O'tkir burchakli	Остроугольный
Obtuse-angled	O'tmas burchakli	Тупоугольный
Equiangular	Teng burchakli	Равноугольный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angles of any triangle add to:

The proof draws through a vertex a line:

Two angles are 90° and 35° ; the third is:

A triangle can have at most how many right angles?

1

2

3

any number

An exterior angle equals:

the adjacent angle

the sum of the two remote angles

the sum of all three

180°

An equiangular triangle has angles of:

45°

60°

90°

30°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 50° and 60° : the third angle

ANSWER 70°

2. 90° and 35°

ANSWER 55°

3. 120° and 40°

ANSWER 20°

4. Each angle of an equiangular triangle

ANSWER 60°

5. Isosceles, apex 40° : the base angles

ANSWER 70° each

6. Isosceles, base angle 50° : the apex

ANSWER 80°

7. Exterior 110° : the adjacent interior angle

ANSWER 70°

Medium · the standard question

7 PROBLEMS

1. Angles in the ratio $2 : 3 : 4$

ANSWER $40^\circ, 60^\circ, 80^\circ$

2. Angles in the ratio $1 : 2 : 3$

ANSWER $30^\circ, 60^\circ, 90^\circ$

3. Exterior 120° , one remote angle 45° : the other

ANSWER 75°

4. A right triangle with one angle 37° : the third

ANSWER 53°

5. Can a triangle have two obtuse angles?

ANSWER No — they already exceed 180°

6. The angles are $x, 2x, 3x$: find x

ANSWER $x = 30^\circ$

7. Exterior angles of a triangle add to

ANSWER 360°

Hard · stretch and reason

7 PROBLEMS

1. One angle is 20° more than the second and 40° more than the third: find all three

ANSWER $80^\circ, 60^\circ, 40^\circ$

2. An exterior angle is 3 times its adjacent interior angle: find both

ANSWER 135° and 45°

3. An isosceles triangle has an exterior angle of 100° at the apex: find all angles

ANSWER Apex 80° , base angles 50° each

4. The bisectors of $\angle B$ and $\angle C$ meet at I : find $\angle BIC$ if $\angle A = 60^\circ$

ANSWER 120°

5. Prove that a triangle has at most one obtuse angle

ANSWER Two obtuse angles exceed 180° on their own

6. Angles $2x - 10$, $x + 30$, $3x + 10$: find x

ANSWER $x = 25$

7. Why does the proof need the axiom of parallels?

ANSWER It uses the unique parallel through the vertex

07 Homework

Homework — 5 tasks

Check every answer by adding your three angles back to 180° .

- Two angles are 72° and 43° . Find the third.
- The angles are in the ratio $3 : 4 : 5$. Find them.
- Write out the proof of the angle sum in statement-and-reason form.
- An exterior angle is 115° and one remote interior angle is 52° . Find the other two interior angles.
- Explain why a triangle cannot have two right angles.

Right-angled triangles

Acute angles that add to 90° , the hypotenuse as the longest side, and the 30° half-side rule.

LESSON 88–89

2 × 40 MIN

MATEMATIKA 7, §41

S8 5.2

LESSON CLOCK

9'

19'

23'

24'

5'

Legs and hypotenuse	9 min
The acute angles	19 min
The 30° rule	23 min
Congruence tests	24 min
Homework	5 min

LEARNING OBJECTIVES

- Name the legs and the hypotenuse and show that the hypotenuse is longest.
- Use the fact that the two acute angles add to 90° .
- Prove and apply the 30° rule: the leg opposite 30° is half the hypotenuse.
- Use the congruence tests special to right-angled triangles.

TEXTBOOK

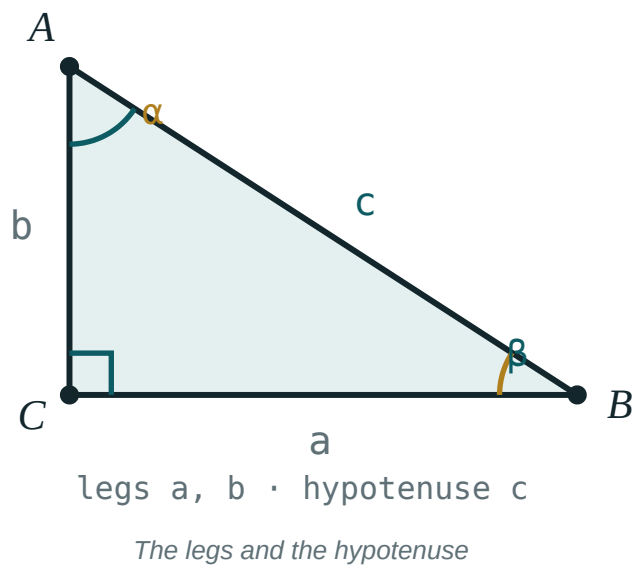
National	pp. 254–261
Cambridge	Stage 8, pp. 48–51
Workbook	Exercise 5.2

01

Explanation

Legs and hypotenuse

In a right-angled triangle the side opposite the right angle is the **hypotenuse**; the other two are the **legs**.



SIDE	OPPOSITE	LENGTH
hypotenuse	90°	the longest
each leg	an acute angle	shorter than the hypotenuse

WHY THE HYPOTENUSE MUST BE LONGEST

The larger angle faces the longer side, and the right angle is the largest of the three. The hypotenuse therefore beats both legs — before any calculation is done.

The acute angles

The three angles add to 180° and one of them is 90° , so the other two add to 90° : they are **complementary**.

$$\angle A + \angle B = 90^\circ \text{ when } \angle C = 90^\circ$$

ONE ACUTE ANGLE	THE OTHER
30°	60°
45°	45°
37°	53°
72°	18°

COMPLEMENTARY, NOT SUPPLEMENTARY

The two acute angles add to 90° , not 180° . Subtracting from 180° here is the commonest slip of the topic and always gives an impossible answer.

The 30° rule

In a right-angled triangle with an angle of 30° , the leg opposite it is exactly half the hypotenuse — and the statement runs backwards too.

$$\angle A = 30^\circ \iff BC = \frac{1}{2} \cdot AB$$

STATEMENT	REASON
reflect the triangle in the leg AC	construction
the two copies form a triangle with three 60° angles	$30^\circ + 30^\circ = 60^\circ$
that triangle is equilateral	equiangular $\implies$ equilateral
the doubled leg equals the hypotenuse	equal sides
$BC = \frac{1}{2} \cdot AB$	halving

HYPOTENUSE	LEG OPPOSITE 30°
10 cm	5 cm
7 cm	3.5 cm
24 cm	12 cm

THIS IS WHERE A RAMP'S RULE COMES FROM

A ramp at 30° rises one metre for every two metres of ramp. Builders quote it as a ratio; it is this theorem.

Congruence tests

Because one angle is already known to be equal, right-angled triangles need less data than others.

TEST	WHAT MUST MATCH	WHICH GENERAL CRITERION IT IS
two legs	both legs	the first — the right angle is included
a leg and an acute angle	one leg, one acute angle	the second
hypotenuse and an acute angle	hypotenuse, one acute angle	the second
hypotenuse and a leg	hypotenuse, one leg	special to right triangles

TWO ANGLES ARE NEVER ENOUGH

Knowing the right angle and one acute angle fixes the shape but not the size: the triangles are the same shape at any scale. At least one side must be given.

02**Worked examples**

EXAMPLE 1 One acute angle of a right-angled triangle is 37° . Find the other.

① The acute angles are complementary.

② $90^\circ - 37^\circ = 53^\circ$

ANSWER 53°

EXAMPLE 2 The hypotenuse is 16 cm and one angle is 30° . Find the leg opposite it.

① The leg opposite 30° is half the hypotenuse.

② $16 \div 2 = 8$

③ The leg is 8 cm .

ANSWER 8 cm

EXAMPLE 3 A leg is 6 cm and the hypotenuse is 12 cm. Find all three angles.

- 1 The leg is half the hypotenuse.
- 2 By the converse of the 30° rule, the angle opposite it is 30° .
- 3 The other acute angle is $90^\circ - 30^\circ = 60^\circ$.
- 4 The angles are $30^\circ, 60^\circ, 90^\circ$.

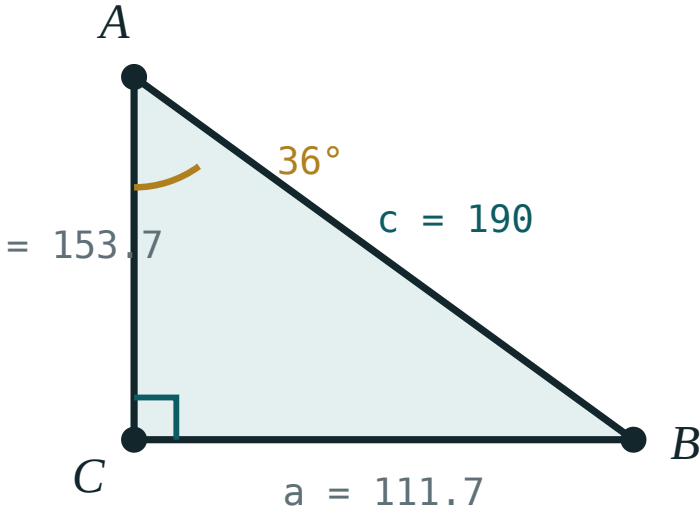
ANSWER

$30^\circ, 60^\circ, 90^\circ$

03 Interactive model

Give each pair a triangle with hypotenuse 10 cm and an angle of 30° and ask them to measure the short leg; every group reports 5 cm, and the theorem is already believed before it is proved.

The two acute angles always add to 90°



α 36°	β 54°	a (opposite) 111.7	b (adjacent) 153.7	c (hyp) 190	$\sin \alpha$ 0.59
$\cos \alpha$ 0.81	$\tan \alpha$ 0.73	$a^2 + b^2$ 36100	c^2 36100		

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Right-angled triangle	To'g'ri burchakli uchburchak	Прямоугольный треугольник
Hypotenuse	Gipotenuza	Гипотенуза
Leg	Katet	Катет
Acute angle	O'tkir burchak	Острый угол
Complementary	Yig'indisi 90° bo'lgan	Дополнительные до 90°
Opposite	Qarshi	Противолежащий
Half	Yarim	Половина
Congruence test	Tenglik alomati	Признак равенства

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The side opposite the right angle is the:

leg

hypotenuse

median

base

The hypotenuse is:

the shortest side

the longest side

equal to a leg

any side

The two acute angles add to:

45°

90°

180°

360°

One acute angle is 72° ; the other is:

18°

28°

108°

118°

The leg opposite 30° is:

the hypotenuse

half the hypotenuse

twice the hypotenuse

equal to the other leg

Two right-angled triangles with two equal angles are:

congruent

not necessarily congruent

equal in area

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. One acute angle 37° : the other

ANSWER 53°

2. One acute angle 45° : the other

ANSWER 45°

3. One acute angle 72° : the other

ANSWER 18°

4. Hypotenuse 10 cm , angle 30° : the opposite leg

ANSWER 5 cm

5. Hypotenuse 7 cm , angle 30° : the opposite leg

ANSWER 3.5 cm

6. The longest side of a right-angled triangle

ANSWER the hypotenuse

7. Name the sides beside the right angle

ANSWER the legs

Medium · the standard question

7 PROBLEMS

1. Leg 6, hypotenuse 12: the angles

ANSWER $30^\circ, 60^\circ, 90^\circ$

2. Hypotenuse 24 cm, angle 60° : the shorter leg

ANSWER 12 cm

3. One acute angle is twice the other: find both

ANSWER 30° and 60°

4. One acute angle is 10° more than the other

ANSWER 40° and 50°

5. Can a right-angled triangle be obtuse-angled?

ANSWER No

6. Can a right-angled triangle be isosceles?

ANSWER Yes — the acute angles are 45° each

7. A ramp at 30° is 6 m long: its rise

ANSWER 3 m

Hard · stretch and reason

7 PROBLEMS

1. One acute angle is 4 times the other: find both

ANSWER 18° and 72°

2. The hypotenuse is 2 times a leg: find the angles

ANSWER 30° , 60° , 90°

3. A leg is 9 cm and it faces 30° : find the hypotenuse

ANSWER 18 cm

4. Which test lets two right triangles be congruent from a hypotenuse and a leg?

ANSWER The test special to right-angled triangles

5. Prove the 30° rule by reflection

ANSWER The two copies form an equilateral triangle

6. The acute angles are $2x$ and $3x$: find them

ANSWER 36° and 54°

7. Why are two equal angles not enough for congruence here?

ANSWER The size is not fixed — only the shape

07 Homework

Homework — 5 tasks

Subtract from 90° , never from 180° , when finding the second acute angle.

1. One acute angle is 28° . Find the other.
2. The hypotenuse is 18 cm and one angle is 30° . Find the opposite leg.
3. A leg is 7 cm and the hypotenuse 14 cm. Find all three angles.
4. The acute angles are in the ratio 1 : 5. Find them.
5. List the four congruence tests for right-angled triangles and say which general criterion each is.

The property of the bisector of an angle

Every point of a bisector is the same distance from the two arms — and only those points are.

LESSON 90–91

2 × 40 MIN

MATEMATIKA 7, §42

S8 5.3

LESSON CLOCK

9'

23'

19'

24'

5'

What distance means here	9 min
The property	23 min
The converse	19 min
The construction and its use	24 min
Homework	5 min

LEARNING OBJECTIVES

- State the bisector property and its converse.
- Prove the property using congruent right-angled triangles.
- Construct the bisector of an angle with compasses.
- Use the property to find the point equidistant from two lines.

TEXTBOOK

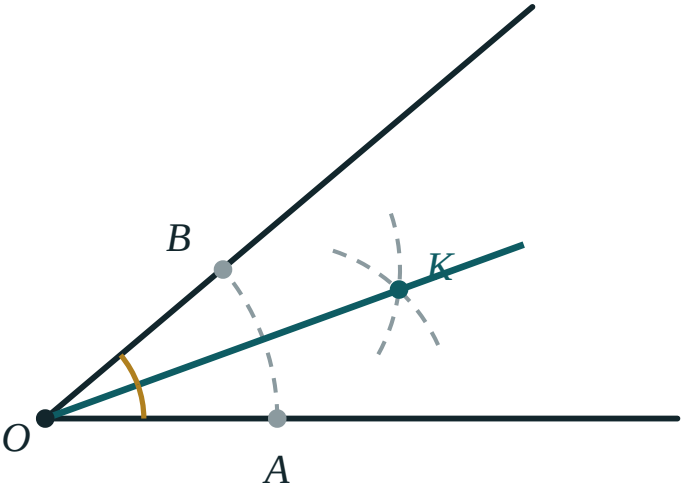
National	pp. 262–268
Cambridge	Stage 8, pp. 52–56
Workbook	Exercise 5.3

01

Explanation

What distance means here

The distance from a point to a line is the length of the perpendicular dropped from the point to the line — not any slanting segment.



Dropping perpendiculars to both arms

SEGMENT FROM M TO THE LINE	IS IT THE DISTANCE?
perpendicular	yes — and it is the shortest
slanting	no — it is longer

MEASURE TO THE ARM, NOT TO THE VERTEX

The distance from a point on the bisector to the vertex is not what the theorem is about. Every statement below concerns the perpendiculars to the two arms.

The property

Let M lie on the bisector of $\angle AOB$, and drop perpendiculars MP and MQ to the arms.

STATEMENT	REASON
$\angle MOP = \angle MOQ$	OM is the bisector
$\angle MPO = \angle MQO = 90^\circ$	construction
$OM = OM$	common hypotenuse
$\triangle MPO \equiv \triangle MQO$	hypotenuse and an acute angle
$MP = MQ$	corresponding sides of congruent triangles

M on the bisector $\implies MP = MQ$

THE RIGHT-ANGLE CONGRUENCE TEST EARNS ITS KEEP

The proof needs exactly the test met last lesson: a common hypotenuse and one equal acute angle. That is why the two topics stand next to each other in the plan.

The converse

The statement also runs backwards: a point inside the angle that is equidistant from both arms lies on the bisector.

DIRECTION	GIVEN	CONCLUSION
property	M	$MP = MQ$
converse	$MP = MQ$	M

Taken together the two say: the bisector is **exactly** the set of points inside the angle equidistant from the arms — a locus, not merely a line with a nice property.

WHERE THE THREE BISECTORS MEET

Each bisector of a triangle is the locus of points equidistant from two sides, so the point common to two of them is equidistant from all three — and therefore lies on the third. That point is the incentre, the centre of the inscribed circle.

The construction and its use

STEP	WHAT TO DO
1	with centre O , cut both arms at A and B
2	with the same radius and centres A and B , draw two arcs
3	call their intersection M
4	draw OM — it bisects the angle

Why it works: $OA = OB$ and $AM = BM$ by construction, and OM is common, so $\triangle OAM \equiv \triangle OBM$ by the third criterion and the two angles at O are equal.

PROBLEM	THE POINT WANTED
a well equidistant from two straight roads	on the bisector of the angle between them
a lamp lighting two walls equally	on the bisector of the corner
the centre of a circle touching both arms	on the bisector

02 Worked examples

EXAMPLE 1 *M* lies on the bisector of $\angle AOB$ and its distance to one arm is 3.5 cm . Find its distance to the other.

- ① The bisector property applies.
- ② Both perpendiculars are equal.
- ③ So the distance is 3.5 cm .

ANSWER

3.5 cm

EXAMPLE 2 A point inside an angle is 4 cm from each arm. What can be said about it?

- ① It is equidistant from both arms.
- ② By the converse, it lies on the bisector.

ANSWER

It lies on the bisector

EXAMPLE 3 Prove that the compass construction really bisects the angle.

- ① $OA = OB$
Same radius from O .

- ② $AM = BM$
Same radius from A and B .

- ③ $OM = OM$
Common side.

- ④ $\triangle OAM \cong \triangle OBM$ by the third criterion, so $\angle AOM = \angle BOM$.

ANSWER

Proved by SSS

03 Interactive model

Ask where to dig a well equidistant from two roads before any theory; the class reaches for the bisector on its own, and the proof then explains a choice they have already made.

Distance, bisector, locus

The distance from a point to a line is measured:

any way

along the perpendicular

to the nearest end

to the vertex

A point on a bisector is equidistant from:

the two arms

the vertex and an arm

two vertices

nothing

The proof of the property uses:

the third criterion

hypotenuse and an acute angle

the angle sum

the axiom of parallels

A point equidistant from both arms lies:

at the vertex

on the bisector

anywhere inside

outside

The bisector is the set of such points — that makes it a:

locus

median

transversal

tangent

The three bisectors of a triangle meet at the:

centroid

incentre

circumcentre

orthocentre

The compass construction is proved by:

SSS

SAS

ASA

measuring

A well equidistant from two roads should be:

at the crossing

on the bisector

on either road

anywhere

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bisector	Bissektrisa	Биссектриса
Arm of an angle	Burchak tomoni	Сторона угла
Equidistant	Teng uzoqlikda	Равноудалённый
Distance to a line	Nuqtadan chiziqqa masofa	Расстояние до прямой
Converse	Teskari teorema	Обратная теорема
Locus	Geometrik o'rin	Геометрическое место точек
Compasses	Sirkul	Циркуль
Incentre	Ichki markaz	Центр вписанной окружности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A bisector divides an angle into:

two equal parts

two unequal parts

three parts

a right angle

Distance to a line is measured along the:

bisector

perpendicular

arm

diagonal

On a bisector, the two perpendicular distances are:

equal

in ratio 2 : 1

unrelated

zero

The converse concludes that the point:

is the vertex

lies on the bisector

is outside

is the midpoint

The proof uses which congruence test?

two legs

hypotenuse and an acute angle

three angles

two angles only

The incentre is equidistant from:

the vertices

the three sides

two sides

the midpoints

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A bisector divides an angle into

ANSWER two equal angles

2. Distance from a point to a line is measured along the

ANSWER perpendicular

3. M on the bisector, one distance 3.5 cm : the other

ANSWER 3.5 cm

4. A point 4 cm from each arm lies on the

ANSWER bisector

5. The bisector of 80° makes two angles of

ANSWER 40°

6. The three bisectors of a triangle meet at the

ANSWER incentre

7. The incentre is the centre of the

ANSWER inscribed circle

Medium · the standard question

7 PROBLEMS

1. The bisector of $\angle AOB = 116^\circ$: each half

ANSWER 58°

2. Two bisectors of adjacent supplementary angles meet at

ANSWER 90°

3. Name the congruence test used in the proof

ANSWER Hypotenuse and an acute angle

4. Name the congruence test used for the construction

ANSWER The third criterion (SSS)

5. Where should a well equidistant from two roads be dug?

ANSWER On the bisector of the angle between them

6. Is the vertex equidistant from both arms?

ANSWER Yes — both distances are zero

7. A point is 3 cm from one arm and 5 cm from the other: is it on the bisector?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. Prove that the bisector property and its converse together define a locus

ANSWER Every point on it qualifies and every point that qualifies is on it

2. Why do the three bisectors of a triangle meet at one point?

ANSWER Two give a point equidistant from all three sides, so the third passes through it

3. In a triangle with $\angle A = 60^\circ$, the bisectors of $\angle B$ and $\angle C$ meet at I : find $\angle BIC$

ANSWER 120°

4. The bisector of the apex of an isosceles triangle is also its

ANSWER median and altitude

5. A circle of radius 2 cm touches both arms: where is its centre?

ANSWER On the bisector, 2 cm from each arm

6. Explain why the compass construction cannot fail for an angle under 180°

ANSWER The two equal arcs always meet inside the angle

7. The bisectors of two vertical angles form

ANSWER one straight line

07 Homework

Homework — 5 tasks

Leave your construction arcs on the page — they are the working.

- Construct the bisector of an angle of 70° with compasses and check it with a protractor.
- Write the proof of the bisector property in statement-and-reason form.
- A point inside an angle is 2.5 cm from each arm. Where does it lie, and why?
- Construct the three bisectors of a triangle and mark the incentre.
- Explain why the bisector is called the locus of points equidistant from the arms.

Relations between the sides and the angles of a triangle

The longer side faces the larger angle — and the triangle inequality follows from the same idea.

LESSON 92–93

2 × 40 MIN

MATEMATIKA 7, §43

S8 5

LESSON CLOCK

14'

19'

23'

19'

5'

The larger angle faces the longer side	14 min
Reading the order both ways	19 min
The triangle inequality	23 min
The shortest path to a line	19 min
Homework	5 min

LEARNING OBJECTIVES

- State that the larger angle faces the longer side, and its converse.
- Order the sides of a triangle from its angles, and the angles from its sides.
- Use the triangle inequality to decide whether three lengths make a triangle.
- Show that the perpendicular is the shortest segment from a point to a line.

TEXTBOOK

National	pp. 269–276
Cambridge	Stage 8, pp. 44–56
Workbook	Exercise 5.4

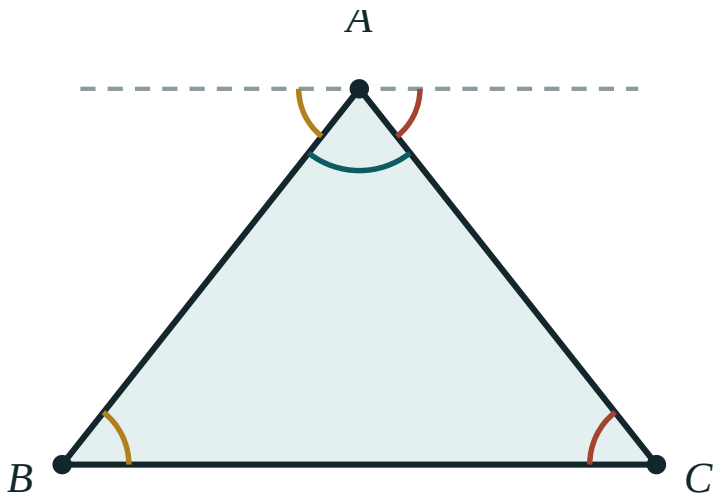
01

Explanation

The larger angle faces the longer side

In any triangle, the longer of two sides is opposite the larger of two angles — and the statement runs backwards as well.

$$AB > AC \iff \angle C > \angle B$$



the three angles fill 180°

Each side faces the angle of matching rank

TRIANGLE	LARGEST ANGLE	LONGEST SIDE
$\angle A = 90^\circ, \angle B = 60^\circ, \angle C = 30^\circ$	$\angle A$	BC
$\angle A = 40^\circ, \angle B = 40^\circ, \angle C = 100^\circ$	$\angle C$	AB
equilateral	all equal	all equal

A SIDE AND THE ANGLE AT ITS OWN END ARE NOT PARTNERS

The side BC is matched with $\angle A$, the angle it faces, not with $\angle B$ or $\angle C$. Reading the label off the wrong vertex reverses every answer.

Reading the order both ways

Ranking the angles ranks the sides in exactly the same order, and the other way round.

GIVEN	ORDER OF ANGLES	ORDER OF SIDES
$\angle A = 80^\circ, \angle B = 60^\circ, \angle C = 40^\circ$	$\angle A > \angle B > \angle C$	$BC > AC > AB$
$a = 5, b = 7, c = 9$	$\angle C > \angle B > \angle A$	$c > b > a$
isosceles with $AB = AC$	$\angle B = \angle C$	$AC = AB$

THE ISOSCELES THEOREM IS THIS THEOREM

Equal sides $\iff$ equal base angles is the borderline case of the same statement: when neither side is longer, neither angle is larger.

The triangle inequality

Each side of a triangle is shorter than the sum of the other two — the straight path from one vertex to another beats any detour through a third.

$$a < b + c, b < a + c, c < a + b$$

LENGTHS	CHECK THE LONGEST	A TRIANGLE?
3, 4, 5	$3 + 4 > 5$	yes
5, 8, 14	$5 + 8 < 14$	no
6, 9, 13	$6 + 9 > 13$	yes
2, 3, 5	$2 + 3 = 5$	no — the three points are in a line

ONLY THE LONGEST SIDE NEEDS CHECKING

If the longest side is beaten by the sum of the other two, the other two inequalities hold automatically. One line of working settles the question.

The shortest path to a line

Drop a perpendicular MP from a point M to a line, and take any other segment MQ to that line.

STATEMENT	REASON
$\angle MPQ = 90^\circ$	construction
$\angle MPQ$	the largest angle of $\triangle MPQ$
$MQ > MP$	the longer side faces the larger angle

So the perpendicular is shorter than every slant segment — which is exactly why distance to a line was defined along the perpendicular two lessons ago.

A DEFINITION JUSTIFIED AFTER THE EVENT

Calling the perpendicular “the distance” was a choice; this theorem shows it was the only sensible one, since it is the minimum.

02 Worked examples

EXAMPLE 1 In $\triangle ABC$, $\angle A = 80^\circ$, $\angle B = 60^\circ$. Order the sides.

① $\angle C = 180^\circ - 140^\circ = 40^\circ$

② $\angle A > \angle B > \angle C$

③ Each side faces the angle of matching rank.

④ $BC > AC > AB$

ANSWER

$BC > AC > AB$

EXAMPLE 2 Can 5, 8, 14 be the sides of a triangle? And 6, 9, 13?

① Longest of the first is 14.

② $5 + 8 = 13 < 14$ — no.

③ Longest of the second is 13.

④ $6 + 9 = 15 > 13$ — yes.

ANSWER

No; yes

EXAMPLE 3 Two sides of a triangle are 7 and 10. Between what values must the third lie?

- ① It must beat the difference: $x > 10 - 7 = 3$.
- ② It must be beaten by the sum: $x < 10 + 7 = 17$.
- ③ $3 < x < 17$

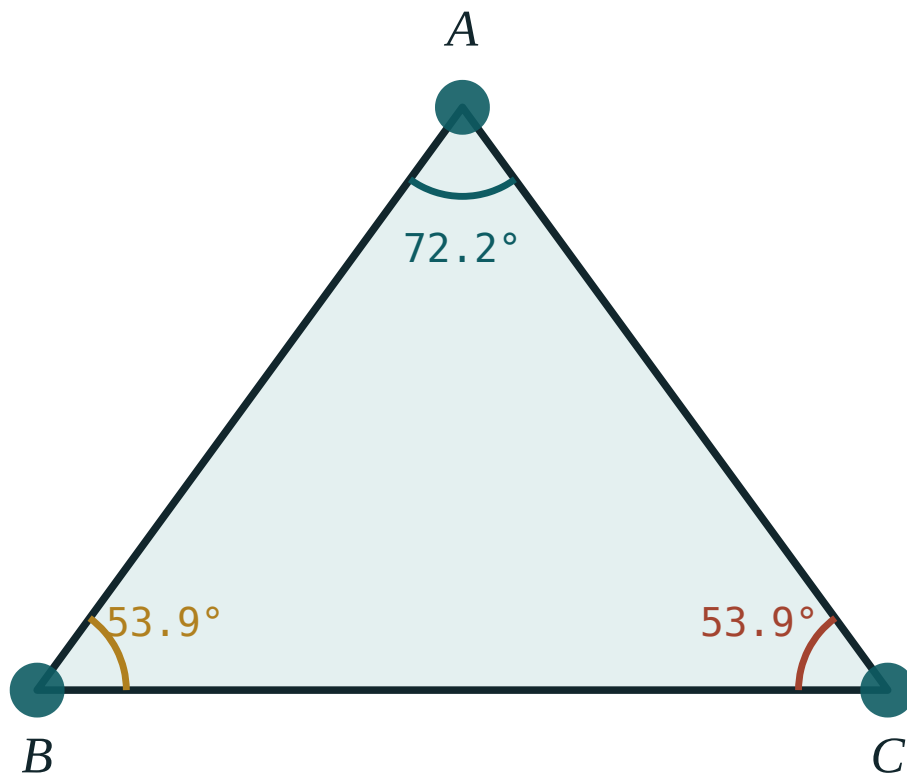
ANSWER

$$3 < x < 17$$

03 Interactive model

Hand out strips of 5, 8 and 14 cm and ask the class to build the triangle; the failure is visible and the inequality is never forgotten.

Watch the longest side and the largest angle move together



$\angle A$ 72.2°

$\angle B$ 53.9°

$\angle C$ 53.9°

sum 180°

longest side **BC**

largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Opposite side	Qarshi tomon	Противолежащая сторона
The larger	Kattaroq	Больший
The smaller	Kichikroq	Меньший
To order	Tartiblash	Упорядочить
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника
Perpendicular	Perpendikulyar	Перпендикуляр
Slant segment	Og'ma	Наклонная
Shortest	Eng qisqa	Кратчайший

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The longest side faces:

the smallest angle

the largest angle

the right angle only

any angle

In a right-angled triangle the longest side is:

a leg

the hypotenuse

either

undetermined

$\angle A = 80^\circ$, $\angle B = 60^\circ$, $\angle C = 40^\circ$ gives:

$AB > BC > AC$

$BC > AC > AB$

$AC > AB > BC$

all equal

Sides 5, 8, 14 make:

a triangle

no triangle

a right triangle

an isosceles triangle

To test three lengths you need to check:

all three sums

the longest side only

the shortest side

the perimeter

The shortest segment from a point to a line is:

any slant

the perpendicular

the bisector

the median

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The longest side faces the

ANSWER largest angle

2. The shortest side faces the

ANSWER smallest angle

3. 3, 4, 5: a triangle?

ANSWER Yes

4. 5, 8, 14: a triangle?

ANSWER No

5. 2, 3, 5: a triangle?

ANSWER No

6. In a right triangle the longest side is the

ANSWER hypotenuse

7. The shortest segment to a line is the

ANSWER perpendicular

Medium · the standard question

7 PROBLEMS

1. $\angle A = 80^\circ$, $\angle B = 60^\circ$: order the sides

ANSWER $BC > AC > AB$

2. $a = 5$, $b = 7$, $c = 9$: order the angles

ANSWER $\angle C > \angle B > \angle A$

3. Sides 7 and 10: the range of the third

ANSWER $3 < x < 17$

4. Sides 4 and 9: the range of the third

ANSWER $5 < x < 13$

5. An obtuse triangle: which side is longest?

ANSWER The one facing the obtuse angle

6. Isosceles with $AB = AC$: which angles are equal?

ANSWER $\angle B = \angle C$

7. 6, 9, 13: a triangle?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. The perimeter of a triangle with sides 7, 10, x is a whole number: how many values of x are whole?

ANSWER 13 — from 4 to 16

2. Prove that the perpendicular is shorter than any slant segment

ANSWER It faces the largest angle of the right triangle

3. In $\triangle ABC$, $\angle A = 90^\circ$: prove BC is the longest side

ANSWER It faces the largest angle

4. Two sides are 8 and 15 and the third is a whole number: the smallest possible

ANSWER 8

5. Can a triangle have sides 1, 2, 3?

ANSWER No — $1 + 2 = 3$

6. If $\angle A > \angle B$ and $\angle B > \angle C$, order the sides

ANSWER $a > b > c$

7. Why is checking the longest side enough?

ANSWER The other two inequalities then hold automatically

07 Homework

Homework — 5 tasks

For a triangle test, find the longest side first and check only that one.

- In $\triangle ABC$, $\angle A = 95^\circ$ and $\angle B = 45^\circ$. Order the three sides.
- Decide whether 4, 6, 11 and 4, 6, 9 make triangles.
- Two sides are 9 and 12. Find the range of the third.
- Explain in two lines why the perpendicular is the shortest path to a line.

5. A triangle has $a = 6$, $b = 6$, $c = 10$. Order its angles.

Practical exercises

— area of triangles and quadrilaterals

A Cambridge insert: one height, one base, and a compound shape cut into pieces you already know.

LESSON 94–95

2 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 15

LESSON CLOCK

9'

23'

24'

19'

5'

The four formulae	9 min
Choosing the right height	23 min
Compound shapes	24 min
Units and estimation	19 min
Homework	5 min

LEARNING OBJECTIVES

- Use the area formulae for a rectangle, a triangle, a parallelogram and a trapezium.
- Identify the height that belongs to a chosen base.
- Find the area of a compound shape by cutting or by subtracting.
- Work with the units of area and convert between them.

TEXTBOOK

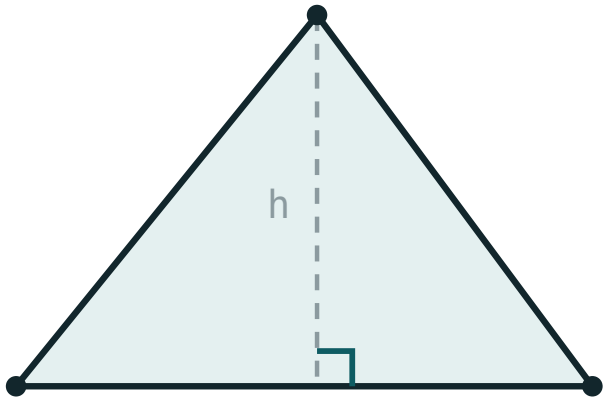
National	pp. 277–281
Cambridge	Stage 8, pp. 148–156
Workbook	Exercise 15.2

01

Explanation

The four formulae

$$S = ab \quad S = \frac{1}{2}ah \quad S = ah \quad S = \frac{a + b}{2} \cdot h$$



a

$$S = \frac{1}{2} \cdot a \cdot h$$

Base and perpendicular height

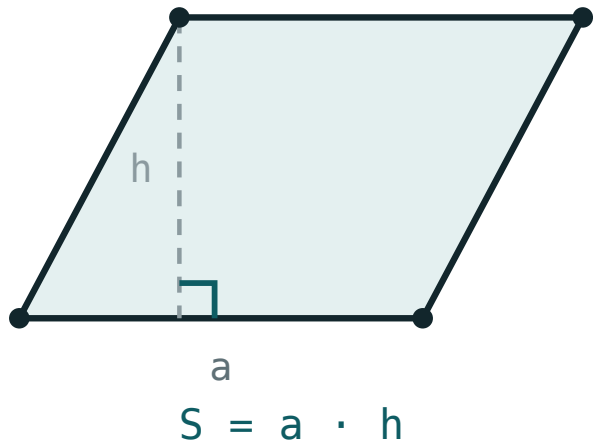
SHAPE	FORMULA	WHAT <i>h</i> MEANS
rectangle	ab	the second side
triangle	$\frac{1}{2}ah$	perpendicular from the base to the opposite vertex
parallelogram	ah	perpendicular between the two parallel sides
trapezium	$\frac{a+b}{2} \cdot h$	perpendicular between the parallel sides

THREE FORMULAE, ONE IDEA

A parallelogram is two copies of a triangle, and a trapezium is a parallelogram whose two parallel sides have been averaged. Learning $S = \frac{1}{2}ah$ and the reason for it makes the other two memorable rather than memorised.

Choosing the right height

The height must be perpendicular to **the base you chose**. A slanting side is never a height, and in an obtuse triangle the height may fall outside the figure.



The height is perpendicular, not slanting

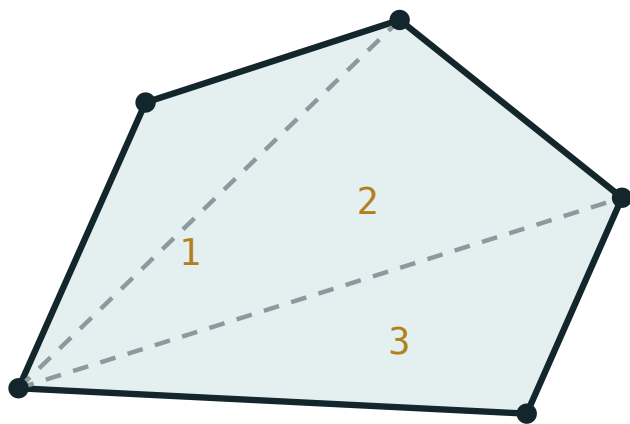
FIGURE	BASE	HEIGHT	AREA
triangle	10 cm	6 cm	30 cm ²
parallelogram	8 cm	5 cm	40 cm ²
trapezium	6 cm and 10 cm	4 cm	32 cm ²
rectangle	7 cm	3 cm	21 cm ²

THE SLANTING SIDE IS NOT THE HEIGHT

A parallelogram with sides 8 and 6 and height 5 has area 40, not 48. Using the second side instead of the perpendicular is the single commonest area error in the paper.

Compound shapes

Two routes work for any compound figure: cut it into known pieces and add, or enclose it in a rectangle and subtract what is not wanted.



split into triangles, then add the areas

Cutting a compound shape into known pieces

SHAPE	METHOD	WORKING	AREA
an L-shape 10×8 with a 4×3 corner removed	subtract	$80 - 12$	68 cm^2
a house outline: 6×4 rectangle under a triangle of height 3	add	$24 + 9$	33 cm^2
a path 1 m wide round a 5×4 plot	subtract	$42 - 20$	22 m^2

LABEL THE PIECES BEFORE YOU CALCULATE

Write the area of each piece next to it on the sketch. Nearly every lost mark on a compound shape is a piece counted twice or left out, not an arithmetic slip.

Units and estimation

CONVERSION	FACTOR	WHY
$1 \text{ cm}^2 = 100 \text{ mm}^2$	10^2	both sides scaled by 10
$1 \text{ m}^2 = 10\,000 \text{ cm}^2$	100^2	both sides scaled by 100
$1 \text{ ha} = 10\,000 \text{ m}^2$	100×100	a square 100 m on a side
$1 \text{ km}^2 = 100 \text{ ha}$	1000^2	a square 1 km on a side

LENGTHS SCALE BY k , AREAS BY k^2

A metre is *100* centimetres, but a square metre is *10 000* square centimetres. Using *100* for the area conversion is wrong by a factor of a hundred.

02 Worked examples

EXAMPLE 1 A triangle has base *10 cm* and height *6 cm*. Find its area.

$$① \quad S = \frac{1}{2}ah$$

$$② \quad S = \frac{1}{2} \cdot 10 \cdot 6$$

$$③ \quad S = 30 \text{ cm}^2$$

ANSWER

30 cm²

EXAMPLE 2 A trapezium has parallel sides *6 cm* and *10 cm* and height *4 cm*. Find its area.

$$① \quad S = \frac{a+b}{2} \cdot h$$

$$② \quad S = \frac{6+10}{2} \cdot 4$$

$$③ \quad S = 8 \cdot 4$$

$$④ \quad S = 32 \text{ cm}^2$$

ANSWER

32 cm²

EXAMPLE 3 A rectangular plot 5 m by 4 m has a path 1 m wide all round it. Find the area of the path.

- ① The outer rectangle is 7 m by 6 m .
One metre on each side.

② $7 \cdot 6 = 42\text{ m}^2$

③ $5 \cdot 4 = 20\text{ m}^2$

④ $42 - 20 = 22\text{ m}^2$

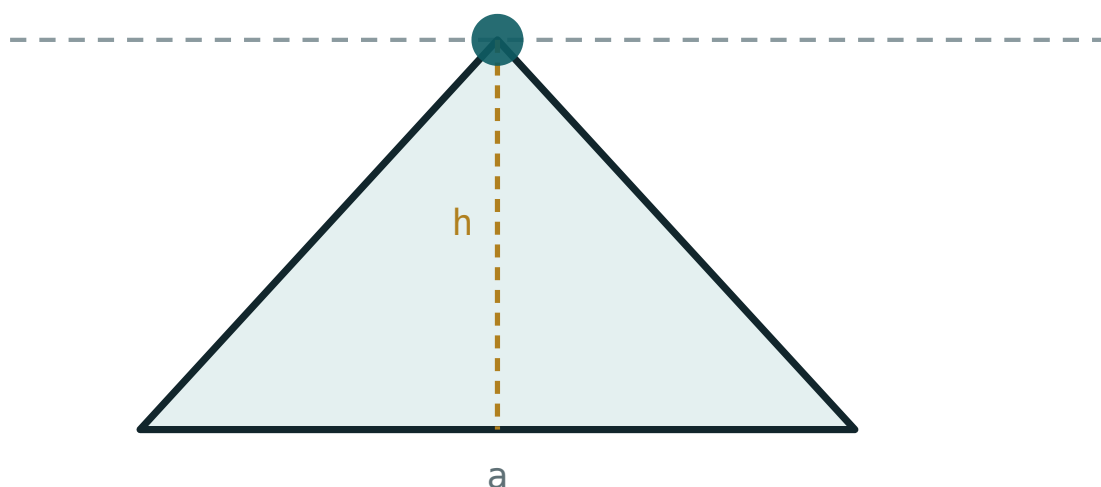
ANSWER

22 m^2

03 Interactive model

Give one parallelogram with all three lengths marked — base, slanting side and height — and ask for the area; the marks separate those who read the figure from those who multiply the first two numbers.

Slide the apex — the area does not move



base a **220**

height h **120**

area $\frac{1}{2}ah$ **13200**

perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Area	Yuza	Площадь
Base	Asos	Основание
Height	Balandlik	Высота
Perpendicular height	Perpendikulyar balandlik	Перпендикулярная высота
Compound shape	Murakkab shakl	Составная фигура
Trapezium	Trapetsiya	Трапеция
Square centimetre	Kvadrat santimetr	Квадратный сантиметр
Hectare	Gektar	Гектар

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area of a triangle is:

The area of a parallelogram is:

In the formulae, h must be:

A trapezium with parallel sides 6 and 10 and height 4 has area:

1 m^2 equals:

1 ha equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Triangle, base 10, height 6

ANSWER 30 cm^2

2. Rectangle 7×3

ANSWER 21 cm^2

3. Parallelogram, base 8, height 5

ANSWER 40 cm^2

4. Trapezium 6 and 10, height 4

ANSWER 32 cm^2

5. Triangle, base 12, height 5

ANSWER 30 cm^2

6. 1 cm^2 in mm^2

ANSWER 100

7. 1 m^2 in cm^2

ANSWER 10 000

Medium · the standard question

7 PROBLEMS

1. Parallelogram with sides 8 and 6 and height 5

ANSWER 40 cm^2

2. A triangle of area 24 has base 8: find the height

ANSWER 6

3. A trapezium of area 45, parallel sides 6 and 9: find the height

ANSWER 6

4. L-shape: 10×8 with a 4×3 corner removed

ANSWER 68 cm^2

5. House outline: 6×4 rectangle plus a triangle of height 3

ANSWER 33 cm^2

6. A 1 m path round a 5×4 plot

ANSWER 22 m^2

7. 3 ha in m^2

ANSWER 30 000

Hard · stretch and reason

7 PROBLEMS

1. A rhombus with diagonals 10 and 12

ANSWER 60 cm^2

2. A triangle with base a and height h has its base doubled: the new area

ANSWER *twice the old*

3. Both dimensions of a rectangle are trebled: the area is multiplied by

ANSWER 9

4. A field of 2.5 ha in m^2

ANSWER $25\,000$

5. A trapezium has parallel sides x and $x + 6$ and height 4 : its area

ANSWER $4x + 12$

6. Two triangles have the same base and lie between the same parallels: their areas are

ANSWER *equal*

7. A 12×9 rectangle contains a triangle on the 12 side with apex on the opposite side: its area

ANSWER 54 cm^2

07 Homework

Homework — 5 tasks

Sketch every shape and write the area of each piece beside it.

- Find the area of a triangle with base 14 cm and height 9 cm .
- A parallelogram has sides 12 and 7 and height 5 . Find its area and say which number is not used.
- Find the area of a trapezium with parallel sides 8 and 14 and height 6 .
- A 2 m path runs round a 10 m by 6 m garden. Find the area of the path.

5. Convert 4.5 ha to square metres and $250\,000\text{ cm}^2$ to square metres.

Control work 6, and work on the mistakes

Parallel lines, the angle sum and right-angled triangles in one paper.

LESSON 96–97

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 6

S8 5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Name a pair of angles at a transversal before quoting any rule about it.
- Use the property when parallelism is given and the criterion when it is wanted.
- Find angles of triangles, including from an exterior angle.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 226–281
Cambridge	Stage 8, pp. 44–56
Workbook	Control paper M6

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Name the three pairs of angles at a transversal and say what each does when the lines are parallel	3	L82–85
2	$a \parallel b$ and one angle is 112° : find all eight angles	4	L84–85
3	Co-interior angles are $4x$ and $5x$: find both	4	L84–85
4	The angles of a triangle are in the ratio $2 : 3 : 4$: find them	4	L86–87
5	An exterior angle is 125° and one remote interior angle is 47° : find the other two interior angles	5	L86–87
6	Hypotenuse 16 cm , one angle 30° : find the opposite leg and the third angle	5	L88–89

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the word “co-interior” alone; Q2 one for noticing that only two sizes appear; Q5 two for using the *remote* pair and one for the check against 180° ; Q6 one for the 30° rule being quoted by name.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
pair misnamed	“alternate” for two angles on the same side	check the position first
co-interior treated as equal	$4x = 5x$	$4x + 5x = 180^\circ$
criterion used as a property	“alternate, so parallel, so equal”	parallelism was given — use the property
all three angles added to the exterior angle	$125^\circ = 47^\circ + b + c$	the two <i>remote</i> angles only
acute angles subtracted from 180°	$180^\circ - 30^\circ = 150^\circ$	$90^\circ - 30^\circ = 60^\circ$
30° rule inverted	the leg is twice the hypotenuse	it is half
no check	three angles that do not total 180°	add them back

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

BLOCK	THE SENTENCE
parallel lines	no common point, and exactly one parallel through an outside point
the criteria	an angle fact proves parallelism
the properties	parallelism proves an angle fact
the angle sum	180°
the exterior angle	equal to the two remote interior angles
right-angled triangles	the acute angles add to 90°
the 30° rule	that leg is half the hypotenuse

LOOKING FORWARD

The rest of Quarter III is algebra: the abridged multiplication formulae, algebraic fractions and linear equations. Geometry returns in Quarter IV with construction problems.

02 Worked examples

EXAMPLE 1 Model answer, Q3: co-interior angles $4x$ and $5x$.

- 1 Co-interior angles on parallel lines add to 180° .
- 2 $4x + 5x = 180^\circ$
- 3 $9x = 180^\circ$, so $x = 20^\circ$.
- 4 The angles are 80° and 100° .

ANSWER

80° and 100°

EXAMPLE 2 Model answer, Q5: exterior 125° , one remote interior angle 47° .

- 1 The exterior angle equals the sum of the two remote angles.
- 2 $125^\circ - 47^\circ = 78^\circ$
The second remote angle.
- 3 The adjacent interior angle is $180^\circ - 125^\circ = 55^\circ$.
- 4 Check: $47^\circ + 78^\circ + 55^\circ = 180^\circ$ ✓

ANSWER

78° and 55°

EXAMPLE 3 Model answer, Q6: hypotenuse 16 cm , one angle 30° .

① The leg opposite 30° is half the hypotenuse.

② $16 \div 2 = 8\text{ cm}$

③ The acute angles add to 90° .

④ $90^\circ - 30^\circ = 60^\circ$

ANSWER

8 cm and 60°

03 Interactive model

Return Q1 first and have the class read the three names aloud; most of the lost marks in Q2 and Q3 turn out to be naming errors carried forward.

The chapter in twelve questions

Parallel lines have how many common points?

Parallels through a point off a line:

Equal alternate angles prove:

On parallel lines, co-interior angles:

are equal

add to 90°

add to 180°

add to 360°

With one angle of 112° , the eight angles take:

one size

two sizes

four sizes

eight sizes

Co-interior $4x$ and $5x$ give x equal to:

10

20

36

45

The angles of a triangle add to:

90°

180°

270°

360°

Ratio 2 : 3 : 4 gives:

$40^\circ, 60^\circ, 80^\circ$

$30^\circ, 60^\circ, 90^\circ$

$45^\circ, 60^\circ, 75^\circ$

$20^\circ, 30^\circ, 40^\circ$

An exterior angle equals:

the adjacent angle

the two remote angles

all three

90°

The acute angles of a right triangle add to:

45°

90°

180°

270°

Hypotenuse 16, angle 30° : the opposite leg is:

4

8

16

32

The longest side of any triangle faces:

the smallest angle

the largest angle

the right angle only

any angle

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Transversal	Kesuvchi	Секущая
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Angle sum	Burchaklar yig'indisi	Сумма углов
Exterior angle	Tashqi burchak	Внешний угол
Hypotenuse	Gipotenuza	Гипотенуза
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 asks first for:

a calculation

the names of the pairs

a construction

a proof

In Q2 the eight angles take:

one size

two sizes

four sizes

eight sizes

Q3 uses:

equal co-interior angles

co-interior angles adding to 180°

the angle sum

the exterior angle

Q4 sets up:

$$9x = 180$$

$$9x = 360$$

$$x = 180$$

$$3x = 180$$

Q5 uses:

all three interior angles

the two remote angles

the adjacent angle only

the angle sum alone

Q6 quotes:

the angle sum

the 30° rule

the criterion

the triangle inequality

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. On parallel lines, alternate angles are

ANSWER equal

2. On parallel lines, co-interior angles add to

ANSWER 180°

3. One angle is 112° : its supplement

ANSWER 68°

4. Angles of a triangle add to

ANSWER 180°

5. Acute angles of a right triangle add to

ANSWER 90°

6. Hypotenuse 16, angle 30° : the opposite leg

ANSWER 8

7. Exterior 125° : the adjacent interior angle

ANSWER 55°

Medium · the standard question

7 PROBLEMS

1. Co-interior $4x$ and $5x$

ANSWER 80° and 100°

2. Ratio $2 : 3 : 4$

ANSWER $40^\circ, 60^\circ, 80^\circ$

3. Exterior 125° , one remote angle 47° : the other

ANSWER 78°

4. $a \parallel b$ and one angle 112° : the alternate angle

ANSWER 112°

5. $a \parallel b$ and one angle 112° : the co-interior angle

ANSWER 68°

6. Right triangle with an angle of 58° : the third angle

ANSWER 32°

7. Sides 4, 6, 11: a triangle?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. Alternate angles $6x - 5$ and $4x + 25$ on parallel lines

ANSWER $x = 15$

2. An exterior angle is 4 times its adjacent interior angle

ANSWER 144° and 36°

3. A right triangle has acute angles in the ratio 2 : 7

ANSWER 20° and 70°

4. Co-interior $2x + 30$ and $x + 60$

ANSWER $x = 30$

5. One angle of a triangle is the sum of the other two: what kind of triangle is it?

ANSWER Right-angled

6. A transversal cuts two parallel lines at 90° : all eight angles are

ANSWER 90°

7. Angles x , $x + 20$, $x + 40$

ANSWER 40° , 60° , 80°

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Draw a transversal across two parallel lines, mark 118° and label all eight angles.
3. Co-interior angles are $7x$ and $2x$. Find both.
4. An exterior angle is 132° and one remote interior angle is 61° . Find the other two.
5. Write one sentence for each of the seven blocks on the chapter map.

The square of a sum and of a difference

Two formulae worth knowing by heart — and the middle term that everybody forgets.

LESSON 98–99

2 × 40 MIN

MATEMATIKA 7, §44

S8 2.3

LESSON CLOCK

14'

19'

23'

19'

5'

Where the formulae come from	14 min
The square picture	19 min
Using them both ways	23 min
Squaring numbers in the head	19 min
Homework	5 min

LEARNING OBJECTIVES

- Derive $(a + b)^2$ and $(a - b)^2$ by multiplying out.
- Apply both formulae in either direction.
- Explain the middle term with a square picture.
- Use the formulae to square numbers mentally.

TEXTBOOK

National	pp. 282–289
Cambridge	Stage 8, pp. 18–24
Workbook	Exercise 2.3

01

Explanation

Where the formulae come from

Nothing new is being claimed: both formulae are ordinary multiplications, done once and then remembered.

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

STEP	WORKING
write it out	$(a + b)(a + b)$
four products	$a^2 + ab + ba + b^2$
collect	$a^2 + 2ab + b^2$

$(A + B)^2$ IS NOT $A^2 + B^2$

Try $a = 3, b = 4$: the left side is 49, the right 25. The missing $2ab = 24$ is the whole difference, and it is missing from more answers than any other term in the course.

The square picture

A square of side $a + b$ splits into four pieces: two squares and two equal rectangles.

PIECE	DIMENSIONS	AREA
large square	$a \times a$	a^2
rectangle	$a \times b$	ab
rectangle	$b \times a$	ab
small square	$b \times b$	b^2

Total: $a^2 + 2ab + b^2$. The two rectangles are the middle term, and they are impossible to overlook once the picture has been drawn.

THE SIGN IS THE ONLY DIFFERENCE

In $(a - b)^2$ the middle term is subtracted, but b^2 is still added: a square is never negative. Both formulae end in $+ b^2$.

Using them both ways

EXPAND	ANSWER
$(x + 5)^2$	$x^2 + 10x + 25$
$(x - 7)^2$	$x^2 - 14x + 49$
$(3a + 2)^2$	$9a^2 + 12a + 4$
$(2x - 5y)^2$	$4x^2 - 20xy + 25y^2$

RECOGNISE A PERFECT SQUARE	AS A SQUARE
$x^2 + 10x + 25$	$(x + 5)^2$
$x^2 - 14x + 49$	$(x - 7)^2$
$9a^2 + 12a + 4$	$(3a + 2)^2$
$x^2 + 6x + 10$	not a perfect square

THE TEST FOR A PERFECT SQUARE

Take the square roots of the first and last terms, double their product, and compare with the middle term. For $x^2 + 6x + 10$ that gives $2 \cdot x \cdot \sqrt{10}$, which is not $6x$ — so it is not one.

Squaring numbers in the head

NUMBER	SPLIT	WORKING	ANSWER
51^2	$(50 + 1)^2$	$2500 + 100 + 1$	2601
49^2	$(50 - 1)^2$	$2500 - 100 + 1$	2401
102^2	$(100 + 2)^2$	$10000 + 400 + 4$	10404
98^2	$(100 - 2)^2$	$10000 - 400 + 4$	9604

SPLIT AT THE NEAREST ROUND NUMBER

Choose a to be the ten or hundred nearest to the number; then b is small, b^2 is tiny, and the whole calculation fits in the head.

02 Worked examples

EXAMPLE 1 Expand $(2x - 5y)^2$.

① $a = 2x, b = 5y$.

② $a^2 = 4x^2$

③ $2ab = 2 \cdot 2x \cdot 5y = 20xy$

Subtracted, because of the minus.

④ $b^2 = 25y^2$, so the answer is $4x^2 - 20xy + 25y^2$.

ANSWER

$$4x^2 - 20xy + 25y^2$$

EXAMPLE 2 Is $9a^2 + 12a + 4$ a perfect square?

① $\sqrt{9a^2} = 3a$ and $\sqrt{4} = 2$.

② $2 \cdot 3a \cdot 2 = 12a$

It matches the middle term.

③ So it is $(3a + 2)^2$.

ANSWER

$$(3a + 2)^2$$

EXAMPLE 3 Find 49^2 mentally.

① $49 = 50 - 1$

② $50^2 = 2500$

③ $2 \cdot 50 \cdot 1 = 100$, subtracted.

④ $2500 - 100 + 1 = 2401$

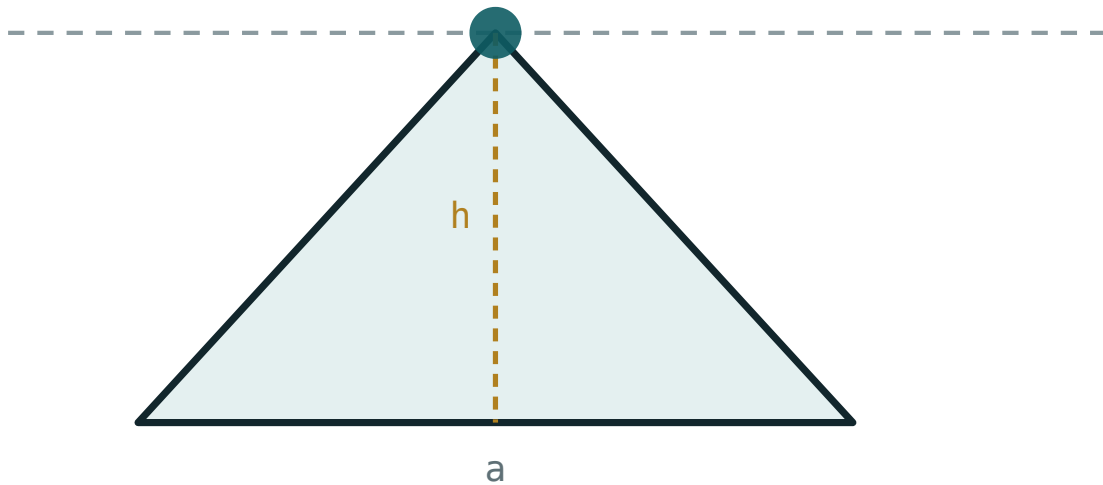
ANSWER

2401

03 Interactive model

Ask the class for $(a + b)^2$ before teaching anything; collect the wrong $a^2 + b^2$ answers on the board, then draw the square picture beside them.

The square cut into four pieces

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Square of a sum	Yig'indining kvadrati	Квадрат суммы
Square of a difference	Ayirmaning kvadrati	Квадрат разности
Abridged multiplication	Qisqa ko'paytirish	Сокращённое умножение
Middle term	O'rta had	Средний член
Double product	Ikkilangan ko'paytma	Удвоенное произведение
Identity	Ayniyat	Тождество
Perfect square	To'la kvadrat	Полный квадрат
Mental arithmetic	Og'zaki hisoblash	Устный счёт

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(a + b)^2$ equals:

$$a^2 + b^2$$

$$a^2 + 2ab + b^2$$

$$a^2 - 2ab + b^2$$

$$2a + 2b$$

$(a - b)^2$ equals:

$$a^2 - b^2$$

$$a^2 - 2ab + b^2$$

$$a^2 + 2ab - b^2$$

$$a^2 - 2ab - b^2$$

$(x + 5)^2$ equals:

$$x^2 + 25$$

$$x^2 + 10x + 25$$

$$x^2 + 5x + 25$$

$$x^2 + 10x + 10$$

The middle term of $(3a + 2)^2$ is:

$6a$

$12a$

$5a$

$9a$

$x^2 + 6x + 10$ is:

a perfect square

not a perfect square

$(x + 3)^2$

$(x + 5)^2$

49^2 by the formula is:

2401

2409

2601

2500

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x + 3)^2$

ANSWER $x^2 + 6x + 9$

2. $(x - 4)^2$

ANSWER $x^2 - 8x + 16$

3. $(x + 5)^2$

ANSWER $x^2 + 10x + 25$

4. $(a + 1)^2$

ANSWER $a^2 + 2a + 1$

5. $(2x + 1)^2$

ANSWER $4x^2 + 4x + 1$

6. $(y - 6)^2$

ANSWER $y^2 - 12y + 36$

7. 51^2

ANSWER 2601

Medium · the standard question**7 PROBLEMS**

1. $(3a + 2)^2$

ANSWER $9a^2 + 12a + 4$

2. $(2x - 5y)^2$

ANSWER $4x^2 - 20xy + 25y^2$

3. $(4m - 3n)^2$

ANSWER $16m^2 - 24mn + 9n^2$

4. $x^2 + 10x + 25$ as a square

ANSWER $(x + 5)^2$

5. $x^2 - 14x + 49$ as a square

ANSWER $(x - 7)^2$

6. 98^2

ANSWER 9604

7. 102^2

ANSWER 10404

Hard · stretch and reason

7 PROBLEMS

1. $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

2. $(a + b)^2 + (a - b)^2$

ANSWER $2a^2 + 2b^2$

3. If $a + b = 7$ and $ab = 12$, find $a^2 + b^2$

ANSWER 25

4. If $a - b = 3$ and $ab = 10$, find $a^2 + b^2$

ANSWER 29

5. $(x + \frac{1}{x})^2$

ANSWER $x^2 + 2 + \frac{1}{x^2}$

6. For which k is $x^2 + kx + 36$ a perfect square?

ANSWER $k = \pm 12$

7. $(2a - 3)^2 - (2a + 3)^2$

ANSWER $-24a$ **07 Homework****Homework — 5 tasks**

Write the middle term before the last one; it is the term that gets forgotten.

1. Expand $(x + 8)^2$, $(x - 9)^2$ and $(5a + 2)^2$.
2. Expand $(3x - 4y)^2$.
3. Write $x^2 + 12x + 36$ and $4a^2 - 20a + 25$ as squares.
4. Find 52^2 and 97^2 mentally, showing the split.

5. If $a + b = 9$ and $ab = 20$, find $a^2 + b^2$.

The difference of two squares

The one product whose middle term cancels — and the fastest factorisation in algebra.

LESSON 100–101

2 × 40 MIN

MATEMATIKA 7, §45

S8 2.4

LESSON CLOCK

13'

23'

18'

22'

4'

The vanishing middle term	13 min
Factorising in one step	23 min
What does not work	18 min
Two uses	22 min
Homework	4 min

LEARNING OBJECTIVES

- Derive $(a - b)(a + b) = a^2 - b^2$ and explain why the middle term vanishes.
- Factorise a difference of two squares in one step.
- Recognise that a sum of two squares does not factorise this way.
- Use the formula for mental arithmetic and for cancelling fractions.

TEXTBOOK

National	pp. 290–296
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The vanishing middle term

$$(a - b)(a + b) = a^2 - b^2$$

STEP	WORKING
four products	$a^2 + ab - ba - b^2$
$+ab$	cancels $-ba$
what is left	$a^2 - b^2$

WHY THIS PAIR AND NO OTHER

The two middle products are equal and opposite only because one bracket has $+b$ and the other $-b$. Change either sign and the middle term returns.

Factorising in one step

Read the identity from right to left: any difference of two squares factorises immediately.

EXPRESSION	A	B	FACTORISED
$x^2 - 9$	x	3	$(x - 3)(x + 3)$
$4a^2 - 25$	$2a$	5	$(2a - 5)(2a + 5)$
$49 - y^2$	7	y	$(7 - y)(7 + y)$
$16x^2 - 81y^2$	$4x$	$9y$	$(4x - 9y)(4x + 9y)$

TAKE THE COMMON FACTOR OUT FIRST

$2x^2 - 18$ is not a difference of two squares as it stands. Take out the 2 first: $2(x^2 - 9) = 2(x - 3)(x + 3)$. Factorising is always finished, never stopped halfway.

What does not work

EXPRESSION	FACTORISES LIKE THIS?	WHY
$x^2 - 9$	yes	a difference of squares
$x^2 + 9$	no	a sum of squares
$x^2 - 5$	not over whole numbers	$\sqrt{5}$
$x^3 - 8$	not by this formula	cubes, not squares

$A^2 + B^2$ DOES NOT FACTORISE

Writing $x^2 + 9 = (x + 3)(x + 3)$ is wrong twice over: that product is $x^2 + 6x + 9$. A sum of two squares has no factorisation with real numbers at all.

Two uses

PRODUCT	AS A DIFFERENCE	ANSWER
$51 \cdot 49$	$50^2 - 1^2$	2499
$102 \cdot 98$	$100^2 - 2^2$	9996
$37 \cdot 43$	$40^2 - 3^2$	1591

And in a fraction the factorisation is what makes the cancelling possible:

$$\frac{x^2 - 9}{x + 3} = \frac{(x - 3)(x + 3)}{x + 3} = x - 3$$

THIS IS WHY FACTORISING COMES BEFORE FRACTIONS

Nothing can be cancelled from $x^2 - 9$ as it stands, because cancelling needs factors. Three lessons from now every algebraic fraction will begin with exactly this step.

02

Worked examples

EXAMPLE 1 Factorise $16x^2 - 81y^2$.

1

$\sqrt{16x^2} = 4x$

2

$\sqrt{81y^2} = 9y$

3

$= (4x - 9y)(4x + 9y)$

4

Check: the middle terms cancel ✓

ANSWER

$(4x - 9y)(4x + 9y)$

EXAMPLE 2 Factorise $2x^2 - 18$ completely.

① Common factor first: $2(x^2 - 9)$.

② $x^2 - 9 = (x - 3)(x + 3)$

③ $= 2(x - 3)(x + 3)$

ANSWER

$$2(x - 3)(x + 3)$$

EXAMPLE 3 Find $37 \cdot 43$ mentally.

① Both are 3 from 40.

② $37 \cdot 43 = (40 - 3)(40 + 3)$

③ $= 1600 - 9$

④ $= 1591$

ANSWER

$$1591$$

03 Interactive model

Put $51 \cdot 49$ and $50 \cdot 50$ side by side on the board and ask which is bigger; the difference of exactly 1 is the formula arriving before its name.

Factorise, then cancel

$$\frac{x^2 - 9}{x + 3}$$

① $x^2 - 9 = (x - 3)(x + 3)$

A
difference
of
two
squares.

② $\frac{(x - 3)(x + 3)}{x + 3}$

Now
there
is a
common
factor.

③ $x - 3$

Cancel
 $x +$
 $3 -$
and
note
 $x \neq$
 -3 .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Difference of two squares	Kvadratlar ayirmasi	Разность квадратов
Sum of two squares	Kvadratlar yig'indisi	Сумма квадратов
Conjugate pair	Qo'shma ifodalar	Сопряжённые выражения
To cancel	Qisqartirish	Сокращать
Factorisation	Ko'paytuvchilarga ajratish	Разложение на множители
Square root	Kvadrat ildiz	Квадратный корень
One step	Bir qadamda	В один шаг
Identity	Ayniyat	Тождество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(a - b)(a + b)$ equals:

$x^2 - 9$ factorises to:

$x^2 + 9$ factorises to:

$4a^2 - 25$ factorises to:

$2x^2 - 18$ factorises completely to:

$51 \cdot 49$ equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x^2 - 9$

ANSWER $(x - 3)(x + 3)$

2. $x^2 - 16$

ANSWER $(x - 4)(x + 4)$

3. $a^2 - 1$

ANSWER $(a - 1)(a + 1)$

4. $49 - y^2$

ANSWER $(7 - y)(7 + y)$

5. $4a^2 - 25$

ANSWER $(2a - 5)(2a + 5)$

6. $(x - 2)(x + 2)$

ANSWER $x^2 - 4$

7. $51 \cdot 49$

ANSWER 2499

Medium · the standard question

7 PROBLEMS

1. $16x^2 - 81y^2$

ANSWER $(4x - 9y)(4x + 9y)$

2. $2x^2 - 18$

ANSWER $2(x - 3)(x + 3)$

3. $x^4 - 16$

ANSWER $(x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$

4. $102 \cdot 98$

ANSWER 9996

5. $37 \cdot 43$

ANSWER 1591

6.
$$\frac{x^2 - 9}{x + 3}$$

ANSWER $x - 3$

7. Does $x^2 + 25$ factorise?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $9a^2 - 4b^2$

ANSWER $(3a - 2b)(3a + 2b)$

2. $(x + 1)^2 - 4$

ANSWER $(x - 1)(x + 3)$

3. $a^4 - b^4$

ANSWER $(a - b)(a + b)(a^2 + b^2)$

4.
$$\frac{4x^2 - 25}{2x + 5}$$

ANSWER $2x - 5$

5. $123^2 - 122^2$

ANSWER 245

6. Factorise $50^2 - 49^2$ and evaluate

ANSWER 99

7. $x^2 - \frac{1}{4}$

ANSWER $(x - \frac{1}{2})(x + \frac{1}{2})$

07 Homework**Homework — 5 tasks**

Always take out the common factor before looking for a difference of squares.

- Factorise $x^2 - 36$, $9a^2 - 16$ and $25 - 4y^2$.
- Factorise $3x^2 - 27$ completely.
- Find $61 \cdot 59$ and $104 \cdot 96$ by the formula.

4. Simplify $\frac{x^2 - 25}{x - 5}$.

5. Explain in one sentence why $x^2 + 16$ does not factorise.

The cube of a sum. The cube of a difference

Four terms, coefficients 1, 3, 3, 1, and the signs that alternate in the second formula.

LESSON 102–103

2 × 40 MIN

MATEMATIKA 7, §46

BEYOND STAGE 8

LESSON CLOCK

14'

19'

23'

19'

5'

From the square to the cube	14 min
The two formulae	19 min
Expanding with coefficients	23 min
Reading a cube backwards	19 min
Homework	5 min

LEARNING OBJECTIVES

- Derive $(a + b)^3$ from $(a + b)^2$ by one more multiplication.
- Write both cube formulae from memory, with the right signs.
- Expand cubes of binomials with coefficients and two letters.
- Recognise a perfect cube and cube numbers mentally.

TEXTBOOK

National	pp. 297–303
Cambridge	Stage 8 extension
Workbook	Extension sheet 2A

01

Explanation

From the square to the cube

$(a + b)^3$ is $(a + b)^2$ multiplied by one more bracket — nothing more mysterious than that.

STEP	WORKING
the square	$a^2 + 2ab + b^2$
times a	$a^3 + 2a^2b + ab^2$
times b	$a^2b + 2ab^2 + b^3$
collect	$a^3 + 3a^2b + 3ab^2 + b^3$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

THE COEFFICIENTS ARE 1, 3, 3, 1

The square had 1, 2, 1 and the cube has 1, 3, 3, 1. Each row of this triangle of numbers is built from the one above it — a pattern you will meet again in Grade 11 under the name of the binomial theorem.

The two formulae

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

TERM	IN $(A + B)^3$	IN $(A - B)^3$
a^3	+	+
$3a^2b$	+	−
$3ab^2$	+	+
b^3	+	−

IN THE DIFFERENCE THE SIGNS ALTERNATE, AND THE LAST ONE IS MINUS

Unlike $(a - b)^2$, which ends in $+ b^2$, the cube ends in $- b^3$ — because an odd power of a negative number stays negative. Ending a cube with a plus is the standard slip.

Expanding with coefficients

EXPRESSION	A	B	EXPANSION
$(x + 2)^3$	x	2	$x^3 + 6x^2 + 12x + 8$
$(x - 1)^3$	x	1	$x^3 - 3x^2 + 3x - 1$
$(2x + 1)^3$	$2x$	1	$8x^3 + 12x^2 + 6x + 1$
$(a - 2b)^3$	a	$2b$	$a^3 - 6a^2b + 12ab^2 - 8b^3$

THE WHOLE TERM IS CUBED, COEFFICIENT INCLUDED

In $(2x + 1)^3$ the first term is $(2x)^3 = 8x^3$, not $2x^3$. Every power in the expansion applies to the bracketed term as a whole.

Reading a cube backwards

Four terms with coefficients in the ratio $1 : 3 : 3 : 1$ and matching cube roots make a perfect cube.

EXPRESSION	AS A CUBE
$x^3 + 6x^2 + 12x + 8$	$(x + 2)^3$
$x^3 - 3x^2 + 3x - 1$	$(x - 1)^3$
$8x^3 + 12x^2 + 6x + 1$	$(2x + 1)^3$
$x^3 + 3x^2 + 3x + 2$	not a cube — the last term should be 1

CUBING NUMBERS IN THE HEAD

$21^3 = (20 + 1)^3 = 8000 + 1200 + 60 + 1 = 9261$, and $19^3 = (20 - 1)^3 = 8000 - 1200 + 60 - 1 = 6859$. The alternating signs are easiest to see in exactly these calculations.

02 Worked examples

EXAMPLE 1 Expand $(2x + 1)^3$.

① $a = 2x, b = 1.$

② $a^3 = 8x^3$

The coefficient is cubed too.

③ $3a^2b = 3 \cdot 4x^2 \cdot 1 = 12x^2$

④ $3ab^2 = 3 \cdot 2x \cdot 1 = 6x$, and $b^3 = 1.$

ANSWER

$$8x^3 + 12x^2 + 6x + 1$$

EXAMPLE 2 Expand $(a - 2b)^3$.

① a^3

② $-3a^2(2b) = -6a^2b$

③ $+3a(2b)^2 = 12ab^2$

④ $-(2b)^3 = -8b^3$

ANSWER

$$a^3 - 6a^2b + 12ab^2 - 8b^3$$

EXAMPLE 3 Find 19^3 mentally.

① $19 = 20 - 1$

② $20^3 = 8000$

③ $-3 \cdot 400 \cdot 1 = -1200$ and $+3 \cdot 20 \cdot 1 = +60$.

④ $8000 - 1200 + 60 - 1 = 6859$

ANSWER

6859

03 Interactive model

Build the row 1, 3, 3, 1 from 1, 2, 1 on the board by adding neighbours; the class then predicts 1, 4, 6, 4, 1 without being told, and the pattern is theirs.

Signs and coefficients of a cube

The coefficients of $(a + b)^3$ are:

1, 2, 1

1, 3, 3, 1

1, 1, 1, 1

1, 4, 6, 4, 1

How many terms does a cube of a binomial have?

2

3

4

6

The last term of $(a - b)^3$ is:

$+ b^3$

$- b^3$

$+ b^2$

$- b^2$

The last term of $(a - b)^2$ is:

The first term of $(2x + 1)^3$ is:

$(x - 1)^3$ equals:

19^3 equals:

The row after 1, 3, 3, 1 is:

1, 4, 4, 1

1, 4, 6, 4, 1

1, 3, 4, 3, 1

1, 5, 10, 5, 1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Cube of a sum	Yig‘indining kubi	Куб суммы
Cube of a difference	Ayirmaning kubi	Куб разности
Cube	Kub	Куб
Coefficient	Koeffitsiyent	Коэффициент
Alternating signs	Almashinuvchi ishoralar	Чередующиеся знаки
Perfect cube	To‘la kub	Полный куб
Expansion	Yoyilma	Разложение
Pattern	Qonuniyat	Закономерность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The coefficients in $(a + b)^3$ are:

1, 2, 1

1, 3, 3, 1

1, 1, 1, 1

1, 4, 6, 4, 1

$(a - b)^3$ ends in:

$+ b^3$

$- b^3$

$+ b^2$

$- b^2$

$(x - 1)^3$ equals:

$x^3 - 1$

$x^3 - 3x^2 + 3x - 1$

$x^3 + 3x^2 + 3x + 1$

$x^3 - 3x + 1$

The first term of $(2x + 1)^3$ is:

The middle terms of $(x + 2)^3$ are:

21^3 equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x + 1)^3$

ANSWER $x^3 + 3x^2 + 3x + 1$

2. $(x - 1)^3$

ANSWER $x^3 - 3x^2 + 3x - 1$

3. $(x + 2)^3$

ANSWER $x^3 + 6x^2 + 12x + 8$

4. $(x - 2)^3$

ANSWER $x^3 - 6x^2 + 12x - 8$

5. $(a + 3)^3$

ANSWER $a^3 + 9a^2 + 27a + 27$

6. The coefficients of a cube

ANSWER $1, 3, 3, 1$

7. 11^3

ANSWER 1331

Medium · the standard question

7 PROBLEMS

1. $(2x + 1)^3$

ANSWER $8x^3 + 12x^2 + 6x + 1$

2. $(3a - 1)^3$

ANSWER $27a^3 - 27a^2 + 9a - 1$

3. $(a - 2b)^3$

ANSWER $a^3 - 6a^2b + 12ab^2 - 8b^3$

4. $x^3 + 6x^2 + 12x + 8$ as a cube

ANSWER $(x + 2)^3$

5. $8x^3 + 12x^2 + 6x + 1$ as a cube

ANSWER $(2x + 1)^3$

6. 21^3

ANSWER 9261

7. 19^3

ANSWER 6859

Hard · stretch and reason

7 PROBLEMS

1. $(x + \frac{1}{x})^3$

ANSWER $x^3 + 3x + \frac{3}{x} + \frac{1}{x^3}$

2. $(a + b)^3 - (a - b)^3$

ANSWER $6a^2b + 2b^3$

3. $(a + b)^3 + (a - b)^3$

ANSWER $2a^3 + 6ab^2$

4. If $a + b = 5$ and $ab = 6$, find $a^3 + b^3$

ANSWER 35

5. $(2a - 3b)^3$

ANSWER $8a^3 - 36a^2b + 54ab^2 - 27b^3$

6. Is $x^3 + 3x^2 + 3x + 2$ a perfect cube?

ANSWER No — the last term should be 1

7. The next row after 1, 3, 3, 1

ANSWER 1, 4, 6, 4, 1

07 Homework

Homework — 5 tasks

Write the four signs before you write any numbers.

1. Expand $(x + 4)^3$ and $(x - 3)^3$.
2. Expand $(3x + 2)^3$.
3. Expand $(2a - b)^3$.
4. Write $x^3 - 9x^2 + 27x - 27$ as a cube.

5. Find 31^3 and 29^3 by the formulae.

The difference and the sum of two cubes

Two factorisations that a sum of squares never had — and the trinomial that will not factorise further.

LESSON 104–105

2 × 40 MIN

MATEMATIKA 7, §47

BEYOND STAGE 8

LESSON CLOCK

14'

19'

23'

19'

5'

The two formulae	14 min
Why the middle terms vanish	19 min
Factorising	23 min
Where they are used	19 min
Homework	5 min

LEARNING OBJECTIVES

- Verify $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ by expanding.
- Factorise sums and differences of cubes.
- Distinguish the incomplete square $a^2 + ab + b^2$ from $(a + b)^2$.
- Use the formulae in numerical work and in cancelling fractions.

TEXTBOOK

National	pp. 304–310
Cambridge	Stage 8 extension
Workbook	Extension sheet 2B

01

Explanation

The two formulae

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

FORMULA	SIGN IN THE BRACKET OF TWO	SIGN OF THE MIDDLE TERM
$a^3 - b^3$	–	$+ab$
$a^3 + b^3$	+	$-ab$

THE SIGNS ARE ALWAYS OPPOSITE

Whatever sign stands between a and b in the short bracket, the middle term of the long bracket takes the other one. Getting these two the same way round is the only real difficulty in the topic.

Why the middle terms vanish

Expand $(a - b)(a^2 + ab + b^2)$ in full and watch four of the six terms cancel.

PRODUCT	TERMS
$a \cdot (a^2 + ab + b^2)$	$a^3 + a^2b + ab^2$
$-b \cdot (a^2 + ab + b^2)$	$-a^2b - ab^2 - b^3$
sum	$a^3 - b^3$

$A^2 + AB + B^2$ IS NOT $(A + B)^2$

The square has $2ab$ in the middle; this trinomial has just ab . It is called the **incomplete square** for exactly that reason, and it does not factorise any further.

Factorising

EXPRESSION	A	B	FACTORISED
$x^3 - 8$	x	2	$(x - 2)(x^2 + 2x + 4)$
$x^3 + 27$	x	3	$(x + 3)(x^2 - 3x + 9)$
$8a^3 - 1$	$2a$	1	$(2a - 1)(4a^2 + 2a + 1)$
$27x^3 + 64y^3$	$3x$	$4y$	$(3x + 4y)(9x^2 - 12xy + 16y^2)$

A SUM OF CUBES DOES FACTORISE

$a^2 + b^2$ was a dead end, but $a^3 + b^3$ is not — the odd power makes all the difference. That is worth saying out loud, because the two look alike on the page.

Where they are used

TASK	WORKING	ANSWER
$\frac{x^3 - 8}{x - 2}$	$\frac{(x - 2)(x^2 + 2x + 4)}{x - 2}$	$x^2 + 2x + 4$
$\frac{x^3 + 27}{x + 3}$	cancel $x + 3$	$x^2 - 3x + 9$
$10^3 - 1$	$9 \cdot (100 + 10 + 1)$	999

WHY $N^3 - 1$ IS ALWAYS DIVISIBLE BY $N - 1$

The factorisation shows it directly: one factor is $n - 1$. The same argument explains why 999, 9999 and every repunit-like number of that shape has an obvious divisor.

02 Worked examples

EXAMPLE 1 Factorise $8a^3 - 1$.

- ① $a = 2a$ and $b = 1$ — the cube roots.
- ② Short bracket: $(2a - 1)$.
- ③ Long bracket: $(2a)^2 + 2a \cdot 1 + 1^2 = 4a^2 + 2a + 1$.
Middle sign is the opposite.
- ④ $= (2a - 1)(4a^2 + 2a + 1)$

ANSWER

$$(2a - 1)(4a^2 + 2a + 1)$$

EXAMPLE 2 Factorise $27x^3 + 64y^3$.

① $a = 3x, b = 4y.$

② Short bracket: $(3x + 4y).$

③ Long bracket: $9x^2 - 12xy + 16y^2.$
Plus outside, minus inside.

④ $= (3x + 4y)(9x^2 - 12xy + 16y^2)$

ANSWER

$$(3x + 4y)(9x^2 - 12xy + 16y^2)$$

EXAMPLE 3 Simplify $\frac{x^3 - 8}{x - 2}$.

① $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$

② The factor $x - 2$ cancels.

③ $= x^2 + 2x + 4$
For $x \neq 2$.

ANSWER

$$x^2 + 2x + 4$$

03 Interactive model

Ask the class to expand $(a - b)(a^2 + ab + b^2)$ themselves before the formula is given; the six terms collapsing to two is more convincing than any statement of the rule.

A cube factorised, then cancelled

$$\frac{x^3 - 8}{x - 2}$$

① $a = x$

and $b = 2$

The
cube
roots
of
the
two
terms.

② $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$

Minus
outside,
plus
inside.

③ $x^2 + 2x + 4$

Cancel
 $x -$
2;
note
 $x \neq$
2.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Difference of two cubes	Kublar ayirmasi	Разность кубов
Sum of two cubes	Kublar yig'indisi	Сумма кубов
Incomplete square	To'liqsiz kvadrat	Неполный квадрат
Cube root	Kub ildiz	Кубический корень
Trinomial	Uch had	Трёхчлен
Opposite signs	Qarama-qarshi ishoralar	Противоположные знаки
To verify	Tekshirmoq	Проверить
Factorisation	Ko'paytuvchilarga ajratish	Разложение на множители

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a^3 - b^3$ equals:

$$(a - b)(a^2 + ab + b^2)$$

$$(a - b)^3$$

$$(a - b)(a^2 - ab + b^2)$$

$$(a - b)(a + b)^2$$

$a^3 + b^3$ equals:

$$(a + b)^3$$

$$(a + b)(a^2 - ab + b^2)$$

$$(a + b)(a^2 + ab + b^2)$$

it does not factorise

$x^3 - 8$ factorises to:

$$(x - 2)(x^2 + 2x + 4)$$

$$(x - 2)^3$$

$$(x - 2)(x^2 - 2x + 4)$$

$$(x - 8)(x^2 + 1)$$

$a^2 + ab + b^2$ is:

$(a + b)^2$

the incomplete square

$(a - b)^2$

a difference of squares

$x^3 + 27$ factorises to:

$(x + 3)(x^2 - 3x + 9)$

$(x + 3)^3$

$(x + 3)(x^2 + 3x + 9)$

it does not

Compared with $a^2 + b^2$, a sum of cubes:

also fails to factorise

does factorise

is a square

is negative

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x^3 - 8$

ANSWER $(x - 2)(x^2 + 2x + 4)$

2. $x^3 + 8$

ANSWER $(x + 2)(x^2 - 2x + 4)$

3. $x^3 - 1$

ANSWER $(x - 1)(x^2 + x + 1)$

4. $x^3 + 1$

ANSWER $(x + 1)(x^2 - x + 1)$

5. $x^3 + 27$

ANSWER $(x + 3)(x^2 - 3x + 9)$

6. $x^3 - 64$

ANSWER $(x - 4)(x^2 + 4x + 16)$

7. The name of $a^2 + ab + b^2$

ANSWER the incomplete square

Medium · the standard question

7 PROBLEMS

1. $8a^3 - 1$

ANSWER $(2a - 1)(4a^2 + 2a + 1)$

2. $27x^3 + 64y^3$

ANSWER $(3x + 4y)(9x^2 - 12xy + 16y^2)$

3. $125 - a^3$

ANSWER $(5 - a)(25 + 5a + a^2)$

4.
$$\frac{x^3 - 8}{x - 2}$$

ANSWER $x^2 + 2x + 4$

5.
$$\frac{x^3 + 27}{x + 3}$$

ANSWER $x^2 - 3x + 9$

6. $10^3 - 1$ factorised

ANSWER $9 \cdot 111 = 999$

7. Does $a^3 + b^3$ factorise?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. $64x^3 - 27y^3$

ANSWER $(4x - 3y)(16x^2 + 12xy + 9y^2)$

2. $2x^3 + 16$

ANSWER $2(x + 2)(x^2 - 2x + 4)$

3. $a^6 - b^6$ as a product of four factors

ANSWER $(a - b)(a + b)(a^2 + ab + b^2)(a^2 - ab + b^2)$

4. Why is $n^3 - 1$ always divisible by $n - 1$?

ANSWER It is one of the factors

5. $x^3 - \frac{1}{8}$

ANSWER $(x - \frac{1}{2})(x^2 + \frac{x}{2} + \frac{1}{4})$

6. If $a + b = 5$ and $ab = 6$, find $a^3 + b^3$ using the formula

ANSWER 35

7. Factorise $x^6 - 64$ as a difference of squares first

ANSWER $(x^3 - 8)(x^3 + 8)$

07 Homework**Homework — 5 tasks**

Write the short bracket first; its sign decides the middle sign of the long one.

1. Factorise $x^3 - 125$ and $x^3 + 64$.

2. Factorise $27a^3 - 8b^3$.

3. Simplify $\frac{x^3 + 1}{x + 1}$.

4. Factorise $5x^3 - 40$ completely.

5. Explain in one sentence the difference between $a^2 + ab + b^2$ and $(a + b)^2$.

Methods of factorising

Four methods in a fixed order — common factor, formula, grouping, and the trinomial split.

LESSON 106–107

2 × 40 MIN

MATEMATIKA 7, §48

S8 2.4

LESSON CLOCK

9'

23'

24'

19'

5'

The order of the methods	9 min
Splitting the middle term	23 min
Combining methods	24 min
When to stop	19 min
Homework	5 min

LEARNING OBJECTIVES

- Apply the four methods of factorising in the right order.
- Split the middle term of a quadratic trinomial into two.
- Combine methods where one alone is not enough.
- Recognise when a polynomial is already fully factorised.

TEXTBOOK

National	pp. 311–318
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The order of the methods

Always work down this list. Taking the steps out of order is what makes a factorisation look impossible.

STEP	METHOD	LOOK FOR
1	common factor	something in every term
2	an abridged formula	two terms, or a perfect square
3	grouping	four terms
4	splitting the middle term	a quadratic trinomial

STEP 1 IS NEVER SKIPPED

$2x^2 - 18$ looks like nothing until the 2 comes out; $3x^2 + 12x + 12$ becomes $3(x + 2)^2$ the same way. Take the common factor first, every single time.

Splitting the middle term

For $x^2 + px + q$, find two numbers whose **product** is q and whose **sum** is p . Split the middle term into those two and group.

TRINOMIAL	PRODUCT	SUM	THE TWO NUMBERS	FACTORISED
$x^2 + 7x + 12$	12	7	3, 4	$(x + 3)(x + 4)$
$x^2 - 5x + 6$	6	-5	-2, -3	$(x - 2)(x - 3)$
$x^2 + x - 12$	-12	1	4, -3	$(x + 4)(x - 3)$
$x^2 - 2x - 15$	-15	-2	-5, 3	$(x - 5)(x + 3)$

THE SIGN OF THE PRODUCT COMES FIRST

A negative product means the two numbers have opposite signs; a positive product means they match, and then the sign of the middle term tells you which sign both take. Deciding the signs before hunting for the numbers halves the work.

Combining methods

EXPRESSION	STEPS	RESULT
$2x^2 - 18$	common factor, then difference of squares	$2(x - 3)(x + 3)$
$3x^2 + 12x + 12$	common factor, then a perfect square	$3(x + 2)^2$
$x^3 - x$	common factor, then difference of squares	$x(x - 1)(x + 1)$
$2x^2 + 10x + 12$	common factor, then split the middle term	$2(x + 2)(x + 3)$
$a^3 + a^2 - a - 1$	grouping, then difference of squares	$(a - 1)(a + 1)^2$

ONE METHOD IS RARELY THE WHOLE ANSWER

Stopping at $2(x^2 - 9)$ loses the marks that the second step carries. Ask after every step whether any bracket can still be broken down.

When to stop

EXPRESSION	FULLY FACTORISED?	WHY
$2(x - 3)(x + 3)$	yes	each bracket is linear
$2(x^2 - 9)$	no	$x^2 - 9$
$(x + 1)(x^2 + 1)$	yes	a sum of squares does not factorise
$(x - 2)(x^2 + 2x + 4)$	yes	the incomplete square does not factorise
$x^2 + x + 1$	yes — it is prime	no two whole numbers give the product and the sum

CHECK BY EXPANDING, ALWAYS

A factorisation is finished when no bracket can be broken further, and it is correct when expanding it returns the original polynomial exactly. Both checks take seconds.

02 Worked examples

EXAMPLE 1 Factorise $x^2 + x - 12$.

- ① Product -12 , sum 1 .
A negative product: opposite signs.
- ② The numbers are 4 and -3 .
- ③ $x^2 + 4x - 3x - 12$
- ④ $x(x + 4) - 3(x + 4) = (x + 4)(x - 3)$

ANSWER

$$(x + 4)(x - 3)$$

EXAMPLE 2 Factorise $2x^2 + 10x + 12$ completely.

- ① Common factor 2: $2(x^2 + 5x + 6)$.
- ② Product 6, sum 5: the numbers are 2 and 3.
- ③ $= 2(x + 2)(x + 3)$
- ④ Check by expanding ✓

ANSWER

$$2(x + 2)(x + 3)$$

EXAMPLE 3 Factorise $a^3 + a^2 - a - 1$.

- ① Group: $a^2(a + 1) - (a + 1)$.
- ② $= (a + 1)(a^2 - 1)$
- ③ $a^2 - 1 = (a - 1)(a + 1)$
Do not stop here.
- ④ $= (a - 1)(a + 1)^2$

ANSWER

$$(a - 1)(a + 1)^2$$

03 Interactive model

Put five mixed expressions on the board and ask only which method each needs, without factorising; choosing the method is the skill, and it can be practised on its own.

Which method does this one need? $6x^2 + 9x$ needs: $x^2 - 25$ needs: $ax + ay + bx + by$ needs:

$x^2 + 7x + 12$ needs:

a common factor

a formula

grouping

splitting

$2x^2 - 18$ needs:

a common factor only

a formula only

a common factor then a formula

grouping

For $x^2 - 5x + 6$ the two numbers are:

2, 3

-2, -3

-1, -6

1, 6

A negative product means the numbers:

are both positive

are both negative

have opposite signs

are equal

$2(x^2 - 9)$ is:

fully factorised

not fully factorised

wrong

prime

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Method	Usul	Способ
Common factor	Umumiy ko'paytuvchi	Общий множитель
Abridged formula	Qisqa ko'paytirish formulasi	Формула сокращённого умножения
Grouping	Guruhlash	Группировка
Trinomial	Uch had	Трёхчлен
To split a term	Hadni ajratish	Разбить член
Completely factorised	To'liq ajratilgan	Полностью разложен
Order of steps	Qadamlar tartibi	Порядок действий

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first method to try is always:

a formula

the common factor

grouping

splitting

Grouping is used when there are:

two terms

three terms

four terms

one term

For $x^2 + px + q$ the two numbers have product:

p

q

$p + q$

pq

$x^2 + x - 12$ factorises to:

$x^3 - x$ factorises to:

A factorisation is checked by:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x^2 + 7x + 12$

ANSWER $(x + 3)(x + 4)$

2. $x^2 - 5x + 6$

ANSWER $(x - 2)(x - 3)$

3. $x^2 + x - 12$

ANSWER $(x + 4)(x - 3)$

4. $x^2 - 2x - 15$

ANSWER $(x - 5)(x + 3)$

5. $x^2 - 9$

ANSWER $(x - 3)(x + 3)$

6. $6x^2 + 9x$

ANSWER $3x(2x + 3)$

7. $x^2 + 6x + 9$

ANSWER $(x + 3)^2$

Medium · the standard question**7 PROBLEMS**

1. $2x^2 - 18$

ANSWER $2(x - 3)(x + 3)$

2. $3x^2 + 12x + 12$

ANSWER $3(x + 2)^2$

3. $x^3 - x$

ANSWER $x(x - 1)(x + 1)$

4. $2x^2 + 10x + 12$

ANSWER $2(x + 2)(x + 3)$

5. $x^2 - 8x + 15$

ANSWER $(x - 3)(x - 5)$

6. $x^2 + 2x - 8$

ANSWER $(x + 4)(x - 2)$

7. $ax + ay + bx + by$

ANSWER $(x + y)(a + b)$

Hard · stretch and reason

7 PROBLEMS

1. $a^3 + a^2 - a - 1$

ANSWER $(a - 1)(a + 1)^2$

2. $x^4 - 5x^2 + 4$

ANSWER $(x - 1)(x + 1)(x - 2)(x + 2)$

3. $3x^3 - 27x$

ANSWER $3x(x - 3)(x + 3)$

4. $x^2 + x + 1$

ANSWER Prime — it does not factorise

5. $2x^3 + 16$

ANSWER $2(x + 2)(x^2 - 2x + 4)$

6. $x^2 - 10x + 25 - y^2$

ANSWER $(x - 5 - y)(x - 5 + y)$

7. $4x^2 - 12x + 9$

ANSWER $(2x - 3)^2$

07 Homework**Homework — 5 tasks**

Take the common factor out first, and expand your answer back at the end.

- Factorise $x^2 + 9x + 20$ and $x^2 - 7x + 10$.
- Factorise $x^2 - 3x - 10$.
- Factorise $5x^2 - 45$ completely.
- Factorise $x^3 + 2x^2 - x - 2$ by grouping.
- Say which method each of $4a^2 - 9$, $6ab + 9a$ and $x^2 - x - 6$ needs.

Applying the abridged multiplication formulae

The five formulae at work — mental arithmetic, simplification, proofs and identities.

LESSON 108–109

2 × 40 MIN

MATEMATIKA 7, §49

S8 2.4

LESSON CLOCK

9'

19'

23'

24'

5'

The five formulae in one table

9 min

Mental arithmetic

19 min

Simplifying

23 min

Proving something

24 min

Homework

5 min

LEARNING OBJECTIVES

- Choose the right formula from the shape of an expression.
- Use the formulae to calculate quickly without a calculator.
- Simplify expressions in which several formulae appear together.
- Prove a simple divisibility statement by factorising.

TEXTBOOK

National

pp. 319–326

Cambridge

Stage 8, pp. 25–30

Workbook

Exercise 2.4

01

Explanation

The five formulae in one table

NAME	FORMULA	RECOGNISE IT BY
square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	three terms, middle $2ab$
square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$	three terms, middle $-2ab$
difference of squares	$a^2 - b^2 = (a - b)(a + b)$	two square terms, a minus
cube of a sum or difference	$(a \pm b)^3$	four terms, $1, 3, 3, 1$
sum or difference of cubes	$a^3 \pm b^3$	two cube terms

COUNT THE TERMS FIRST

Two terms point to a difference of squares or a sum of cubes; three to a square; four to a cube or to grouping. Counting decides the formula before any algebra is attempted.

Mental arithmetic

TASK	FORMULA	WORKING	ANSWER
103^2	$(a + b)^2$	$10000 + 600 + 9$	10609
97^2	$(a - b)^2$	$10000 - 600 + 9$	9409
$63 \cdot 57$	$a^2 - b^2$	$3600 - 9$	3591
$75^2 - 25^2$	$a^2 - b^2$	$50 \cdot 100$	5000
31^3	$(a + b)^3$	$27000 + 2700 + 90 + 1$	29791

75² – 25² IN ONE LINE

Factorised it is $(75 - 25)(75 + 25) = 50 \cdot 100$. Squaring both numbers first takes four times as long and invites an arithmetic slip.

Simplifying

EXPRESSION	SIMPLIFIED
$(a + b)^2 - (a - b)^2$	$4ab$
$(a + b)^2 + (a - b)^2$	$2a^2 + 2b^2$
$(x + 3)^2 - (x - 3)^2$	$12x$
$(x + 2)(x - 2) - x^2$	-4
$\frac{x^2 - 16}{x + 4}$	$x - 4$

EXPAND BOTH BRACKETS BEFORE SUBTRACTING

$(a + b)^2 - (a - b)^2$ is not $(2b)^2$. Subtracting the brackets as if they were single numbers is the standard slip; expand each one and then subtract term by term.

Proving something

A factorised form can prove a statement that the expanded form hides completely.

CLAIM	FACTORISED	WHY IT FOLLOWS
$n^2 + n$	$n(n + 1)$	consecutive numbers — one is even
$n^2 - 1$	$(n - 1)(n + 1)$	for odd n , both factors are even
$(n + 1)^2 - n^2$	$2n + 1$	always odd
$n^3 - n$	$(n - 1)n(n + 1)$	three consecutive numbers

WHY THE DIFFERENCE OF TWO CONSECUTIVE SQUARES IS ODD

$(n + 1)^2 - n^2 = 2n + 1$, which is odd for every whole n . It also explains why every odd number is a difference of two squares — $9 = 5^2 - 4^2$, $11 = 6^2 - 5^2$, and so on.

02 Worked examples

EXAMPLE 1 Find $75^2 - 25^2$ without a calculator.

1 A difference of squares.

2 $= (75 - 25)(75 + 25)$

3 $= 50 \cdot 100$

4 $= 5000$

ANSWER

5000

EXAMPLE 2 Simplify $(x + 3)^2 - (x - 3)^2$.

$$1 \quad (x + 3)^2 = x^2 + 6x + 9$$

$$2 \quad (x - 3)^2 = x^2 - 6x + 9$$

$$3 \quad \text{Subtract: } 12x.$$

The squares and the 9s cancel.

ANSWER

$12x$

EXAMPLE 3 Prove that $n^2 - 1$ is divisible by 8 for every odd n .

$$1 \quad n^2 - 1 = (n - 1)(n + 1)$$

$$2 \quad \text{For odd } n \text{ both factors are even.}$$

$$3 \quad \text{They are consecutive even numbers, so one is a multiple of 4.}$$

$$4 \quad \text{The product is therefore a multiple of } 2 \cdot 4 = 8.$$

ANSWER

Divisible by 8

03 Interactive model

Give the class $25 \cdot 35$, $51 \cdot 49$ and 98^2 as a race against the calculators; the formulae win, and the point needs no further argument.

Which formula fits? $x^2 - 49$ fits: $x^2 + 10x + 25$ fits: $x^3 - 27$ fits:

$x^3 - 6x^2 + 12x - 8$ fits:

difference of cubes

cube of a difference

square of a difference

none

97^2 is easiest as:

$(100 - 3)^2$

$(90 + 7)^2$

$97 \cdot 97$

$(100 + 3)^2$

$63 \cdot 57$ equals:

3591

3600

3609

3549

$(a + b)^2 - (a - b)^2$ equals:

$2b^2$

$4ab$

$2a^2$

0

$(n + 1)^2 - n^2$ is:

always even

always odd

always prime

sometimes zero

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Abridged multiplication	Qisqa ko'paytirish	Сокращённое умножение
To apply	Qo'llash	Применять
To simplify	Soddalashtirish	Упростить
Identity	Ayniyat	Тождество
Divisible	Bo'linadi	Делится
Consecutive	Ketma-ket	Последовательные
Even number	Juft son	Чётное число
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Three terms with a middle of $2ab$ point to:

a square

a difference of squares

a cube

grouping

Two square terms with a minus point to:

a square of a sum

a difference of squares

a sum of cubes

nothing

$75^2 - 25^2$ equals:

2500

5000

5625

10000

$(x + 3)^2 - (x - 3)^2$ equals:

$12x$

18

$6x$

0

$n^2 + n$ is always:

odd

even

prime

square

103^2 equals:

10609

10309

10906

10009

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 103^2

ANSWER 10609

2. 97^2

ANSWER 9409

3. $63 \cdot 57$

ANSWER 3591

4. $75^2 - 25^2$

ANSWER 5000

5. $(x + 3)^2 - (x - 3)^2$

ANSWER $12x$

6. $(x + 2)(x - 2) - x^2$

ANSWER -4

7.
$$\frac{x^2 - 16}{x + 4}$$

ANSWER $x - 4$ **Medium · the standard question**

7 PROBLEMS

1. $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

2. $(a + b)^2 + (a - b)^2$

ANSWER $2a^2 + 2b^2$

3. 31^3

ANSWER 29791

4. $45^2 - 35^2$

ANSWER 800

5. $(2x + 1)^2 - (2x - 1)^2$

ANSWER $8x$

6. Prove $n^2 + n$ is even

ANSWER $n(n + 1)$ — one factor is even

7. $(n + 1)^2 - n^2$

ANSWER $2n + 1$

Hard · stretch and reason

7 PROBLEMS

1. Prove $n^2 - 1$ is divisible by 8 for odd n

ANSWER Two consecutive even factors, one a multiple of 4

2. $n^3 - n$ factorised

ANSWER $(n - 1)n(n + 1)$

3. Write 11 as a difference of two squares

ANSWER $6^2 - 5^2$

4. $(a + b)^3 - (a - b)^3$

ANSWER $6a^2b + 2b^3$

5. If $a + b = 10$ and $a - b = 4$, find $a^2 - b^2$

ANSWER 40

6. $1000^2 - 999^2$

ANSWER 1999

7. Simplify $\frac{x^3 - 27}{x - 3}$

ANSWER $x^2 + 3x + 9$

07 Homework

Homework — 5 tasks

Count the terms, name the formula, then calculate.

- Find 104^2 , 96^2 and $72 \cdot 68$ by the formulae.
- Find $85^2 - 15^2$ in one line.
- Simplify $(x + 5)^2 - (x - 5)^2$ and $(3a + 1)^2 - (3a - 1)^2$.
- Prove that $n^3 - n$ is divisible by 6 for every whole n .

5. Write 15 and 21 each as a difference of two squares.

Control work 7, and work on the mistakes

The five abridged multiplication formulae, in both directions, in one paper.

LESSON 110–111

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 7

S8 2 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Expand squares and cubes of binomials accurately under time.
- Factorise by common factor, formula, grouping and splitting.
- Use the formulae for mental calculation.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 282–326
Cambridge	Stage 8, pp. 18–30
Workbook	Control paper M7

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Expand $(x + 6)^2$, $(2a - 5)^2$ and $(x - 1)^3$	5	L98–103
2	Factorise $x^2 - 49$ and $9a^2 - 16b^2$	4	L100–101
3	Factorise $2x^2 - 50$ completely	4	L106–107
4	Factorise $x^2 + 2x - 15$ and $x^3 + 8$	5	L104–107
5	Find 98^2 and $64 \cdot 56$ without a calculator	4	L108–109
6	Simplify $(a + b)^2 - (a - b)^2$	3	L108–109

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the middle term of each square and one for the four signs of the cube; Q3 two for taking the 2 out before anything else; Q5 one for naming the formula used; Q6 two for expanding both brackets rather than subtracting them whole.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
middle term dropped	$(x + 6)^2 = x^2 + 36$	$x^2 + 12x + 36$
coefficient not squared	$(2a - 5)^2 = 2a^2 - 20a + 25$	$4a^2 - 20a + 25$
cube ended with a plus	$(x - 1)^3 = x^3 - 3x^2 + 3x + 1$	$\dots - 1$
sum of squares “factorised”	$x^2 + 49 = (x + 7)^2$	it does not factorise
common factor left in	$2(x^2 - 25)$	$2(x - 5)(x + 5)$
cube formula signs matched	$x^3 + 8 = (x + 2)(x^2 + 2x + 4)$	$(x + 2)(x^2 - 2x + 4)$
brackets subtracted whole	$(a + b)^2 - (a - b)^2 = (2b)^2$	expand both first

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

FORMULA	READ LEFT TO RIGHT	READ RIGHT TO LEFT
$(a \pm b)^2$	expanding	recognising a perfect square
$a^2 - b^2$	a quick product	the fastest factorisation
$(a \pm b)^3$	expanding	recognising a perfect cube
$a^3 \pm b^3$	checking	factorising
splitting the middle term	—	factorising a trinomial

LOOKING FORWARD

Every one of these formulae is about to earn its keep: algebraic fractions cancel only when numerator and denominator are factorised, and that is the next four lessons.

02 Worked examples

EXAMPLE 1 Model answer, Q1: expand $(2a - 5)^2$.

① $a = 2a, b = 5.$

② $(2a)^2 = 4a^2$

The coefficient is squared too.

③ $-2 \cdot 2a \cdot 5 = -20a$

④ $+25$, so the answer is $4a^2 - 20a + 25$.

ANSWER

$$4a^2 - 20a + 25$$

EXAMPLE 2 Model answer, Q3: factorise $2x^2 - 50$ completely.

① Common factor 2: $2(x^2 - 25).$

② $x^2 - 25 = (x - 5)(x + 5)$

Do not stop at the previous line.

③ $= 2(x - 5)(x + 5)$

ANSWER

$$2(x - 5)(x + 5)$$

EXAMPLE 3 Model answer, Q5: find $64 \cdot 56$.

① Both are 4 from 60.

② $= (60 + 4)(60 - 4)$
A difference of squares.

③ $= 3600 - 16$

④ $= 3584$

ANSWER

3584

03 Interactive model

Hand back Q1 and Q2 together: expanding and factorising are the same five formulae read in opposite directions, and the paper is easier to learn from that way.

The chapter in twelve questions

$(a + b)^2$ equals:

$$a^2 + b^2$$

$$a^2 + 2ab + b^2$$

$$a^2 - 2ab + b^2$$

$$2a + 2b$$

$(a - b)^2$ ends in:

$$+ b^2$$

$$- b^2$$

$$+ b^3$$

$$- b$$

$(x + 6)^2$ equals:

$$x^2 + 36$$

$$x^2 + 12x + 36$$

$$x^2 + 6x + 36$$

$$x^2 + 12x + 6$$

$(2a - 5)^2$ begins with:

$(x - 1)^3$ ends in:

$x^2 - 49$ factorises to:

$x^2 + 49$ factorises to:

$2x^2 - 50$ factorises completely to:

$x^2 + 2x - 15$ factorises to:

$x^3 + 8$ factorises to:

98^2 equals:

$(a + b)^2 - (a - b)^2$ equals:

$4ab$

$2b^2$

$4b^2$

0

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Square of a sum	Yig'indining kvadrati	Квадрат суммы
Difference of two squares	Kvadratlar ayirmasi	Разность квадратов
Cube of a difference	Ayirmaning kubi	Куб разности
Sum of two cubes	Kublar yig'indisi	Сумма кубов
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Completely	To'liq	Полностью
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 loses a mark most often through:

the first term

the middle term

the sign of b^2

the brackets

Q2 needs:

a common factor

a difference of squares

grouping

splitting

Q3 begins with:

the formula

the common factor

grouping

splitting

In Q4, $x^3 + 8$ has middle sign:

plus

minus

either

none

Q5 uses $64 \cdot 56$ as:

$(60 + 4)(60 - 4)$

$(64 - 8)^2$

$60 \cdot 60$

$(60 + 4)^2$

Q6 must begin by:

subtracting the brackets

expanding both brackets

factorising

substituting

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x + 6)^2$

ANSWER $x^2 + 12x + 36$

2. $(2a - 5)^2$

ANSWER $4a^2 - 20a + 25$

3. $(x - 1)^3$

ANSWER $x^3 - 3x^2 + 3x - 1$

4. $x^2 - 49$

ANSWER $(x - 7)(x + 7)$

5. $9a^2 - 16b^2$

ANSWER $(3a - 4b)(3a + 4b)$

6. 98^2

ANSWER 9604

7. $64 \cdot 56$

ANSWER 3584

Medium · the standard question**7 PROBLEMS**

1. $2x^2 - 50$

ANSWER $2(x - 5)(x + 5)$

2. $x^2 + 2x - 15$

ANSWER $(x + 5)(x - 3)$

3. $x^3 + 8$

ANSWER $(x + 2)(x^2 - 2x + 4)$

4. $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

5. $3x^2 + 18x + 27$

ANSWER $3(x + 3)^2$

6. $x^2 - 7x + 12$

ANSWER $(x - 3)(x - 4)$

7. Does $x^2 + 49$ factorise?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $x^4 - 81$

ANSWER $(x - 3)(x + 3)(x^2 + 9)$

2. $2x^3 - 54$

ANSWER $2(x - 3)(x^2 + 3x + 9)$

3. $(3a + 1)^2 - (3a - 1)^2$

ANSWER $12a$

4. $x^2 + 6x + 9 - y^2$

ANSWER $(x + 3 - y)(x + 3 + y)$

5. $125^2 - 75^2$

ANSWER 10000

6. $a^3 - a$

ANSWER $a(a - 1)(a + 1)$

7. Simplify $\frac{x^2 - 36}{x - 6}$

ANSWER $x + 6$

07 Homework**Homework — 5 tasks**

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write out the five formulae with one worked example of each.
3. Factorise $3x^2 - 75$ and $x^3 - 27$ completely.
4. Find 106^2 and $73 \cdot 67$ by the formulae.

5. Simplify $(x + 4)^2 - (x - 4)^2$ and say which slip you were avoiding.

The algebraic fraction.

Cancelling fractions

A fraction whose parts are polynomials — and the one rule that only factors may be cancelled.

LESSON 112–113

2 × 40 MIN

MATEMATIKA 7, §50

S8 7

LESSON CLOCK

14'

19'

23'

19'

5'

What an algebraic fraction is	14 min
Where it has no value	19 min
Cancelling	23 min
Factors, never terms	19 min
Homework	5 min

LEARNING OBJECTIVES

- Define an algebraic fraction and state where it has no value.
- Factorise numerator and denominator before cancelling.
- Cancel only common factors, never terms.
- Write the values of the letter that must be excluded.

TEXTBOOK

National	pp. 327–334
Cambridge	Stage 8, pp. 66–78
Workbook	Exercise 7.1

01

Explanation

What an algebraic fraction is

An **algebraic fraction** is a quotient of two polynomials, exactly as an ordinary fraction is a quotient of two whole numbers.

$$\frac{P(x)}{Q(x)} \text{ with } Q(x) \neq 0$$

EXPRESSION	AN ALGEBRAIC FRACTION?
$\frac{x + 1}{x - 2}$	yes
$\frac{x^2}{3}$	yes — the denominator may be a number
$\frac{5}{x}$	yes
$x^2 + 1$	a polynomial, not a fraction

EVERYTHING YOU KNOW ABOUT ORDINARY FRACTIONS CARRIES OVER

Cancelling, common denominators, adding, multiplying, dividing — the rules are the same ones learnt in Quarter I. Only the objects in the numerator and denominator have changed.

Where it has no value

A denominator of zero makes the fraction meaningless, so those values of the letter must be excluded and written down.

FRACTION	DENOMINATOR ZERO WHEN	EXCLUDED
$\frac{5}{x}$	$x = 0$	$x \neq 0$
$\frac{x + 1}{x - 2}$	$x = 2$	$x \neq 2$
$\frac{3}{x^2 - 9}$	$x = \pm 3$	$x \neq 3, x \neq -3$
$\frac{x}{x^2 + 1}$	never	nothing to exclude

FIND THE EXCLUDED VALUES BEFORE CANCELLING

In $\frac{x^2 - 9}{x - 3}$ the answer is $x + 3$, but $x = 3$ is still forbidden — the original fraction had no value there.

Cancelling does not make a forbidden value legal.

Cancelling

Factorise both parts fully, then divide out every factor they share.

FRACTION	FACTORISED	SIMPLIFIED
$\frac{12a^3}{4a}$	$\frac{4a \cdot 3a^2}{4a}$	$3a^2$
$\frac{x^2 - 9}{x + 3}$	$\frac{(x - 3)(x + 3)}{x + 3}$	$x - 3$
$\frac{x^2 + 5x + 6}{x + 2}$	$\frac{(x + 2)(x + 3)}{x + 2}$	$x + 3$
$\frac{2x + 6}{x^2 - 9}$	$\frac{2(x + 3)}{(x - 3)(x + 3)}$	$\frac{2}{x - 3}$

THIS IS WHY THE LAST CHAPTER CAME FIRST

Every one of these cancellations begins with a factorisation from the abridged formulae. Without that chapter, none of these fractions could be simplified at all.

Factors, never terms

ATTEMPT	RIGHT?	WHY
$\frac{(x - 3)(x + 3)}{x + 3} = x - 3$	yes	$x + 3$
$\frac{x^2 - 9}{x + 3} = x^2 - 3$	no	the 9 and the 3 are not factors
$\frac{x + 5}{x} = 5$	no	x
$\frac{2x}{2} = x$	yes	2

A TERM THAT IS ADDED CAN NEVER BE CANCELLED

$\frac{x + 5}{x}$ is not 5: try $x = 1$ and the fraction is 6. Only things that are **multiplied** may be divided out.

02 Worked examples

EXAMPLE 1 Simplify $\frac{x^2 + 5x + 6}{x + 2}$ and state the excluded value.

① $x + 2 = 0$ when $x = -2$, so $x \neq -2$.

Written down first.

② $x^2 + 5x + 6 = (x + 2)(x + 3)$

Product 6, sum 5.

③
$$\frac{(x + 2)(x + 3)}{x + 2}$$

④ $= x + 3$

ANSWER

$x + 3, x \neq -2$

EXAMPLE 2 Simplify $\frac{2x + 6}{x^2 - 9}$.

① $2x + 6 = 2(x + 3)$

② $x^2 - 9 = (x - 3)(x + 3)$

③ Cancel $x + 3$.

④ $= \frac{2}{x - 3}, \text{ with } x \neq \pm 3.$

ANSWER

$$\frac{2}{x - 3}$$

EXAMPLE 3 Is $\frac{x+5}{x} = 5$ correct?

- ① Try $x = 1$: the fraction is 6.
- ② So the equality is false.
- ③ x is a term of the numerator, not a factor of it.

ANSWER

No

03 Interactive model

Write $(x + 5)/x = 5$ on the board and ask the class to test it with $x = 1$; one substitution ends the argument for the rest of the year.

Factorise, then cancel

$$\frac{x^2 + 5x + 6}{x + 2}$$

① $x \neq -2$

The
excluded
value,
noted
first.

② $x^2 + 5x + 6 = (x + 2)(x + 3)$

Product
6,
sum
5.

③ $x + 3$

Cancel
the
common
factor
 $x +$
2.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
To cancel	Qisqartirish	Сокращать
Common factor	Umumiy ko'paytuvchi	Общий множитель
Excluded value	Mumkin bo'lmagan qiymat	Недопустимое значение
Undefined	Aniqlanmagan	Не определено
Simplest form	Eng sodda ko'rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An algebraic fraction is a quotient of:

two numbers

two polynomials

two equations

two roots

5 x has no value when:

$x = 5$

$x = 0$

$x = 1$

never

3 $x^2 - 9$ excludes:

$x = 3$

$x = -3$

both

neither

$x^2 - 9$ $x + 3$ simplifies to:

$x + 5$ x simplifies to:

Cancelling is allowed for:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.
$$\frac{12a^3}{4a}$$

ANSWER $3a^2$

2.
$$\frac{x^2 - 9}{x + 3}$$

ANSWER $x - 3$

3.
$$\frac{2x}{2}$$

ANSWER x

4.
$$\frac{6x^2}{3x}$$

ANSWER $2x$

5. $\frac{5}{x}$: the excluded value

ANSWER $x = 0$

6. $\frac{x + 1}{x - 2}$: the excluded value

ANSWER $x = 2$

7.
$$\frac{x^2 + 5x + 6}{x + 2}$$

ANSWER $x + 3$

Medium · the standard question

7 PROBLEMS

1. $\frac{2x + 6}{x^2 - 9}$

ANSWER $\frac{2}{x - 3}$

2. $\frac{x^2 - 4}{x^2 + 4x + 4}$

ANSWER $\frac{x - 2}{x + 2}$

3. $\frac{3}{x^2 - 9}$: the excluded values

ANSWER $x = \pm 3$

4. $\frac{x^2 - 7x + 12}{x - 3}$

ANSWER $x - 4$

5. $\frac{4x^2 - 25}{2x + 5}$

ANSWER $2x - 5$

6. $\frac{x}{x^2 + 1}$: the excluded values

ANSWER None

7. Is $\frac{x + 5}{x} = 5$?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1.
$$\frac{x^3 - 8}{x - 2}$$

ANSWER $x^2 + 2x + 4$

2.
$$\frac{x^2 - 1}{x^2 + 2x + 1}$$

ANSWER $\frac{x - 1}{x + 1}$

3.
$$\frac{2x^2 - 18}{x^2 + 6x + 9}$$

ANSWER $\frac{2(x - 3)}{x + 3}$

4.
$$\frac{a^2 - b^2}{a^2 - 2ab + b^2}$$

ANSWER $\frac{a + b}{a - b}$

5.
$$\frac{x^2 + x - 6}{x^2 - 4}$$

ANSWER $\frac{x + 3}{x + 2}$

6. Why must the excluded values be found before cancelling?

ANSWER The original fraction, not the simplified one, defines them

7.
$$\frac{3 - x}{x - 3}$$

ANSWER -1

07 Homework

Homework — 5 tasks

Write the excluded values first, then simplify.

1. Simplify $\frac{x^2 - 16}{x + 4}$ and state the excluded value.
2. Simplify $\frac{3x + 12}{x^2 - 16}$.
3. Simplify $\frac{x^2 + 7x + 10}{x + 5}$.
4. Find the excluded values of $\frac{7}{x^2 - 4x}$.
5. Explain with one substitution why $\frac{x + 3}{x}$ is not 3.

Reducing algebraic fractions to a common denominator

Factorise every denominator, take each factor to its highest power, and multiply each fraction up.

LESSON 114–115

2 × 40 MIN

MATEMATIKA 7, §51

S8 7

LESSON CLOCK

9'

23'

24'

19'

5'

The rule that makes it legal	9 min
Building the denominator	23 min
The extra factors	24 min
Denominators that differ in sign	19 min
Homework	5 min

LEARNING OBJECTIVES

- Find the lowest common denominator of two algebraic fractions.
- Determine the extra factor each fraction must be multiplied by.
- Multiply numerator and denominator by the same expression without changing the value.
- Handle denominators that differ only in sign.

TEXTBOOK

National	pp. 335–341
Cambridge	Stage 8, pp. 66–78
Workbook	Exercise 7.2

01

Explanation

The rule that makes it legal

Multiplying the numerator and the denominator by the same non-zero expression does not change the value of a fraction — it only changes how it is written.

$$\frac{P}{Q} = \frac{P \cdot R}{Q \cdot R} \text{ for } R \neq 0$$

FRACTION	MULTIPLIED BY	NEW FORM
$\frac{1}{x}$	$x + 1$	$\frac{x + 1}{x(x + 1)}$
$\frac{2}{x - 3}$	$x + 3$	$\frac{2(x + 3)}{x^2 - 9}$
$\frac{3}{a}$	2	$\frac{6}{2a}$

CANCELLING AND BUILDING ARE THE SAME RULE

Last lesson divided both parts by a common factor; this lesson multiplies both parts by an extra one. It is one rule read in two directions.

Building the denominator

Factorise every denominator, then take each distinct factor to the highest power in which it appears anywhere.

DENOMINATORS	FACTORISED	LOWEST COMMON DENOMINATOR
$6x, 8x^2$	$2 \cdot 3 \cdot x, 2^3 \cdot x^2$	$24x^2$
$x - 3, x + 3$	already factorised	$(x - 3)(x + 3)$
$x^2 - 9, x + 3$	$(x - 3)(x + 3), x + 3$	$(x - 3)(x + 3)$
$x^2 + 2x + 1, x^2 - 1$	$(x + 1)^2, (x - 1)(x + 1)$	$(x + 1)^2(x - 1)$

MULTIPLYING THE DENOMINATORS TOGETHER IS NOT WRONG, ONLY WASTEFUL

For $x^2 - 9$ and $x + 3$ the product is $(x - 3)(x + 3)^2$, which works but leaves a factor to cancel at the end. Factorising first gives the lowest one and half the arithmetic.

The extra factors

Each fraction is multiplied by whatever its own denominator is missing.

FRACTION	DENOMINATOR	EXTRA FACTOR	NEW NUMERATOR
$\frac{1}{x-3}$	$x-3$	$x+3$	$x+3$
$\frac{2}{x+3}$	$x+3$	$x-3$	$2x-6$
$\frac{5}{6x}$	$6x$	$4x$	$20x$
$\frac{3}{8x^2}$	$8x^2$	3	9

MULTIPLY THE NUMERATOR OUT

$\frac{2}{x+3}$ becomes $\frac{2x-6}{(x-3)(x+3)}$, not $\frac{2(x-3)}{\dots}$ left unexpanded — the next lesson has to add these numerators, and an unexpanded bracket with a minus in front is where the sign errors start.

Denominators that differ in sign

$3-x$ and $x-3$ are opposites, so one sign change makes them equal.

$\frac{1}{3-x} = -\frac{1}{x-3}$

PAIR	REWRITTEN	COMMON DENOMINATOR
$x-3, 3-x$	$3-x = -(x-3)$	$x-3$
$a-b, b-a$	$b-a = -(a-b)$	$a-b$
$2x-1, 1-2x$	$1-2x = -(2x-1)$	$2x-1$

TREATING THEM AS DIFFERENT FACTORS DOUBLES THE WORK

Building $(x-3)(3-x)$ as a common denominator is legal but pointless — and it hides the fact that the answer will simplify. Take the minus out in front instead.

02 Worked examples

EXAMPLE 1 Reduce $\frac{1}{x-3}$ and $\frac{2}{x+3}$ to a common denominator.

① Neither denominator factorises further.

② The LCD is $(x-3)(x+3) = x^2 - 9$.

③ $\frac{1}{x-3} = \frac{x+3}{x^2-9}$
Extra factor $x+3$.

④ $\frac{2}{x+3} = \frac{2x-6}{x^2-9}$
Extra factor $x-3$.

ANSWER

$$\frac{x+3}{x^2-9} \text{ and } \frac{2x-6}{x^2-9}$$

EXAMPLE 2 Find the lowest common denominator of $\frac{1}{x^2+2x+1}$ and $\frac{1}{x^2-1}$.

① $x^2 + 2x + 1 = (x+1)^2$

② $x^2 - 1 = (x-1)(x+1)$

③ Take $(x+1)$ to its highest power and $(x-1)$ once.

④ LCD = $(x+1)^2(x-1)$.

ANSWER

$$(x+1)^2(x-1)$$

EXAMPLE 3 Reduce $\frac{1}{x-3}$ and $\frac{2}{3-x}$ to a common denominator.

① $3 - x = -(x - 3)$

② $\frac{2}{3-x} = -\frac{2}{x-3}$

③ The common denominator is just $x - 3$.

④ The fractions are $\frac{1}{x-3}$ and $\frac{-2}{x-3}$.

ANSWER

Common denominator $x - 3$

03 Interactive model

Set the pair $1/(x-3)$ and $2/(3-x)$ before teaching the sign rule; the class builds a monstrous denominator, and the one-line alternative then sells itself.

Assembling the lowest common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
$6x$	$2 \cdot 3 \cdot x$	$4x$
$8x^2$	$2^3 \cdot x^2$	3

LOWEST COMMON DENOMINATOR

$$24x^2$$

The numerators become $20x$ and 9 .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Extra factor	Qo'shimcha ko'paytuvchi	Дополнительный множитель
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Highest power	Eng yuqori daraja	Наивысшая степень
Equivalent fraction	Teng kuchli kasr	Равная дробь
Opposite expression	Qarama-qarshi ifoda	Противоположное выражение
Sign change	Ishorani o'zgartirish	Смена знака

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Multiplying both parts of a fraction by the same expression:

changes its value

keeps its value

makes it zero

is not allowed

The LCD of $6x$ and $8x^2$ is:

$48x^3$

$24x^2$

$14x^2$

$24x$

The LCD of $x - 3$ and $x + 3$ is:

$x^2 - 9$

$x^2 + 9$

$x - 3$

$(x - 3)^2$

The LCD of $x^2 - 9$ and $x + 3$ is:

$3 - x$ equals:

The extra factor for 2 $x + 3$ towards $x^2 - 9$ is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. LCD of $6x$ and $8x^2$

ANSWER $24x^2$

2. LCD of $x - 3$ and $x + 3$

ANSWER $x^2 - 9$

3. LCD of $2a$ and $3a$

ANSWER $6a$

4. LCD of x and $x + 1$

ANSWER $x(x + 1)$

5. $\frac{1}{x}$ to denominator $x(x + 1)$

ANSWER $\frac{x + 1}{x(x + 1)}$

6. $\frac{3}{a}$ to denominator $2a$

ANSWER $\frac{6}{2a}$

7. $3 - x$ written with $x - 3$

ANSWER $-(x - 3)$

Medium · the standard question

7 PROBLEMS

1. LCD of $x^2 - 9$ and $x + 3$

ANSWER $x^2 - 9$

2. LCD of $x^2 + 2x + 1$ and $x^2 - 1$

ANSWER $(x + 1)^2(x - 1)$

3. $\frac{2}{x + 3}$ to denominator $x^2 - 9$

ANSWER $\frac{2x - 6}{x^2 - 9}$

4. $\frac{1}{x - 3}$ to denominator $x^2 - 9$

ANSWER $\frac{x + 3}{x^2 - 9}$

5. $\frac{5}{6x}$ to denominator $24x^2$

ANSWER $\frac{20x}{24x^2}$

6. LCD of $x - 3$ and $3 - x$

ANSWER $x - 3$

7. LCD of $a - b$ and $b - a$

ANSWER $a - b$

Hard · stretch and reason

7 PROBLEMS

1. LCD of $x^2 - 4$, $x^2 + 4x + 4$ and $x + 2$

ANSWER $(x - 2)(x + 2)^2$

2. LCD of $x^2 - x$ and $x^2 - 1$

ANSWER $x(x - 1)(x + 1)$

3. $\frac{1}{x^2 - 4}$ to denominator $(x - 2)(x + 2)^2$

ANSWER $\frac{x + 2}{(x - 2)(x + 2)^2}$

4. LCD of $4x^2y$ and $6xy^3$

ANSWER $12x^2y^3$

5. Why is the lowest common denominator preferred?

ANSWER The numbers stay small and nothing has to be cancelled at the end

6. $\frac{2}{3 - x}$ with denominator $x - 3$

ANSWER $\frac{-2}{x - 3}$

7. LCD of $x^2 - 9$, $x^2 - 6x + 9$

ANSWER $(x + 3)(x - 3)^2$

07 Homework

Homework — 5 tasks

Factorise every denominator before you write a single common denominator.

- Find the LCD of $10x$ and $15x^2$.
- Reduce $\frac{1}{x - 5}$ and $\frac{3}{x + 5}$ to a common denominator.
- Find the LCD of $x^2 - 16$ and $x + 4$, and rewrite both fractions.

4. Find the LCD of $x^2 - 25$ and $x^2 + 10x + 25$.

5. Rewrite $\frac{4}{2-x}$ with denominator $x - 2$.

Addition and subtraction of algebraic fractions

One denominator, then the numerators — and the bracket that must be written when subtracting.

LESSON 116–117

2 × 40 MIN

MATEMATIKA 7, §52

S8 7.3

LESSON CLOCK

9'

23'

24'

19'

5'

The same denominator	9 min
Different denominators	23 min
The bracket when subtracting	24 min
Finishing the answer	19 min
Homework	5 min

LEARNING OBJECTIVES

- Add and subtract fractions with the same denominator.
- Reduce to a common denominator first when the denominators differ.
- Bracket the whole numerator that is being subtracted.
- Simplify the answer and state the excluded values.

TEXTBOOK

National	pp. 342–349
Cambridge	Stage 8, pp. 70–74
Workbook	Exercise 7.3

01

Explanation

The same denominator

$$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}$$

The denominator is written once and the numerators are combined — exactly as with ordinary fractions.

SUM	COMBINED	SIMPLIFIED
$\frac{2x}{5} + \frac{3x}{5}$	$\frac{5x}{5}$	x
$\frac{3}{x} + \frac{4}{x}$	$\frac{7}{x}$	$\frac{7}{x}$
$\frac{x}{x+1} + \frac{1}{x+1}$	$\frac{x+1}{x+1}$	1

ALWAYS LOOK AT THE ANSWER AGAIN

The third line collapsed to 1 only because the numerator turned out to be the denominator. Simplifying the result is part of the question, not an extra.

Different denominators

Reduce to the lowest common denominator first — the whole of the previous lesson — and only then combine the numerators.

STEP	WORKING
the sum	$\frac{1}{x-3} + \frac{2}{x+3}$
the LCD	$x^2 - 9$
rewrite both	$\frac{x+3}{x^2-9} + \frac{2x-6}{x^2-9}$
combine	$\frac{3x-3}{x^2-9}$
factorise and check	$\frac{3(x-1)}{(x-3)(x+3)}$

NOTHING CANCELS HERE – AND THAT IS AN ANSWER TOO

$3(x-1)$ shares no factor with $(x-3)(x+3)$, so the fraction is already in its simplest form. Checking is quick; assuming is what costs marks.

The bracket when subtracting

When a whole numerator is subtracted, it goes inside a bracket, and every sign in it changes.

$$\frac{5x-1}{x+2} - \frac{3x+4}{x+2} = \frac{(5x-1) - (3x+4)}{x+2} = \frac{2x-5}{x+2}$$

WRITTEN	RIGHT?
$(5x - 1) - (3x + 4) = 2x - 5$	yes
$5x - 1 - 3x + 4 = 2x + 3$	no — the +4 was not negated

THE MINUS APPLIES TO THE WHOLE NUMERATOR

This single missing bracket accounts for more lost marks in the algebraic-fractions paper than every other slip combined. Write the bracket first and remove it afterwards.

Finishing the answer

STEP	WHAT TO DO
1	factorise the denominators and find the LCD
2	multiply each fraction up by its extra factor
3	combine the numerators, bracketing anything subtracted
4	collect like terms in the numerator
5	factorise the numerator and cancel if possible
6	state the excluded values

EXPRESSION	ANSWER
$\frac{1}{x} + \frac{1}{x+1}$	$\frac{2x+1}{x(x+1)}$
$\frac{1}{x-1} - \frac{1}{x+1}$	$\frac{2}{x^2-1}$
$\frac{x}{x-2} - \frac{2}{x-2}$	1

02 Worked examples

EXAMPLE 1 Simplify $\frac{1}{x-3} + \frac{2}{x+3}$.

① LCD = $x^2 - 9$, with $x \neq \pm 3$.

② $\frac{x+3}{x^2-9} + \frac{2x-6}{x^2-9}$

③ $= \frac{3x-3}{x^2-9}$

④ $= \frac{3(x-1)}{(x-3)(x+3)}$

Nothing cancels.

ANSWER

$$\frac{3x-3}{x^2-9}$$

EXAMPLE 2 Simplify $\frac{5x - 1}{x + 2} - \frac{3x + 4}{x + 2}$.

① The denominators already match.

② $= \frac{(5x - 1) - (3x + 4)}{x + 2}$

The bracket is compulsory.

③ $= \frac{5x - 1 - 3x - 4}{x + 2}$

Both signs changed.

④ $= \frac{2x - 5}{x + 2}$

ANSWER

$$\frac{2x - 5}{x + 2}$$

EXAMPLE 3 Simplify $\frac{1}{x - 1} - \frac{1}{x + 1}$.

① $\text{LCD} = (x - 1)(x + 1) = x^2 - 1$.

② $= \frac{(x + 1) - (x - 1)}{x^2 - 1}$

③ $= \frac{x + 1 - x + 1}{x^2 - 1}$

④ $= \frac{2}{x^2 - 1}$

ANSWER

$$\frac{2}{x^2 - 1}$$

03 Interactive model

Mark one exercise for the bracket alone, ignoring everything else; the class discovers where its marks actually go.

Adding and subtracting, step by step

$$\frac{1}{x-3} + \frac{2}{x+3}$$

① LCD

$= (x - 3)(x + 3)$
 $= (x^2 - 9)$

② $\frac{x+3}{x^2-9} + \frac{2x-6}{x^2-9}$

Each
fraction
multiplied
up.

③ $\frac{3x-3}{x^2-9}$

Collect
the
numerators.

④ $\frac{3(x-1)}{(x-3)(x+3)}$

Factorise
and
check
for a
cancellation
—
there
is
none.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Addition	Qo‘shish	Сложение
Subtraction	Ayirish	Вычитание
Common denominator	Umumiy maxraj	Общий знаменатель
Numerator	Surat	Числитель
Bracket	Qavs	Скобка
Like terms	O‘xshash hadlar	Подобные члены
To simplify	Soddalashtirish	Упростить
Excluded value	Mumkin bo‘lmagan qiymat	Недопустимое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

With the same denominator you combine:

the denominators

the numerators

both

neither

$\frac{3}{x} + \frac{4}{x}$ equals:

$$\frac{7}{x}$$

$$\frac{7}{2x}$$

$$\frac{12}{x}$$

$$\frac{7}{x^2}$$

When subtracting, the second numerator needs:

a bracket

a denominator

nothing

a factor

$(5x - 1) - (3x + 4)$ equals:

$\frac{1}{x-1} - \frac{1}{x+1}$ equals:

\frac{2}{x^2 - 1}"/>

\frac{2x}{x^2 - 1}"/>

\frac{1}{x^2 - 1}"/>

The last step of every such question is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

$$1. \quad \frac{2x}{5} + \frac{3x}{5}$$

ANSWER x

$$2. \quad \frac{3}{x} + \frac{4}{x}$$

ANSWER $\frac{7}{x}$

$$3. \quad \frac{x}{x+1} + \frac{1}{x+1}$$

ANSWER 1

$$4. \quad \frac{5}{a} - \frac{2}{a}$$

ANSWER $\frac{3}{a}$

$$5. \quad \frac{x}{x-2} - \frac{2}{x-2}$$

ANSWER 1

$$6. \quad \frac{7x}{3} - \frac{x}{3}$$

ANSWER $2x$

$$7. \quad (5x - 1) - (3x + 4)$$

ANSWER $2x - 5$

Medium · the standard question

7 PROBLEMS

1.
$$\frac{1}{x-3} + \frac{2}{x+3}$$

ANSWER
$$\frac{3x-3}{x^2-9}$$

2.
$$\frac{1}{x-1} - \frac{1}{x+1}$$

ANSWER
$$\frac{2}{x^2-1}$$

3.
$$\frac{1}{x} + \frac{1}{x+1}$$

ANSWER
$$\frac{2x+1}{x(x+1)}$$

4.
$$\frac{5x-1}{x+2} - \frac{3x+4}{x+2}$$

ANSWER
$$\frac{2x-5}{x+2}$$

5.
$$\frac{2}{3x} + \frac{1}{2x}$$

ANSWER
$$\frac{7}{6x}$$

6.
$$\frac{1}{x} - \frac{1}{x+2}$$

ANSWER
$$\frac{2}{x(x+2)}$$

7.
$$\frac{3}{x-1} + \frac{3}{1-x}$$

ANSWER
$$0$$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{x-2} + \frac{4}{x^2-4}$

ANSWER $\frac{x+6}{x^2-4}$

2. $\frac{x}{x+1} - \frac{1}{x-1}$

ANSWER $\frac{x^2-2x-1}{x^2-1}$

3. $\frac{1}{x} + \frac{1}{x+1} + \frac{1}{x+2}$ as one fraction

ANSWER $\frac{3x^2+6x+2}{x(x+1)(x+2)}$

4. $\frac{2}{x-3} - \frac{2}{x+3}$

ANSWER $\frac{12}{x^2-9}$

5. $\frac{a}{a-b} + \frac{b}{b-a}$

ANSWER 1

6. $\frac{1}{x^2-1} + \frac{1}{x^2+2x+1}$

ANSWER $\frac{2x}{(x-1)(x+1)^2}$

7. Why is a bracket needed when subtracting?

ANSWER The minus applies to every term of that numerator

07 Homework

Homework — 5 tasks

Write the bracket before you remove it, every single time.

1. Simplify $\frac{2}{x} + \frac{3}{x}$ and $\frac{4x}{7} - \frac{x}{7}$.
2. Simplify $\frac{1}{x-5} + \frac{2}{x+5}$.
3. Simplify $\frac{4x+3}{x-1} - \frac{2x+7}{x-1}$.
4. Simplify $\frac{1}{x} - \frac{1}{x+3}$.
5. Simplify $\frac{5}{x-2} + \frac{5}{2-x}$ and explain the answer.

Multiplication and division of algebraic fractions

Straight across for a product; turn the second one over for a quotient — and cancel before you multiply.

LESSON 118–119

2 × 40 MIN

MATEMATIKA 7, §53

S8 7.4–7.5

LESSON CLOCK

14'

19'

23'

19'

5'

Multiplying	14 min
Cancel first	19 min
Dividing	23 min
Powers, and all the excluded values	19 min
Homework	5 min

LEARNING OBJECTIVES

- Multiply algebraic fractions and cancel before multiplying out.
- Divide by multiplying by the reciprocal.
- Raise an algebraic fraction to a power.
- State every excluded value, including those from the divisor.

TEXTBOOK

National	pp. 350–357
Cambridge	Stage 8, pp. 75–78
Workbook	Exercise 7.5

01

Explanation

Multiplying

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

No common denominator is needed — multiplication is the easy operation on fractions, not the hard one.

PRODUCT	STRAIGHT ACROSS	SIMPLIFIED
$\frac{2}{x} \cdot \frac{x}{3}$	$\frac{2x}{3x}$	$\frac{2}{3}$
$\frac{3a}{4} \cdot \frac{2}{9a^2}$	$\frac{6a}{36a^2}$	$\frac{1}{6a}$
$\frac{x+1}{x} \cdot \frac{x}{x+1}$	$\frac{x(x+1)}{x(x+1)}$	1

CONTRAST THIS WITH ADDING

Adding needed a common denominator and brackets; multiplying needs neither. Pupils often import the common denominator into a product out of habit — it is not wrong, just wasted work.

Cancel first

Factorise everything, cancel any factor of a numerator against any factor of a denominator, and only then multiply what is left.

STEP	WORKING
the product	$\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$
factorise	$\frac{(x - 3)(x + 3)}{x + 1} \cdot \frac{x + 1}{x - 3}$
cancel $x + 1$ and $x - 3$	$x + 3$

MULTIPLYING OUT FIRST MAKES THE WORK FAR HARDER

Expanding $(x^2 - 9)(x + 1)$ produces a cubic that then has to be factorised again to cancel. Cancel first: it is the difference between one line and five.

Dividing

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

QUOTIENT	AS A PRODUCT	ANSWER
$\frac{2}{x} \div \frac{4}{x^2}$	$\frac{2}{x} \cdot \frac{x^2}{4}$	$\frac{x}{2}$
$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	$\frac{x^2 - 1}{x} \cdot \frac{x^2}{x + 1}$	$x(x - 1)$
$\frac{a}{b} \div a$	$\frac{a}{b} \cdot \frac{1}{a}$	$\frac{1}{b}$

INVERT THE DIVISOR, NEVER THE DIVIDEND

It is the **second** fraction that is turned over. Inverting the first gives the reciprocal of the right answer, and the mistake is invisible unless the answer is checked with a number.

Powers, and all the excluded values

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

EXPRESSION	ANSWER
$\left(\frac{2}{x}\right)^3$	$\frac{8}{x^3}$
$\left(\frac{3a}{b^2}\right)^2$	$\frac{9a^2}{b^4}$

In a division, three things must be non-zero: both original denominators, and the numerator of the divisor — because it becomes a denominator when the fraction is inverted.

DIVISION	EXCLUDED	WHY
$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	$x \neq 0$	both denominators
$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	$x \neq -1$	it becomes a denominator

THE HIDDEN EXCLUSION

$x = -1$ never appears as a denominator in the question as written, yet it is forbidden. Dividing by zero is what the rule is protecting against, and the zero is in the numerator of the divisor.

02 Worked examples

EXAMPLE 1 Simplify $\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$.

① $x^2 - 9 = (x - 3)(x + 3)$

② Cancel $x + 1$ and $x - 3$.
Before multiplying anything out.

③ $= x + 3$
With $x \neq -1$ and $x \neq 3$.

ANSWER

$x + 3$

EXAMPLE 2 Simplify $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$.

① Invert the second fraction: $\frac{x^2 - 1}{x} \cdot \frac{x^2}{x + 1}$.

② $x^2 - 1 = (x - 1)(x + 1)$

③ Cancel $x + 1$, and x against x^2 .

④ $= x(x - 1)$
With $x \neq 0$ and $x \neq -1$.

ANSWER

$x(x - 1)$

EXAMPLE 3 Simplify $(\frac{3a}{b^2})^2$.

① The power applies to numerator and denominator.

② $(3a)^2 = 9a^2$

③ $(b^2)^2 = b^4$

④ $= \frac{9a^2}{b^4}$

ANSWER

$$\frac{9a^2}{b^4}$$

03 Interactive model

Set the same product twice — once told to expand first, once told to cancel first — and time both halves of the class; the cancelling half finishes before the other has factorised.

Cancel first, multiply second

$$\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$$

① $x^2 - 9 = (x - 3)(x + 3)$

Factorise
before
anything
else.

② $\frac{(x - 3)(x + 3)}{x + 1} \cdot \frac{x + 1}{x - 3}$

③ $x + 3$

Cancel
 $x +$
 1
and
 $x -$
 3 ; x
 $\neq$
 -1 ,
 3 .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Multiplication	Ko'paytirish	Умножение
Division	Bo'lish	Деление
Reciprocal	Teskari son	Обратная дробь
To invert	Ag'darib qo'yish	Перевернуть
Cross-cancelling	Krest-nakrest qisqartirish	Сокращение крест-накрест
Power of a fraction	Kasrning darajasi	Степень дроби
Divisor	Bo'luvchi	Делитель
Excluded value	Mumkin bo'lmagan qiymat	Недопустимое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To multiply two fractions you:

need a common denominator

multiply across

invert the second

add the numerators

To divide by a fraction you:

invert the first

invert the second

invert both

invert neither

2 x ÷ 4 x² equals:

$$\frac{x}{2}$$

$$\frac{8}{x^3}$$

$$\frac{1}{2x}$$

$$\frac{2}{x}$$

In a product you should cancel:

after multiplying out

before multiplying out

never

only numbers

(2 x)³ equals:

$$\frac{6}{x^3}$$

$$\frac{8}{x^3}$$

$$\frac{8}{3x}$$

$$\frac{2}{x^3}$$

In $\frac{a}{b} \div \frac{c}{d}$ which must be non-zero?

b

b and d

b , c and d

none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{2}{x} \cdot \frac{x}{3}$

ANSWER $\frac{2}{3}$

2. $\frac{3a}{4} \cdot \frac{2}{9a^2}$

ANSWER $\frac{1}{6a}$

3. $\frac{x+1}{x} \cdot \frac{x}{x+1}$

ANSWER 1

4. $\frac{2}{x} \div \frac{4}{x^2}$

ANSWER $\frac{x}{2}$

5. $\frac{a}{b} \div a$

ANSWER $\frac{1}{b}$

6. $(\frac{2}{x})^3$

ANSWER $\frac{8}{x^3}$

7. $(\frac{3a}{b^2})^2$

ANSWER $\frac{9a^2}{b^4}$

Medium · the standard question

7 PROBLEMS

1. $\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$

ANSWER $x + 3$

2. $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$

ANSWER $x(x - 1)$

3. $\frac{x}{x + 2} \cdot \frac{x^2 - 4}{x}$

ANSWER $x - 2$

4. $\frac{4}{x^2} \div \frac{2}{x}$

ANSWER $\frac{2}{x}$

5. $\frac{x^2 + 5x + 6}{x + 2} \cdot \frac{1}{x + 3}$

ANSWER 1

6. The excluded values of $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$

ANSWER $x \neq 0$ and $x \neq -1$

7. $\frac{a^2}{b} \cdot \frac{b^2}{a}$

ANSWER ab

Hard · stretch and reason

7 PROBLEMS

1. $\frac{x^2 - 4}{x^2 + 4x + 4} \cdot \frac{x + 2}{x - 2}$

ANSWER 1

2. $\frac{x^3 - 8}{x^2 - 4} \cdot \frac{x + 2}{1}$

ANSWER $x^2 + 2x + 4$

3. $\frac{a^2 - b^2}{a + b} \div \frac{a - b}{a}$

ANSWER a

4. $(\frac{x}{y})^2 \cdot \frac{y^3}{x}$

ANSWER xy

5. $\frac{x^2 - 9}{x^2 - 4} \div \frac{x + 3}{x + 2}$

ANSWER $\frac{x - 3}{x - 2}$

6. Why must the numerator of the divisor be non-zero?

ANSWER It becomes a denominator when the fraction is inverted

7. $\frac{1}{x} \div \frac{1}{x^2} \div x$

ANSWER 1

07 Homework

Homework — 5 tasks

Factorise and cancel before multiplying anything out.

1. Simplify $\frac{3}{x} \cdot \frac{x^2}{6}$ and $\frac{5a}{2} \cdot \frac{4}{15a^2}$.

2. Simplify $\frac{x^2 - 16}{x + 2} \cdot \frac{x + 2}{x - 4}$.

3. Simplify $\frac{x^2 - 4}{x} \div \frac{x + 2}{x^2}$.

4. Simplify $(\frac{4}{a})^2$ and $(\frac{2x}{y^3})^3$.

5. List every excluded value of $\frac{x + 1}{x - 2} \div \frac{x + 3}{x}$.

Control work 8, and work on the mistakes

Algebraic fractions from end to end — cancelling, common denominators, products and quotients.

LESSON 120–121

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 8

S8 7 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Cancel only after factorising, and state the excluded values.
- Build the lowest common denominator and add or subtract correctly.
- Multiply and divide fractions, cancelling first.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 327–357
Cambridge	Stage 8, pp. 66–78
Workbook	Control paper M8

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Simplify $\frac{x^2 - 25}{x + 5}$ and state the excluded value	4	L112–113
2	Find the excluded values of $\frac{3}{x^2 - 4x}$	3	L112–113
3	Find the lowest common denominator of $x^2 - 9$ and $x + 3$	3	L114–115
4	Simplify $\frac{1}{x - 2} + \frac{3}{x + 2}$	5	L116–117
5	Simplify $\frac{4x + 1}{x - 3} - \frac{2x + 7}{x - 3}$	4	L116–117
6	Simplify $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	6	L118–119

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for $x \neq -5$ alone; Q4 two for the extra factors and one for expanding the numerators; Q5 two for the bracket; Q6 two for inverting the right fraction and one for the hidden exclusion $x \neq -1$.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
terms cancelled	$\frac{x^2 - 25}{x + 5} = x^2 - 5$	factorise, then cancel factors
excluded value forgotten	an answer with no condition	$x \neq -5$
only one excluded value found	$x \neq 0$	$x \neq 0$ and $x \neq 4$
denominators multiplied blindly	$(x - 3)(x + 3)^2$	$(x - 3)(x + 3)$
numerator not multiplied up	$\frac{1 + 3}{x^2 - 4}$	$\frac{(x + 2) + 3(x - 2)}{x^2 - 4}$
bracket missing on subtraction	$4x + 1 - 2x + 7$	$(4x + 1) - (2x + 7)$
wrong fraction inverted	$\frac{x}{x^2 - 1} \cdot \frac{x + 1}{x^2}$	invert the divisor

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

OPERATION	WHAT IT NEEDS	WHAT IT DOES NOT NEED
cancelling	both parts factorised	a common denominator
adding	the LCD and brackets	anything inverted
subtracting	the LCD and a bracket on the second numerator	—
multiplying	factorising, then cancelling	a common denominator
dividing	inverting the divisor	a common denominator

LOOKING FORWARD

The last block of the quarter is equations: what a root is, how a linear equation is solved, Al-Khwarizmi's method, and problems written as equations. Fractions will appear inside those equations, so this chapter stays in use.

02 Worked examples

EXAMPLE 1 Model answer, Q2: the excluded values of $\frac{3}{x^2 - 4x}$.

① $x^2 - 4x = x(x - 4)$
Factorise the denominator.

② It is zero when $x = 0$ or $x = 4$.

③ $x \neq 0$ and $x \neq 4$.
Both, not one.

ANSWER

$x \neq 0, x \neq 4$

EXAMPLE 2 Model answer, Q4: $\frac{1}{x-2} + \frac{3}{x+2}$.

① $\text{LCD} = (x-2)(x+2) = x^2 - 4.$

② $\frac{x+2}{x^2-4} + \frac{3x-6}{x^2-4}$

Numerators multiplied out.

③ $= \frac{4x-4}{x^2-4}$

④ $= \frac{4(x-1)}{(x-2)(x+2)}, x \neq \pm 2.$

Nothing cancels.

ANSWER

$$\frac{4x-4}{x^2-4}$$

EXAMPLE 3 Model answer, Q6: $\frac{x^2-1}{x} \div \frac{x+1}{x^2}$.

① Invert the divisor: $\frac{x^2-1}{x} \cdot \frac{x^2}{x+1}$.

② $x^2 - 1 = (x-1)(x+1)$

③ Cancel $x+1$ and one x .

④ $= x(x-1)$, with $x \neq 0$ and $x \neq -1$.

The second one is hidden.

ANSWER

$$x(x-1)$$

03 Interactive model

Mark Q5 for the bracket alone and Q6 for the exclusions alone; the class sees that the algebra was rarely the problem.

The chapter in twelve questions

An algebraic fraction has no value when:

the numerator is zero

the denominator is zero

both are zero

never

$\frac{x^2 - 25}{x + 5}$ equals:

$x^2 - 5$

$x - 5$

$x + 5$

x

Its excluded value is:

$x \neq 5$

$x \neq -5$

$x \neq 0$

none

3 $x^2 - 4x$ excludes:

$x \neq 0$

$x \neq 4$

both

neither

The LCD of $x^2 - 9$ and $x + 3$ is:

$(x - 3)(x + 3)^2$

$x^2 - 9$

$x + 3$

$(x + 3)^2$

Cancelling is allowed for:

terms

factors

both

neither

When subtracting, the second numerator needs:

a bracket

a denominator

a factor

nothing

$(4x + 1) - (2x + 7)$ equals:

To divide by a fraction you invert:

$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$ equals:

\frac{x - 1}{x}"/>

\frac{1}{x}"/>

Its hidden exclusion is:

$$x \neq 1$$

$$x \neq -1$$

$$x \neq 2$$

none

In a product you should cancel:

after expanding

before expanding

never

only numbers

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
To cancel	Qisqartirish	Сокращать
Common denominator	Umumiy maxraj	Общий знаменатель
Reciprocal	Teskari kasr	Обратная дробь
Bracket	Qavs	Скобка
Excluded value	Mumkin bo'lmagan qiymat	Недопустимое значение
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 begins by:

cancelling

factorising

expanding

substituting

Q2 needs how many excluded values?

0

1

2

3

Q3 is answered with:

 $(x - 3)(x + 3)^2$ $x^2 - 9$ $x + 3$ $(x - 3)^2$

Q4 loses marks most often at:

the LCD

multiplying the numerators up

the final answer

the exclusions

Q5 is about:

the LCD

the bracket

the reciprocal

the exclusions

Q6 requires inverting:

the first fraction

the second fraction

both

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.
$$\frac{x^2 - 25}{x + 5}$$

ANSWER $x - 5$

2. Its excluded value

ANSWER $x \neq -5$

3.
$$\frac{3}{x^2 - 4x}$$
: the excluded values

ANSWER $x \neq 0, x \neq 4$

4. LCD of $x^2 - 9$ and $x + 3$

ANSWER $x^2 - 9$

5. $(4x + 1) - (2x + 7)$

ANSWER $2x - 6$

6.
$$\frac{3}{x} + \frac{5}{x}$$

ANSWER $\frac{8}{x}$

7.
$$\frac{2}{x} \cdot \frac{x}{4}$$

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\frac{1}{x-2} + \frac{3}{x+2}$

ANSWER $\frac{4x-4}{x^2-4}$

2. $\frac{4x+1}{x-3} - \frac{2x+7}{x-3}$

ANSWER $\frac{2x-6}{x-3} = 2$

3. $\frac{x^2-1}{x} \div \frac{x+1}{x^2}$

ANSWER $x(x-1)$

4. Its excluded values

ANSWER $x \neq 0, x \neq -1$

5. $\frac{x^2+6x+9}{x+3}$

ANSWER $x+3$

6. LCD of x^2-4 and $x+2$

ANSWER x^2-4

7. $\frac{x}{x+1} \cdot \frac{x^2-1}{x}$

ANSWER $x-1$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{x-1} - \frac{2}{x^2-1}$

ANSWER $\frac{1}{x+1}$

2. $\frac{x^2-4}{x^2-4x+4}$

ANSWER $\frac{x+2}{x-2}$

3. $\frac{a}{a-b} - \frac{b}{a-b}$

ANSWER 1

4. $\frac{x^2-9}{x+2} \div \frac{x-3}{x+2}$

ANSWER $x+3$

5. $\frac{1}{x} + \frac{1}{x+1}$ as one fraction

ANSWER $\frac{2x+1}{x(x+1)}$

6. Why does Q6 exclude $x = -1$?

ANSWER That numerator becomes a denominator on inverting

7. $\frac{x^3+8}{x+2}$

ANSWER $x^2 - 2x + 4$

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.

2. Simplify $\frac{x^2 - 36}{x - 6}$ and state the excluded value.
3. Simplify $\frac{2}{x - 1} + \frac{1}{x + 1}$.
4. Simplify $\frac{x^2 - 4}{x} \div \frac{x - 2}{x^2}$ and list every excluded value.
5. Write one sentence for each row of the chapter map.

The equation and its root

An equality that is true only for certain values — and what it means to check one.

LESSON 122

1 × 40 MIN

MATEMATIKA 7, §54

S8 2.5

LESSON CLOCK

10'

10'

12'

6'

2'

Three kinds of equality	10 min
What a root is	10 min
How many roots	12 min
Equivalent equations	6 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish an equation from an identity and from an expression.
- Define a root and verify one by substitution.
- Say how many roots an equation has, including none and infinitely many.
- Recognise equivalent equations.

TEXTBOOK

National	pp. 358–362
Cambridge	Stage 8, pp. 31–36
Workbook	Exercise 2.5

01

Explanation

Three kinds of equality

WRITTEN	NAME	TRUE FOR
$2x + 3$	an expression	nothing — it makes no claim
$2x + 3 = 11$	an equation	one value, $x = 4$
$(a + b)^2 = a^2 + 2ab + b^2$	an identity	every value

AN EQUATION IS A QUESTION

It asks: for which values of the letter is this equality true? The whole of the rest of the quarter is about answering that question efficiently.

What a root is

A **root** is a value of the unknown that makes the equation a true statement. Substituting it is the only test.

EQUATION	CANDIDATE	LEFT SIDE	RIGHT SIDE	A ROOT?
$2x + 3 = 11$	$x = 4$	11	11	yes
$2x + 3 = 11$	$x = 3$	9	11	no
$x^2 = 9$	$x = -3$	9	9	yes
$x^2 = 9$	$x = 3$	9	9	yes — there are two

CHECKING MEANS SUBSTITUTING INTO THE ORIGINAL EQUATION

Substituting into a line halfway through the working repeats any error made before it. Go back to the equation as it was set.

How many roots

EQUATION	ROOTS	HOW MANY
$2x + 3 = 11$	$x = 4$	one
$x^2 = 9$	$x = \pm 3$	two
$x + 1 = x + 2$	none	none — it says $1 = 2$
$2(x + 1) = 2x + 2$	every number	infinitely many — an identity
$x^2 = -4$	none among the numbers we know	none

“NO ROOTS” IS AN ANSWER

Reaching $0 = 1$ does not mean the working failed; it means the equation is never true. Reaching $0 = 0$ means it is always true. Both are complete answers.

Equivalent equations

Two equations are **equivalent** when they have exactly the same roots. Solving an equation is replacing it by simpler equivalent ones until the answer is visible.

OPERATION	KEEPS THE ROOTS?
add the same number to both sides	yes
multiply both sides by a non-zero number	yes
multiply both sides by 0	no — everything becomes a root
square both sides	not always — new roots can appear

MULTIPLYING BY AN EXPRESSION THAT COULD BE ZERO IS NOT SAFE

Multiplying $x = 1$ by x gives $x^2 = x$, which also has the root 0 . The new equation is not equivalent, and that is why every answer is checked.

02

Worked examples

EXAMPLE 1 Is $x = 4$ a root of $2x + 3 = 11$?

1

Substitute: $2 \cdot 4 + 3$.

2

$= 11$

3

The right side is 11 .

4

The equality holds, so yes.

ANSWER

Yes

EXAMPLE 2 How many roots has $x + 1 = x + 2$?

- ① Subtract x from both sides.

- ② $1 = 2$

- ③ This is false for every value of x .

- ④ The equation has no roots.

ANSWER

None

EXAMPLE 3 How many roots has $2(x + 1) = 2x + 2$?

- ① Expand the left side: $2x + 2$.

- ② $2x + 2 = 2x + 2$

- ③ True for every value of x .

- ④ It is an identity — infinitely many roots.

ANSWER

Every number

03 Interactive model

Give three equations with one root, no roots and infinitely many, unlabelled, and ask the class to sort them; the categories are learnt in one go.

Root, no root, or every number?

Is $x = 4$ a root of $2x + 3 = 11$?

Is $x = 3$ a root of $2x + 3 = 11$?

How many roots has $x^2 = 9$?

How many roots has $x + 1 = x + 2$?

0

1

2

infinitely many

How many roots has $2(x + 1) = 2x + 2$?

0

1

2

infinitely many

$2x + 3$ on its own is:

an equation

an expression

an identity

a root

Reaching $0 = 0$ means:

no roots

one root

every number is a root

an error

Squaring both sides:

always keeps the roots

can add new ones

removes roots

is never allowed

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Equation	Tenglama	Уравнение
Root	Ildiz	Корень
Solution	Yechim	Решение
To solve	Yechmoq	Решить
Unknown	Noma'lum	Неизвестное
Identity	Ayniyat	Тождество
To check	Tekshirmoq	Проверить
Equivalent equations	Teng kuchli tenglamalar	Равносильные уравнения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A root is a value that makes the equation:

A root is checked by:

$x^2 = 9$ has:

An identity is true for:

one value

two values

every value

no value

Reaching $0 = 1$ means:

no roots

every root

one root

an error was made

Equivalent equations have:

the same form

the same roots

the same degree

the same letters

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is $x = 4$ a root of $2x + 3 = 11$?

ANSWER Yes

2. Is $x = 3$ a root of $2x + 3 = 11$?

ANSWER No

3. Is $x = -3$ a root of $x^2 = 9$?

ANSWER Yes

4. The roots of $x^2 = 9$

ANSWER $x = \pm 3$

5. The root of $x + 5 = 12$

ANSWER $x = 7$

6. The root of $3x = 12$

ANSWER $x = 4$

7. $2x + 3$ is an equation or an expression?

ANSWER An expression

Medium · the standard question

7 PROBLEMS

1. How many roots has $x + 1 = x + 2$?

ANSWER None

2. How many roots has $2(x + 1) = 2x + 2$?

ANSWER Infinitely many

3. Is $(a + b)^2 = a^2 + 2ab + b^2$ an equation or an identity?

ANSWER An identity

4. Is $x = 1$ a root of $x^2 = x$?

ANSWER Yes

5. Is $x = 0$ a root of $x^2 = x$?

ANSWER Yes

6. Is $x = 0$ a root of $x = 1$?

ANSWER No

7. What does reaching $0 = 5$ tell you?

ANSWER The equation has no roots

Hard · stretch and reason

7 PROBLEMS

1. Are $x = 1$ and $x^2 = x$ equivalent?

ANSWER No — the second also has the root 0

2. Which operations preserve the roots?

ANSWER Adding the same number, and multiplying by a non-zero number

3. Why is multiplying both sides by x unsafe?

ANSWER x may be zero, and a new root appears

4. How many roots has $x^2 = -4$ among the numbers of Grade 7?

ANSWER None

5. Give an equation with exactly one root

ANSWER e.g. $2x = 6$

6. Give an equation with no roots

ANSWER e.g. $x = x + 1$

7. Give an equation true for every number

ANSWER e.g. $3(x - 1) = 3x - 3$

07 Homework

Homework — 5 tasks

Check every answer by substituting into the equation as it was originally written.

1. Check whether $x = 5$ is a root of $3x - 4 = 11$.
2. Find all the roots of $x^2 = 16$.
3. How many roots has $x + 3 = x + 7$? Explain.
4. How many roots has $5(x - 2) = 5x - 10$? Explain.
5. Write one equation with one root, one with none, and one true for every number.

Linear equations with one unknown

One standard form, four steps, and the three things that can happen at the end.

LESSON 123–124

2 × 40 MIN

MATEMATIKA 7, §55

S8 2.5

LESSON CLOCK

9'

23'

24'

19'

5'

The standard form	9 min
The four steps	23 min
Brackets and fractions	24 min
The three endings	19 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise a linear equation and write it as $ax = b$.
- Solve by collecting the unknowns on one side and the numbers on the other.
- Solve equations containing brackets and fractions.
- Interpret the cases $a = 0, b \neq 0$ and $a = 0, b = 0$.

TEXTBOOK

National	pp. 363–370
Cambridge	Stage 8, pp. 31–36
Workbook	Exercise 2.5

01

Explanation

The standard form

An equation is **linear** in x when, after simplification, it can be written as

$ax = b$

with no x^2 , no x in a denominator and no x under a root.

EQUATION	LINEAR?
$3x - 5 = 7$	yes
$2(x + 1) = 5x - 4$	yes — after expanding
$x^2 = 9$	no
$\frac{1}{x} = 3$	no — the unknown is in the denominator

WHY THE STANDARD FORM MATTERS

Once an equation reads $ax = b$ the answer is $x = \frac{b}{a}$ and nothing is left to decide. Every step of the solution is aimed at reaching that one line.

The four steps

STEP	WHAT TO DO	EXAMPLE: $3x - 5 = x + 7$
1	expand any brackets	none here
2	unknowns to the left, numbers to the right	$3x - x = 7 + 5$
3	collect like terms	$2x = 12$
4	divide by the coefficient	$x = 6$

A TERM THAT CROSSES THE EQUALS SIGN CHANGES ITS SIGN

-5 on the left becomes $+5$ on the right. Moving a term without changing its sign is the commonest error in the whole of equation-solving, and the check catches it every time.

Brackets and fractions

Expand first; clear denominators by multiplying **every** term by the lowest common denominator.

EQUATION	FIRST MOVE	THEN	ROOT
$2(x + 3) = 5x - 6$	expand	$2x + 6 = 5x - 6$	$x = 4$
$\frac{x}{2} + \frac{x}{3} = 5$	multiply by 6	$3x + 2x = 30$	$x = 6$
$\frac{x-1}{4} = \frac{x+1}{6}$	multiply by 12	$3(x-1) = 2(x+1)$	$x = 5$

EVERY TERM IS MULTIPLIED, THE WHOLE-NUMBER ONES TOO

In $\frac{x}{2} + \frac{x}{3} = 5$ the 5 becomes 30. Multiplying only the fractions is the standard slip and gives a wrong root that the check exposes at once.

The three endings

FINAL LINE	MEANING	ANSWER
$2x = 12$	$a \neq 0$	$x = 6$
$0 \cdot x = 5$	$a = 0, b \neq 0$	no roots
$0 \cdot x = 0$	$a = 0, b = 0$	every number is a root

And the check, always into the original equation:

EQUATION	ROOT	LEFT	RIGHT
$3x - 5 = x + 7$	$x = 6$	13	13
$2(x + 3) = 5x - 6$	$x = 4$	14	14

02

Worked examples

EXAMPLE 1 Solve $3x - 5 = x + 7$.

- 1

$3x - x = 7 + 5$
Both terms change sign as they cross.
- 2

$2x = 12$
- 3

$x = 6$
- 4

Check: $3 \cdot 6 - 5 = 13$ and $6 + 7 = 13$ ✓

ANSWER

$x = 6$

EXAMPLE 2 Solve $\frac{x}{2} + \frac{x}{3} = 5$.

- ① Multiply every term by 6.
Including the 5.

② $3x + 2x = 30$

③ $5x = 30$

④ $x = 6$

ANSWER

$x = 6$

EXAMPLE 3 Solve $2(x + 3) = 2x + 5$.

- ① $2x + 6 = 2x + 5$
Expand.

② $2x - 2x = 5 - 6$

③ $0 \cdot x = -1$

- ④ No value of x can satisfy this.

ANSWER

No roots

03 Interactive model

Insist on the check for one full lesson and mark it as a separate step; the number of sign errors handed in falls immediately.

One step at a time

In $3x - 5 = x + 7$, moving x to the left gives:

Moving -5 to the right gives:

So the equation becomes:

And the root is:

$$x = 2$$

$$x = 6$$

$$x = 12$$

$$x = 24$$

To clear $\underline{x}^2 + \underline{x}^3 = 5$ multiply by:

2

3

5

6

The 5 then becomes:

5

10

15

30

$0 \cdot x = 5$ means:

no roots

one root

every number

an error

$0 \cdot x = 0$ means:

no roots

one root

every number is a root

an error

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Linear equation	Chiziqli tenglama	Линейное уравнение
Unknown	Noma'lum	Неизвестное
Coefficient	Koeffitsiyent	Коэффициент
To transpose	Qarama-qarshi tomonga o'tkazish	Перенести
To collect	Ixchamlash	Привести подобные
To clear fractions	Maxrajdan qutulish	Избавиться от знаменателей
Standard form	Standart ko'rinish	Стандартный вид
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A linear equation can be written as:

$$ax^2 = b$$

$$ax = b$$

$$\frac{a}{x} = b$$

$$a + x = 0 \text{ only}$$

A term crossing the equals sign:

keeps its sign

changes its sign

disappears

doubles

$3x - 5 = x + 7$ gives:

$$x = 3$$

$$x = 6$$

$$x = 12$$

$$x = 1$$

To clear fractions, multiply:

the fractions only

every term

the right side only

nothing

$0 \cdot x = 7$ has:

one root

no roots

two roots

every root

The final check uses:

the last line

the original equation

a calculator

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x + 5 = 12$

ANSWER $x = 7$

2. $3x = 12$

ANSWER $x = 4$

3. $2x - 4 = 10$

ANSWER $x = 7$

4. $3x - 5 = x + 7$

ANSWER $x = 6$

5. $5x + 2 = 3x + 10$

ANSWER $x = 4$

6. $7 - x = 3$

ANSWER $x = 4$

7. $4x = 0$

ANSWER $x = 0$

Medium · the standard question

7 PROBLEMS

1. $2(x + 3) = 5x - 6$

ANSWER $x = 4$

2. $\frac{x}{2} + \frac{x}{3} = 5$

ANSWER $x = 6$

3. $\frac{x - 1}{4} = \frac{x + 1}{6}$

ANSWER $x = 5$

4. $3(x - 2) = 2(x + 1)$

ANSWER $x = 8$

5. $2(x + 3) = 2x + 5$

ANSWER No roots

6. $3(x - 1) = 3x - 3$

ANSWER Every number

7. $\frac{2x}{5} = 4$

ANSWER $x = 10$

Hard · stretch and reason

7 PROBLEMS

1. $4(x - 1) - 3(x + 2) = 0$

ANSWER $x = 10$

2. $\frac{x+1}{2} - \frac{x-1}{3} = 1$

ANSWER $x = 1$

3. $5(2x - 3) = 2(5x - 7)$

ANSWER No roots

4. $\frac{3x}{4} - \frac{x}{6} = 7$

ANSWER $x = 12$

5. $ax = 5$ with $a = 0$: how many roots?

ANSWER None

6. $0.2x + 1 = 0.5x - 0.5$

ANSWER $x = 5$

7. $2(3x - 1) - 3(2x - 1) = 1$

ANSWER Every number

07 Homework

Homework — 5 tasks

Write the check as a separate line under every answer.

1. Solve $4x - 7 = 2x + 9$.

2. Solve $3(x + 2) = 4x - 1$.

3. Solve $\frac{x}{3} + \frac{x}{4} = 7$.

4. Solve $\frac{x+2}{3} = \frac{x-1}{2}$.

5. Solve $2(x+4) = 2x+8$ and explain the answer.

Al-Khwarizmi’s method of solving equations

Al-jabr and al-muqabala — the two moves that gave algebra its name, written in Khiva twelve centuries ago.

LESSON 125–126

2 × 40 MIN

MATEMATIKA 7, §56

S8 2.5 HISTORICAL

LESSON CLOCK

9'

19'

19'

28'

5'

Who he was	9 min
Al-jabr — restoring	19 min
Al-muqabala — balancing	19 min
A medieval solution, and the modern one beside it	28 min
Homework	5 min

LEARNING OBJECTIVES

- Explain what al-jabr and al-muqabala mean and do.
- Solve an equation in the medieval style, without negative numbers.
- Connect the two moves to the modern rules for transposing and collecting.
- Place Al-Khwarizmi in the history of mathematics in Central Asia.

TEXTBOOK

National	pp. 371–377
Cambridge	Stage 8, pp. 31–36
Workbook	History sheet 2C

01

Explanation

Who he was

Muhammad ibn Musa **al-Khwarizmi** (c. 780 – c. 850) came from Khorezm, in what is now Uzbekistan, and worked at the House of Wisdom in Baghdad. His book *al-Kitab al-mukhtasar fi hisab al-jabr wa-l-muqabala* gave two words to the world.

WORD	FROM	MEANING TODAY
algebra	<i>al-jabr</i>	the whole subject
algorithm	his own name, Latinised	a finite list of steps

WHY HE WROTE IN WORDS, NOT SYMBOLS

There were no symbols for the unknown, no equals sign and no negative numbers. Every equation was a sentence, and every solution a paragraph. That is the difficulty his two moves were invented to overcome.

Al-jabr — restoring

Al-jabr means restoring: a subtracted quantity is removed from one side by adding it to both. It is how a negative term is got rid of when negative numbers do not exist.

$$5x - 3 = 12 \rightarrow 5x = 12 + 3$$

BEFORE	AFTER AL - JABR
$5x - 3 = 12$	$5x = 15$
$2x - 7 = x + 1$	$2x = x + 8$
$x^2 - 4x = 5$	$x^2 = 4x + 5$

THE SAME MOVE YOU ALREADY MAKE

Modern algebra says “transpose and change the sign”. Al-jabr says “add the missing amount to both sides”. They are the same operation; only the language differs.

Al-muqabala — balancing

Al-muqabala means balancing: like quantities on the two sides are cancelled against each other.

$$5x + 3 = 2x + 12 \rightarrow 3x = 9$$

BEFORE	AFTER AL - MUQABALA
$5x + 3 = 2x + 12$	$3x = 9$
$4x + 7 = x + 19$	$3x = 12$
$x^2 + 5 = 3x + 5$	$x^2 = 3x$

TWO MOVES, AND EVERY LINEAR EQUATION FALLS

Al-jabr removes what is subtracted; al-muqabala removes what is repeated. Applied in turn they reduce any linear equation to $ax = b$ — which is exactly the modern four-step method with older names.

A medieval solution, and the modern one beside it

AL - KHWARIZMI ' S WORDS	MODERN LINE
“Five things less three dirhams equal two things and nine dirhams.”	$5x - 3 = 2x + 9$
“Restore the three: five things equal two things and twelve dirhams.”	$5x = 2x + 12$
“Balance the two things: three things equal twelve dirhams.”	$3x = 12$
“Then one thing is four.”	$x = 4$

HIS TERM	OUR TERM
thing (shay)	x
dirham	the constant term
square (mal)	x^2
al-jabr	transposing a negative term
al-muqabala	collecting like terms

HE ALSO PROVED HIS METHODS GEOMETRICALLY

Where we write $(x + 3)^2 = x^2 + 6x + 9$, he drew the square and its four pieces — the same picture used two chapters ago. Without symbols, the diagram was the proof.

02 Worked examples

EXAMPLE 1 Solve $5x - 3 = 2x + 9$ in the medieval style, naming each move.

- ① Al-jabr: add 3 to both sides.

$$5x = 2x + 12$$

- ② Al-muqabala: cancel $2x$ from both sides.

$$3x = 12$$

- ③ Divide by 3.

$$x = 4$$

- ④ Check: $5 \cdot 4 - 3 = 17$ and $2 \cdot 4 + 9 = 17$ ✓

ANSWER

$$x = 4$$

EXAMPLE 2 Which move turns $4x + 7 = x + 19$ into $3x = 12$?

- ① The x terms are cancelled against each other.

- ② The dirhams are cancelled too.

- ③ That is al-muqabala, balancing.

ANSWER

Al-muqabala

EXAMPLE 3 Write “four things less two dirhams equal ten dirhams” as an equation and solve it.

① $4x - 2 = 10$

② Al-jabr: $4x = 12$.

③ $x = 3$

ANSWER

$x = 3$

03 Interactive model

Have the class solve one equation entirely in words, with no symbols at all; the effort involved explains better than any lecture why notation was worth inventing.

Which move is this?

$5x - 3 = 12$ becomes $5x = 15$ by:

al-jabr

al-muqabala

division

expanding

$5x + 3 = 2x + 12$ becomes $3x = 9$ by:

al-jabr

al-muqabala

division

squaring

The word “algebra” comes from:

al-muqabala

al-jabr

his name

Arabic for “number”

The word “algorithm” comes from:

al-jabr

al-muqabala

his name

Greek

He was born in:

Khorezm

Baghdad

Cairo

Samarkand

In his work, a “thing” means:

x^2

x

the constant

the answer

A “dirham” means:

x

x^2

the constant term

a fraction

He could not use:

addition

negative numbers

multiplication

fractions

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Al-Khwarizmi	Al-Xorazmiy	Аль-Хорезми
Al-jabr — restoring	Al-jabr — to'ldirish	Аль-джабр — восполнение
Al-muqabala — balancing	Al-muqobala — muvozanatlash	Аль-мукабала — сопоставление
Algebra	Algebra	Алгебра
Algorithm	Algoritm	Алгоритм
Manuscript	Qo'lyozma	Рукопись
House of Wisdom	Bayt ul-hikma	Дом мудрости
Heritage	Meros	Наследие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Al-jabr means:

balancing

restoring

dividing

squaring

Al-muqabala means:

restoring

balancing

expanding

factorising

Al-Khwarizmi worked at:

the House of Wisdom

Oxford

Alexandria

Bukhara only

His century was the:

$2x - 7 = x + 1 \rightarrow 2x = x + 8$ is:

His proofs were:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Al-jabr means

ANSWER restoring

2. Al-muqabala means

ANSWER balancing

3. “Algebra” comes from

ANSWER al-jabr

4. “Algorithm” comes from

ANSWER his name

5. A “thing” is

ANSWER x

6. A “dirham” is

ANSWER the constant term

7. $5x - 3 = 12$ after al-jabr

ANSWER $5x = 15$

Medium · the standard question

7 PROBLEMS

1. $5x - 3 = 2x + 9$

ANSWER $x = 4$

2. $4x + 7 = x + 19$

ANSWER $x = 4$

3. “Four things less two dirhams equal ten dirhams”

ANSWER $4x - 2 = 10, x = 3$

4. $2x - 7 = x + 1$: which move first?

ANSWER Al-jabr

5. $5x + 3 = 2x + 12$: which move first?

ANSWER Al-muqabala

6. Why could he not write -3 ?

ANSWER Negative numbers were not in use

7. In which country was he born?

ANSWER In Khorezm, in present-day Uzbekistan

Hard · stretch and reason

7 PROBLEMS

1. Write $7x - 5 = 3x + 11$ as a medieval sentence

ANSWER "Seven things less five dirhams equal three things and eleven dirhams"

2. Solve it, naming both moves

ANSWER Al-jabr, then al-muqabala; $x = 4$

3. Why do the two moves suffice for any linear equation?

ANSWER One removes subtractions, the other removes repetitions

4. What does the modern phrase "change the sign" correspond to?

ANSWER Al-jabr

5. What does "collect like terms" correspond to?

ANSWER Al-muqabala

6. Why did he prove $(x + 3)^2$ with a diagram?

ANSWER There was no symbolic algebra to argue with

7. Name the two words modern mathematics owes him

ANSWER Algebra and algorithm

07 Homework

Homework — 5 tasks

Name the move beside every line, as he would have done.

1. Solve $6x - 4 = 3x + 11$, naming al-jabr and al-muqabala where you use them.
2. Write "six things less one dirham equal three things and eight dirhams" as an equation and solve it.
3. Explain in three sentences why algebra is named after a book written in Baghdad.
4. Write $9x + 2 = 4x + 22$ as a medieval sentence.

5. Find out one more contribution of Al-Khwarizmi and write two sentences about it.

Solving problems with the help of equations

Name the unknown, write the sentence as an equation, solve it, and answer the question that was asked.

LESSON 127–128

2 × 40 MIN

MATEMATIKA 7, §57

S8 2.5

LESSON CLOCK

9'

23'

24'

19'

5'

The five steps	9 min
Choosing the unknown	23 min
Three families of problem	24 min
Checking against the words	19 min
Homework	5 min

LEARNING OBJECTIVES

- Choose the unknown so that the other quantities are easy to express.
- Translate a worded condition into an equation.
- Solve the equation and check the root against the original words.
- Reject a root that is impossible in the situation described.

TEXTBOOK

National	pp. 378–386
Cambridge	Stage 8, pp. 31–36
Workbook	Exercise 2.6

01

Explanation

The five steps

STEP	WHAT TO DO
1	let x be one of the unknown quantities, and say in words which
2	write every other quantity in terms of x
3	find the sentence in the problem that is an equality, and write it as an equation
4	solve the equation
5	answer the question that was asked, in words and units

STEP 5 IS NOT AUTOMATIC

If x is the son's age but the question asks for the father's, the root is not the answer. Read the question again after solving — this is where whole marks are lost for correct algebra.

Choosing the unknown

Let x be the quantity that makes the others simplest to write — usually the smallest one, or the one the others are compared with.

PROBLEM	LET x BE	THEN
a father is 3 times as old as his son	the son's age	the father is $3x$
two numbers differ by 7	the smaller	the larger is $x + 7$
a rectangle is 4 <i>cm</i> longer than it is wide	the width	the length is $x + 4$
a sum of 120 split in the ratio 2 : 3	one part of the ratio	the shares are $2x$ and $3x$

A BAD CHOICE IS NOT WRONG, ONLY SLOWER

Taking the father's age as x makes the son $\frac{x}{3}$ and brings fractions into the equation. The answer is the same; the work is not.

Three families of problem

FAMILY	PROBLEM	EQUATION	ANSWER
ages	a father is 3 times as old as his son; in 12 years he will be twice as old	$3x + 12 = 2(x + 12)$	son 12, father 36
numbers	two numbers differ by 7 and add to 39	$x + (x + 7) = 39$	16
geometry	a rectangle is 4 <i>cm</i> longer than wide, perimeter 36 <i>cm</i>	$2(x + x + 4) = 36$	7 <i>cm</i>
sharing	120 split in the ratio 2 : 3	$2x + 3x = 120$	48

FIND THE EQUALITY SENTENCE

Every problem contains exactly one sentence that says two things are equal — “add to”, “will be twice”, “the perimeter is”. That sentence, and only that one, becomes the equation.

Checking against the words

The check is against the problem, not against the equation. And a root that cannot be true of the situation must be rejected.

ROOT	QUESTION	VERDICT
$x = 12$	the son's age	accepted
$x = -5$	a length	rejected — a length is positive
$x = 7.5$	a number of pupils	rejected — it must be whole
$x = 0$	a speed	usually rejected — nothing moves

REJECTING A ROOT IS PART OF THE ANSWER, NOT A FAILURE

An equation knows nothing about ages and lengths. Deciding which of its roots the situation allows is the mathematician's job, and saying so explicitly earns the mark.

02

Worked examples

EXAMPLE 1 A father is 3 times as old as his son. In 12 years he will be twice as old. How old is each now?

1

Let x be the son's age; the father is $3x$.

2

In 12 years: son $x + 12$, father $3x + 12$.

3

$3x + 12 = 2(x + 12)$
The equality sentence.

4

$3x + 12 = 2x + 24$, so $x = 12$.
Son 12, father 36.

ANSWER

Son 12, father 36

EXAMPLE 2 The perimeter of a rectangle is 36 cm and it is 4 cm longer than it is wide. Find its sides.

- ① Let the width be x ; the length is $x + 4$.
- ② $2(x + x + 4) = 36$
- ③ $4x + 8 = 36$, so $x = 7$.
- ④ Width 7 cm, length 11 cm. Check: $2(7 + 11) = 36$ ✓

ANSWER

7 cm by 11 cm

EXAMPLE 3 Two numbers differ by 7 and add to 39. Find them.

- ① Let the smaller be x ; the larger is $x + 7$.
- ② $x + (x + 7) = 39$
- ③ $2x = 32$, so $x = 16$.
- ④ The numbers are 16 and 23.

ANSWER

16 and 23

03 Interactive model

Ask the class to write only step 1 and step 3 for six problems, without solving any of them; translation is the skill, and separating it from the arithmetic doubles the practice.

Translate the sentence

“A number increased by 5” is:

$$5x$$

$$x + 5$$

$$x - 5$$

$$\frac{x}{5}$$

“Three times a number, less 2” is:

$$3x - 2$$

$$3(x - 2)$$

$$2 - 3x$$

$$x - 6$$

“Three times a number less 2, all doubled” is:

$$2 \cdot 3x - 2$$

$$2(3x - 2)$$

$$6x - 4 \text{ only}$$

$$3x - 4$$

If the son is x and the father is 3 times as old, the father is:

$x + 3$

$3x$

$$\frac{x}{3}$$

$x - 3$

In 12 years the father will be:

$3x$

$3x + 12$

$3(x + 12)$

$3x - 12$

The equation is then:

$3x + 12 = 2(x + 12)$

$3x = 2x + 12$

$3x + 12 = 2x$

$3(x + 12) = 2x$

A rectangle 4 longer than wide has perimeter:

$$2x + 4$$

$$4x + 8$$

$$x(x + 4)$$

$$4x + 4$$

A root of -5 for a length is:

accepted

rejected

doubled

squared

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Word problem	Matnli masala	Текстовая задача
To denote	Belgilamoq	Обозначить
Condition	Shart	Условие
To translate	O'girmoq	Перевести
Total	Jami	Всего
Speed	Tezlik	Скорость
Impossible answer	Mos kelmaydigan javob	Посторонний ответ
To interpret	Talqin qilmoq	Истолковать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first step is:

writing the equation

naming the unknown

solving

checking

The equation comes from:

the question asked

the equality in the problem

the units

the answer

If the son is x and the father $3x$, in 12 years the father is:

$3x$

$3x + 12$

$3(x + 12)$

$3x + 36$

A rectangle 4 *cm* longer than wide with perimeter 36 has width:

Two numbers differ by 7 and add to 39: the smaller is:

A root of 7.5 pupils is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. "A number increased by 5"

ANSWER $x + 5$

2. "Three times a number, less 2"

ANSWER $3x - 2$

3. "Half a number"

ANSWER $\frac{x}{2}$

4. A father is 3 times a son aged x

ANSWER $3x$

5. Two numbers differing by 7, smaller x

ANSWER x and $x + 7$

6. A rectangle of width x and length $x + 4$: its perimeter

ANSWER $4x + 8$

7. 120 in the ratio 2 : 3

ANSWER $2x$ and $3x$

Medium · the standard question

7 PROBLEMS

1. Two numbers differ by 7 and add to 39

ANSWER 16 and 23

2. A rectangle 4 cm longer than wide, perimeter 36 cm

ANSWER 7 cm by 11 cm

3. 120 split in the ratio 2 : 3

ANSWER 48 and 72

4. A father 3 times his son; in 12 years twice as old

ANSWER Son 12, father 36

5. A number is 5 more than another and they total 31

ANSWER 13 and 18

6. Three consecutive numbers add to 72

ANSWER 23, 24, 25

7. A pencil costs x and a pen $3x$; together 8000 sum

ANSWER Pencil 2000, pen 6000

Hard · stretch and reason

7 PROBLEMS

1. A mother is 4 times her daughter; in 20 years twice as old

ANSWER Daughter 10, mother 40

2. The perimeter of a rectangle is 44 and the length is 3 times the width

ANSWER 5.5 by 16.5

3. Three consecutive even numbers add to 90

ANSWER 28, 30, 32

4. A sum of 300 is split so that the second share is 20 more than the first and the third twice the first

ANSWER 70, 90, 140

5. Half a number less 3 equals a third of it plus 1

ANSWER $x = 24$

6. A father was 5 times his son 4 years ago and is 3 times now

ANSWER Son 8, father 24

7. Why must a root of -5 be rejected for a length?

ANSWER A length cannot be negative

07 Homework

Homework — 5 tasks

Write “Let x be ...” in words before any algebra, and answer in words at the end.

- Two numbers differ by 9 and add to 45. Find them.
- A rectangle is 5 cm longer than it is wide and has perimeter 42 cm. Find its sides.
- A mother is 4 times as old as her daughter; in 20 years she will be twice as old. Find both ages.
- Split 210 in the ratio 3 : 4.

5. Three consecutive numbers add to 81. Find them.

Control work 9, and work on the mistakes

Linear equations and the problems behind them — Quarter III closed.

LESSON 129–130

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 9

S8 2.5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Solve linear equations with brackets and with fractions.
- Interpret the cases of no roots and of every number being a root.
- Turn a worded problem into an equation and answer the question asked.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 358–386
Cambridge	Stage 8, pp. 31–36
Workbook	Control paper M9

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Solve $4x - 7 = 2x + 9$	3	L123–124
2	Solve $3(x + 2) = 5x - 8$	4	L123–124
3	Solve $\frac{x}{2} + \frac{x}{5} = 7$	4	L123–124
4	Solve $2(x + 4) = 2x + 8$ and say how many roots it has	4	L122–124
5	Solve $5x - 3 = 2x + 9$ naming al-jabr and al-muqabala	4	L125–126
6	Two numbers differ by 9 and add to 45: find them	6	L127–128

WHERE THE MARKS ACTUALLY GO

Q3 carries two marks for multiplying the 7 by 10 as well as the fractions; Q4 two for the words “every number is a root”; Q6 one for “let x be the smaller number”, two for the equation and one for answering with both numbers.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
sign not changed on transposing	$4x + 2x = 9 - 7$	$4x - 2x = 9 + 7$
bracket expanded partly	$3x + 2 = 5x - 8$	$3x + 6 = 5x - 8$
whole-number term not multiplied	$5x + 2x = 7$	$5x + 2x = 70$
identity read as no roots	$0 = 0$	every number is a root
moves not named	a correct but unlabelled solution	al-jabr, then al-muqabala
unknown not declared	an equation with no “let”	“let x be the smaller number”
only one number given	16	18 and 27

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Quarter III as one map

BLOCK	THE SENTENCE
parallel lines	an angle fact proves parallelism; parallelism proves an angle fact
triangles	the angles add to 180° , and an exterior angle is the two remote ones
right triangles	the acute angles add to 90° ; the leg opposite 30° is half the hypotenuse
abridged formulae	five products worth knowing by heart, in both directions
factorising	common factor, formula, grouping, splitting — in that order
algebraic fractions	cancel factors only; add over the lowest common denominator
equations	reduce to $ax = b$, then check in the original

LOOKING FORWARD

Quarter IV opens with percentages and constructions, then coordinates, functions, systems of two equations and combinatorics — and every one of those uses the equation work of this block.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\frac{x}{2} + \frac{x}{5} = 7$.

- ① The LCD is 10.
- ② Multiply every term: $5x + 2x = 70$.
The 7 is multiplied too.
- ③ $7x = 70$
- ④ $x = 10$. Check: $5 + 2 = 7$ ✓

ANSWER

$$x = 10$$

EXAMPLE 2 Model answer, Q4: $2(x + 4) = 2x + 8$.

- ① $2x + 8 = 2x + 8$
Expand.
- ② $0 \cdot x = 0$
- ③ True for every value of x .
- ④ It is an identity: every number is a root.

ANSWER

Every number

EXAMPLE 3 Model answer, Q6: two numbers differ by 9 and add to 45.

- ① Let x be the smaller; the larger is $x + 9$.
- ② $x + (x + 9) = 45$
- ③ $2x = 36$, so $x = 18$.
- ④ The numbers are 18 and 27. Check: $18 + 27 = 45$ ✓

ANSWER

18 and 27

03 Interactive model

Give back Q6 and ask only whether the final sentence answers the question; the algebra is usually right and the answer usually incomplete.

Quarter III in twelve questions

On parallel lines, co-interior angles:

are equal

add to 90°

add to 180°

are right

Equal alternate angles prove:

parallelism

congruence

nothing

perpendicularity

The angles of a triangle add to:

90°

180°

270°

360°

An exterior angle equals:

the adjacent angle

the two remote angles

all three

90°

In a right triangle the leg opposite 30° is:

the hypotenuse

half the hypotenuse

twice it

equal to the other leg

$(a + b)^2$ equals:

$a^2 + b^2$

$a^2 + 2ab + b^2$

$a^2 - 2ab + b^2$

$2a + 2b$

$x^2 - 49$ factorises to:

$(x - 7)^2$

$(x - 7)(x + 7)$

$(x - 49)(x + 1)$

it does not

The first factorising method to try is:

a formula

the common factor

grouping

splitting

In an algebraic fraction you may cancel:

terms

factors

anything

signs

When subtracting fractions the second numerator needs:

a bracket

a denominator

nothing

a factor

$0 \cdot x = 0$ means:

no roots

one root

every number is a root

an error

Al-jabr means:

balancing

restoring

dividing

checking

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Linear equation	Chiziqli tenglama	Линейное уравнение
Root	Ildiz	Корень
To transpose	O'tkazish	Перенести
To clear fractions	Maxrajdan qutulish	Избавиться от знаменателей
Word problem	Matnli masala	Текстовая задача
Check	Tekshirish	Проверка
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 gives:

$x = 4$

$x = 8$

$x = 1$

$x = 16$

Q2 begins by:

dividing

expanding the bracket

clearing fractions

checking

Q3 is multiplied through by:

2

5

7

10

Q4 has:

no roots

one root

every number as a root

two roots

Q5 asks you to name:

the roots

al-jabr and al-muqabala

the coefficients

the check

Q6 is answered with:

one number

both numbers

the equation

x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $4x - 7 = 2x + 9$

ANSWER $x = 8$

2. $3(x + 2) = 5x - 8$

ANSWER $x = 7$

3. $\frac{x}{2} + \frac{x}{5} = 7$

ANSWER $x = 10$

4. $2(x + 4) = 2x + 8$

ANSWER Every number

5. $5x - 3 = 2x + 9$

ANSWER $x = 4$

6. Al-jabr means

ANSWER restoring

7. Al-muqabala means

ANSWER balancing

Medium · the standard question

7 PROBLEMS

1. Two numbers differ by 9 and add to 45

ANSWER 18 and 27

2. $2(x - 3) = 2x - 5$

ANSWER No roots

3. $\frac{x}{3} - \frac{x}{4} = 1$

ANSWER $x = 12$

4. On parallel lines, co-interior $5x$ and $4x$

ANSWER 100° and 80°

5. $x^2 - 64$

ANSWER $(x - 8)(x + 8)$

6. $\frac{x^2 - 4}{x + 2}$

ANSWER $x - 2$

7. An exterior angle of 130° , one remote angle 60° : the other

ANSWER 70°

Hard · stretch and reason

7 PROBLEMS

1. $\frac{x+2}{3} - \frac{x-1}{4} = 1$

ANSWER $x = 1$

2. A rectangle 5 cm longer than wide, perimeter 42 cm

ANSWER 8 cm by 13 cm

3. Three consecutive numbers add to 81

ANSWER 26, 27, 28

4. $3(2x - 1) - 2(3x - 2) = 1$

ANSWER Every number

5. Factorise $2x^3 - 8x$

ANSWER $2x(x - 2)(x + 2)$

6. $\frac{1}{x-2} + \frac{1}{x+2}$

ANSWER $\frac{2x}{x^2 - 4}$

7. A right triangle with hypotenuse 20 and an angle of 30°

ANSWER The opposite leg is 10

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Solve $5x - 4 = 3x + 10$ and check your root.
3. Solve $\frac{x}{4} + \frac{x}{6} = 5$.

4. Two numbers differ by 11 and add to 47 . Find them.
5. Write one sentence for each of the seven blocks on the Quarter III map.